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Port-Based Physical Modeling

Why do we need it and how does it work ?

About Physical, Port Based Modeling

Introduction

What is Physical Modeling ?

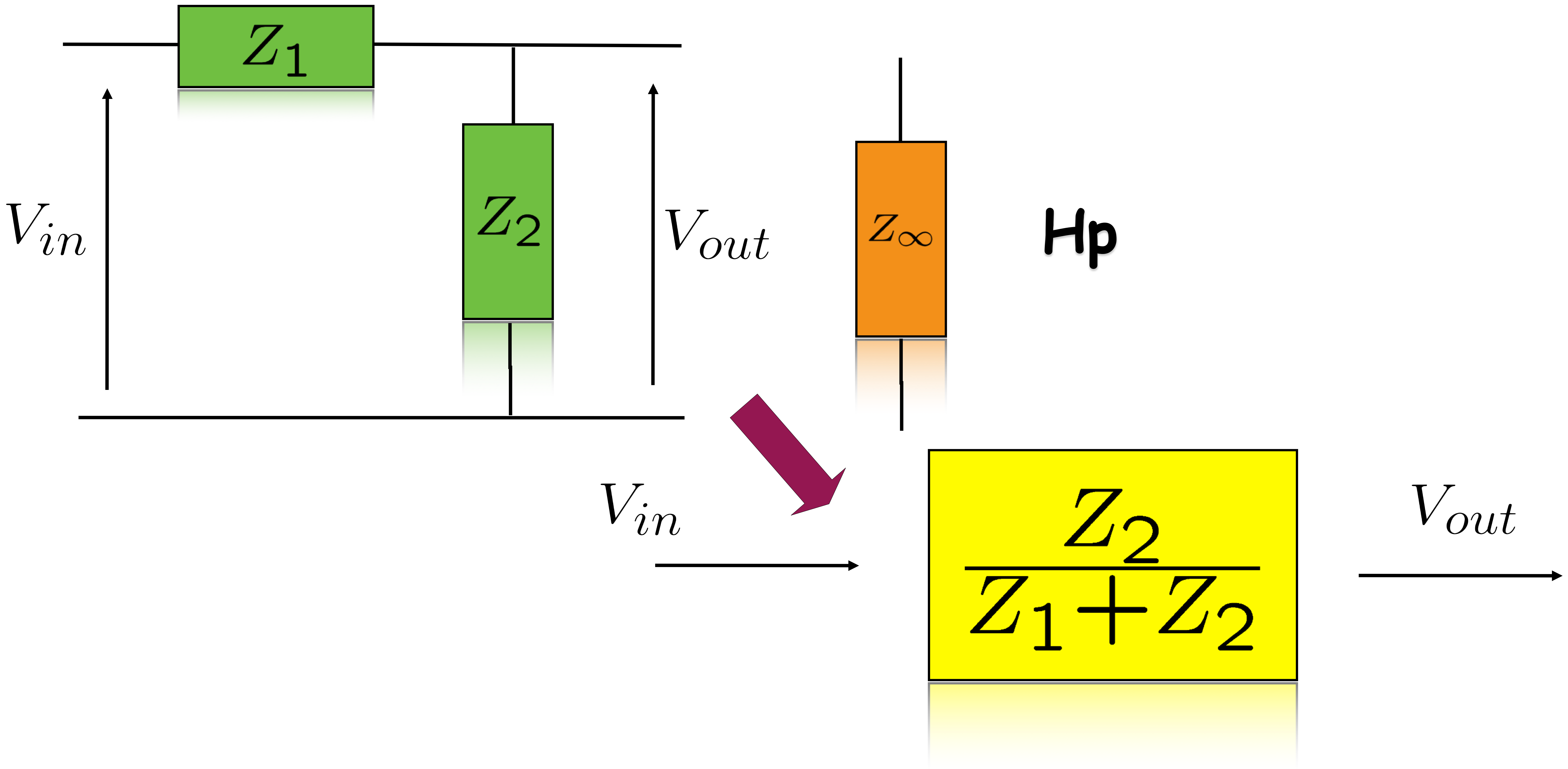
- Physical Modeling is to distill the essence of a physical system and describe it with tools which can show the interesting (dynamic) features of such a system: *model competence*

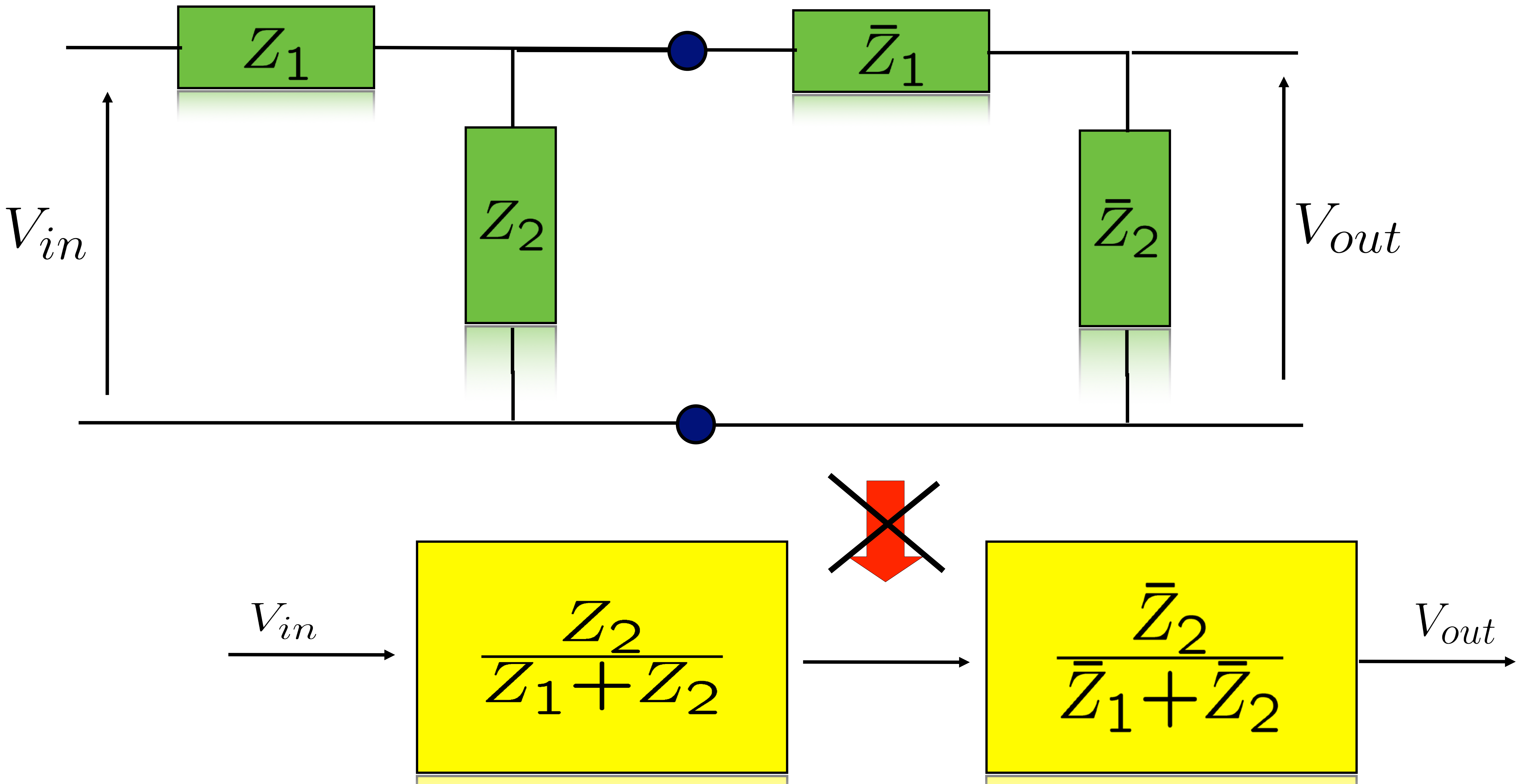
What is Port-Based Modeling ?

- Port-Based modelling is a way to decompose (tearing) the system to be modeled and explicitly model how such composing parts interact with explicit representation of the physical energy exchange among the parts

Conjecture

ANY PHYSICAL SYSTEM, WITH **LUMPED** OR **DISTRIBUTED** PARAMETERS **CAN BE MODELLED BY DECOMPOSING IT VIA PORTS** AND THIS GIVES A LOT OF INSIGHT IN THE SYSTEM DYNAMICS ITSELF AND OFTEN PREVENTS MAKING UNREALISTIC MODELLING CHOICES


Z_1
Z_2
V_{in}
V_{out}
Z_∞
Hp
V_{in}
$\frac{Z_2}{Z_1+Z_2}$
V_{out}



Z_1
Z_2
$\bar{Z}_1$
$\bar{Z}_2$
V_{in}
V_{out}
V_{in}
$\frac{Z_2}{Z_1+Z_2}$
$\frac{\bar{Z}_2}{\bar{Z}_1+\bar{Z}_2}$
V_{out}

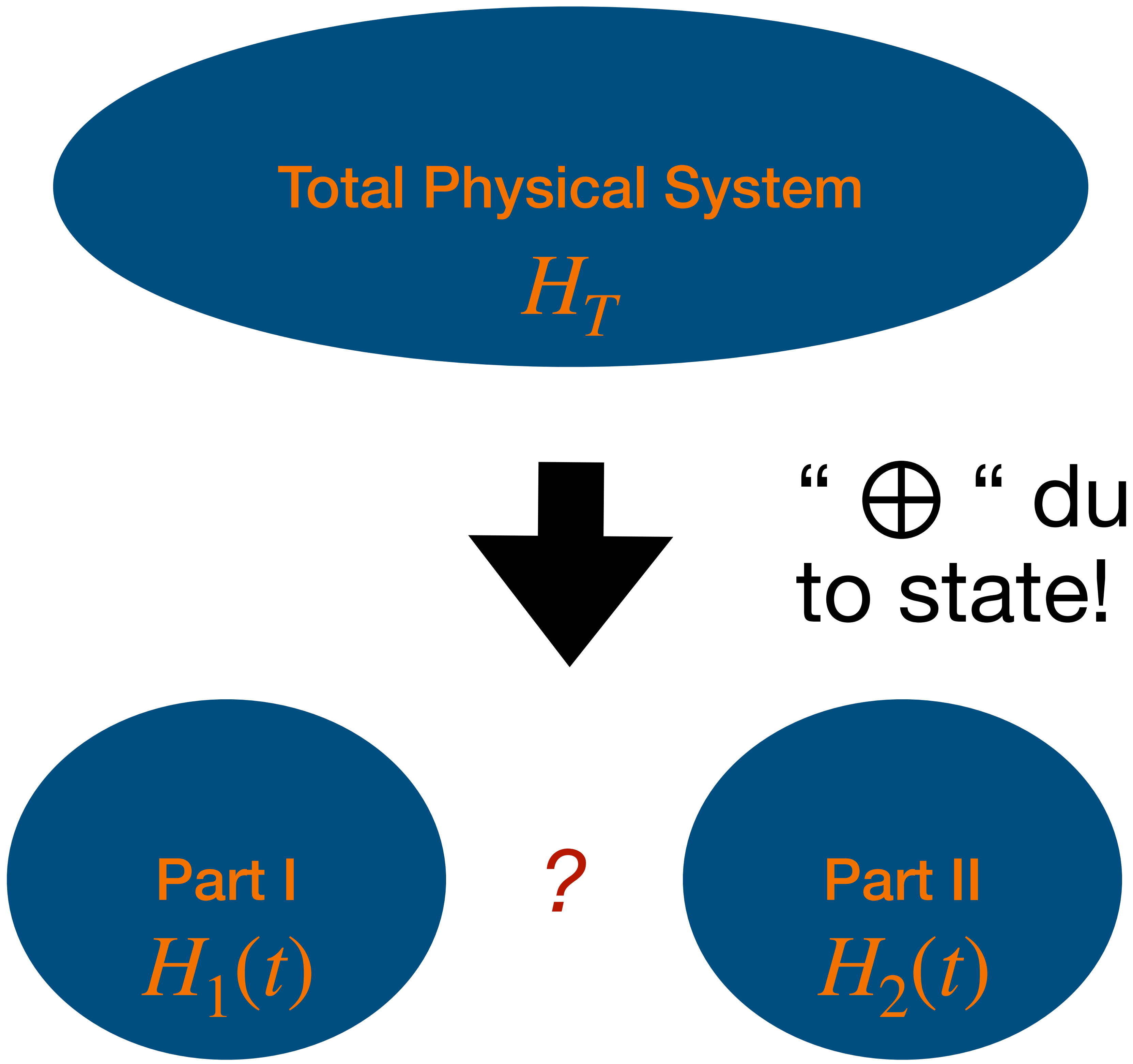
Conclusions from example

Introduction

- With Physical Systems, signal modelling is often not suitable
- Physical Energy governs dynamics
- Always a bi-directional (nonlinear) effect
- To model/control real OPEN systems like robots, signal modeling and block diagrams are NOT the solution and NOT a good tool!
- This is true also between domains: typical example DC motor gyration
- Robotics IS interconnection of multi-domain parts, we need something more !

Decomposition

Energy is the glue of physics: Energy Decomposition



" $\oplus$ " due to state!

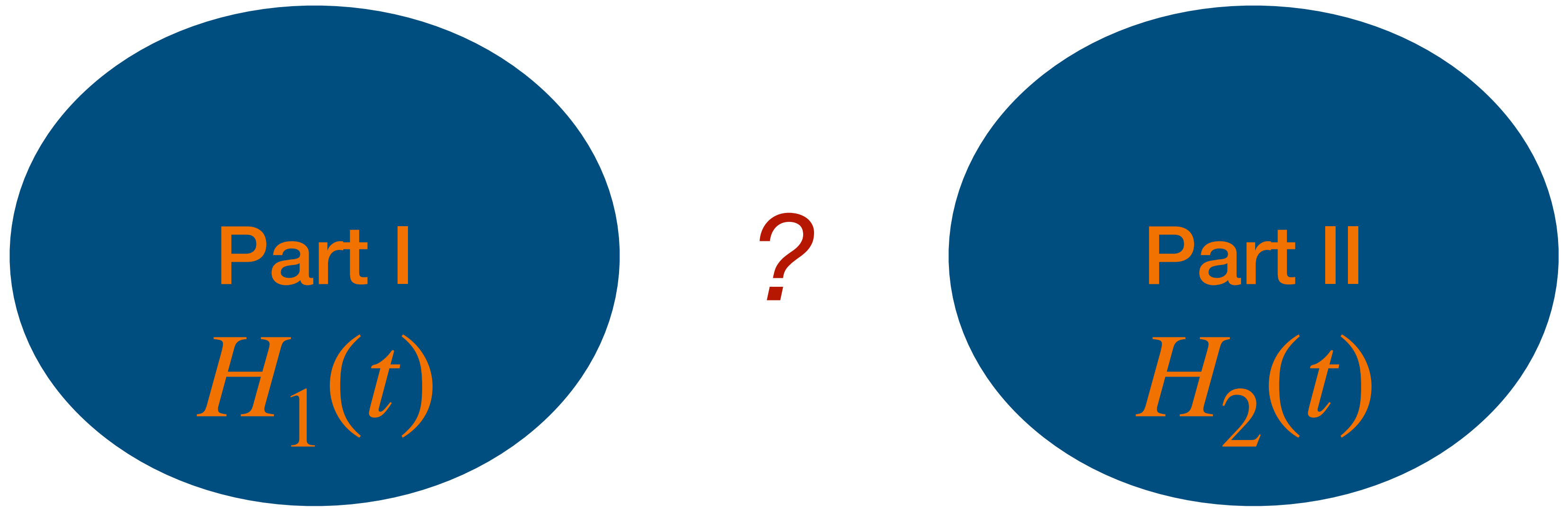
How to describe interconnection to ensure the requirement?

The (classical) Physical System under consideration will have an Energy $H(t) \in \mathbb{R}$ content at each instant of time, independent of its Physical nature! If the system is not interacting with anything else we have that $H(t) = H_T$ is constant.

Splitting the Physical System in two, each of the parts will have an Energy $H_1(t), H_2(t) \in \mathbb{R}$ and we want that for the first principle of thermodynamics, at all times $H_1(t) + H_2(t) = H_T$

Decomposition

Energy is the glue of physics: Energy Decomposition



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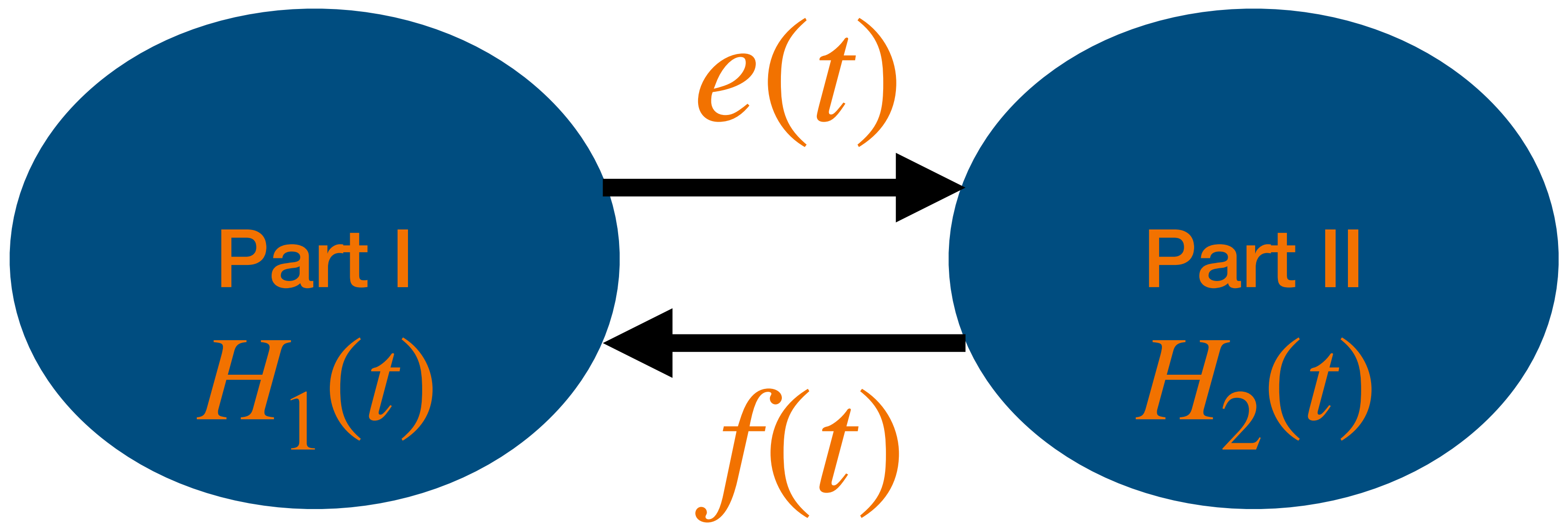
$$H_1(t) + H_2(t) = H_T \Rightarrow \boxed{\dot{H}_1(t) = -\dot{H}_2(t)}$$

Power Transfer!

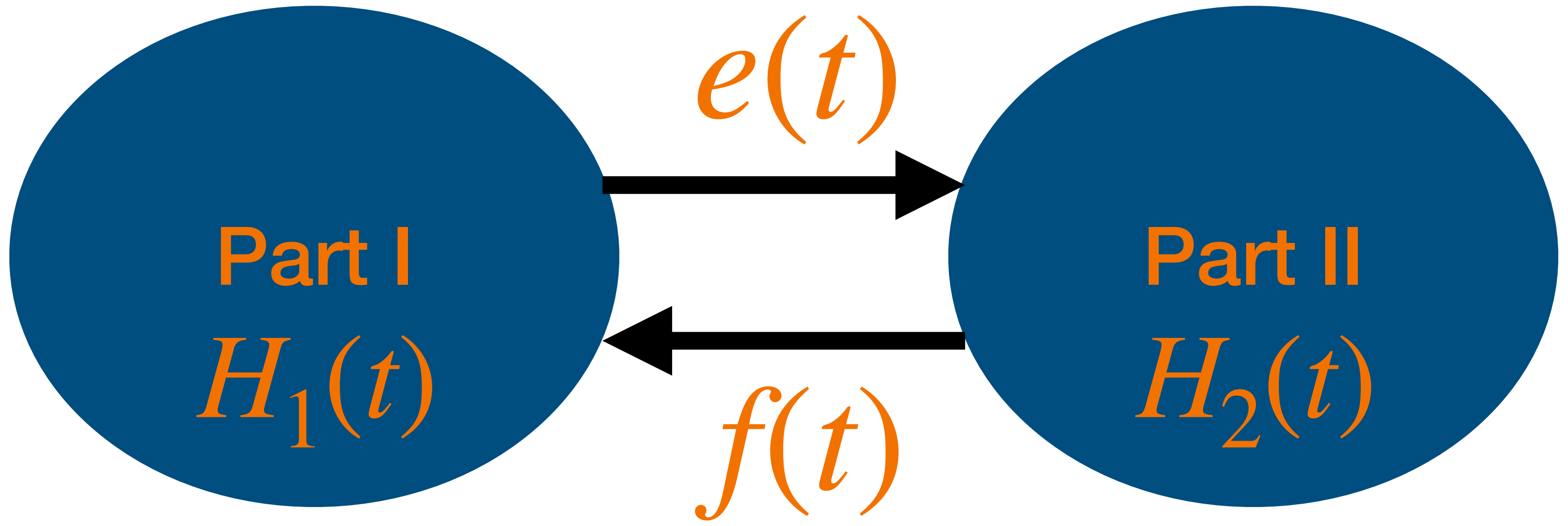


For Physical Reasons, the two parts will influence each other (bilaterally). Let's name $f(t)$ the influence of II on I and $e(t)$ the influence of I on II.

What is "the nature" of $e(t)$ and $f(t)$?

Decomposition

Energy is the glue of physics: Energy Decomposition



$$H_1(t) + H_2(t) = H_T \Rightarrow \boxed{\dot{H}_1(t) = -\dot{H}_2(t)}$$

Power Transfer!

$f(t), e(t)$ should characterise the interaction at each instant of time t and should also take care that they represent a **scalar** power transfer as desired.

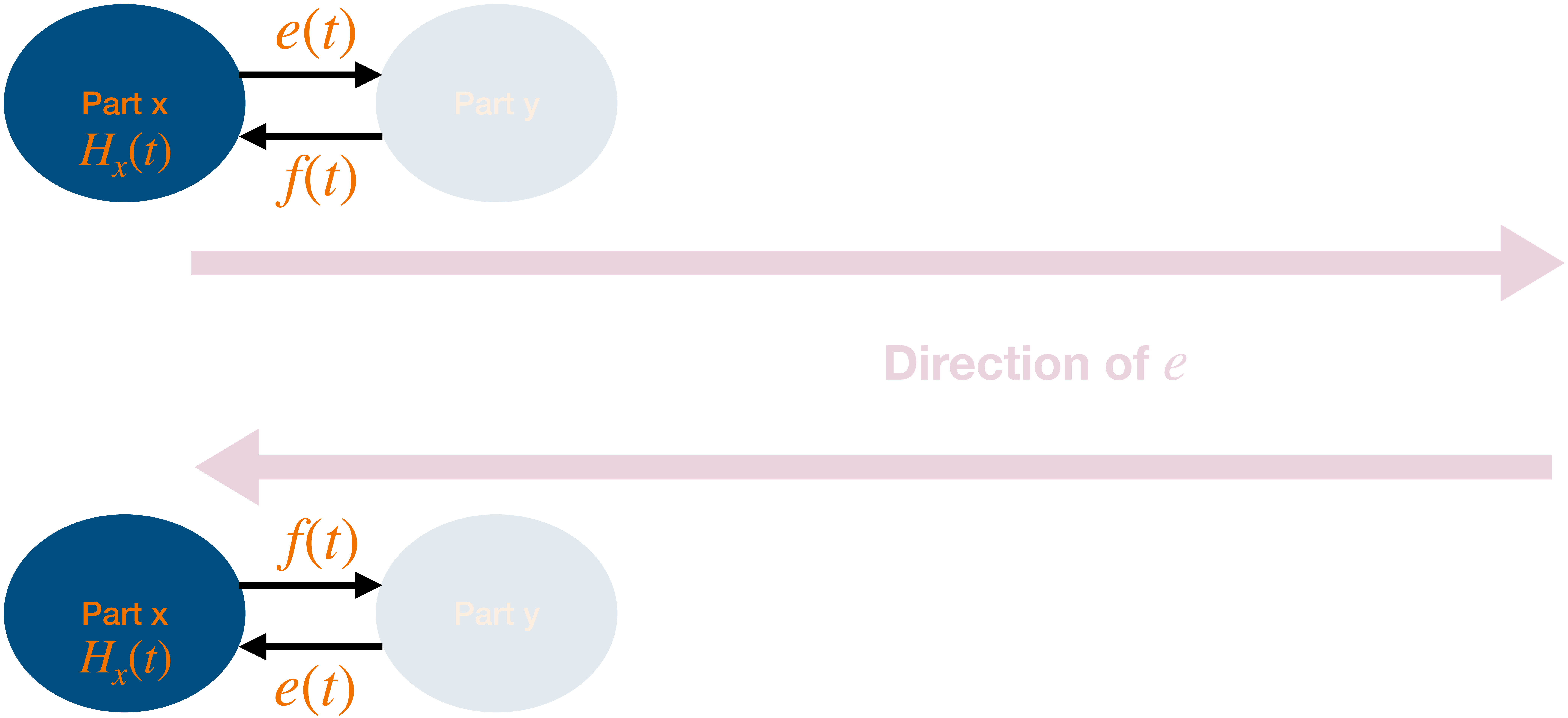
Solution

$f(t), e(t)$ should have **the nature of a vector and a co-vector** and the application of one on the other $e(f)$ will result in a **scalar which will have the value of the power transfer** at that time! Only problem left is that for one of the two systems we need a sign change either in $f(t)$ or in $e(t)$ (more later).

$$f(t) \in \mathscr{V}, e(t) \in \mathscr{V}^* \Rightarrow e(f)(t) \in \mathbb{R}$$

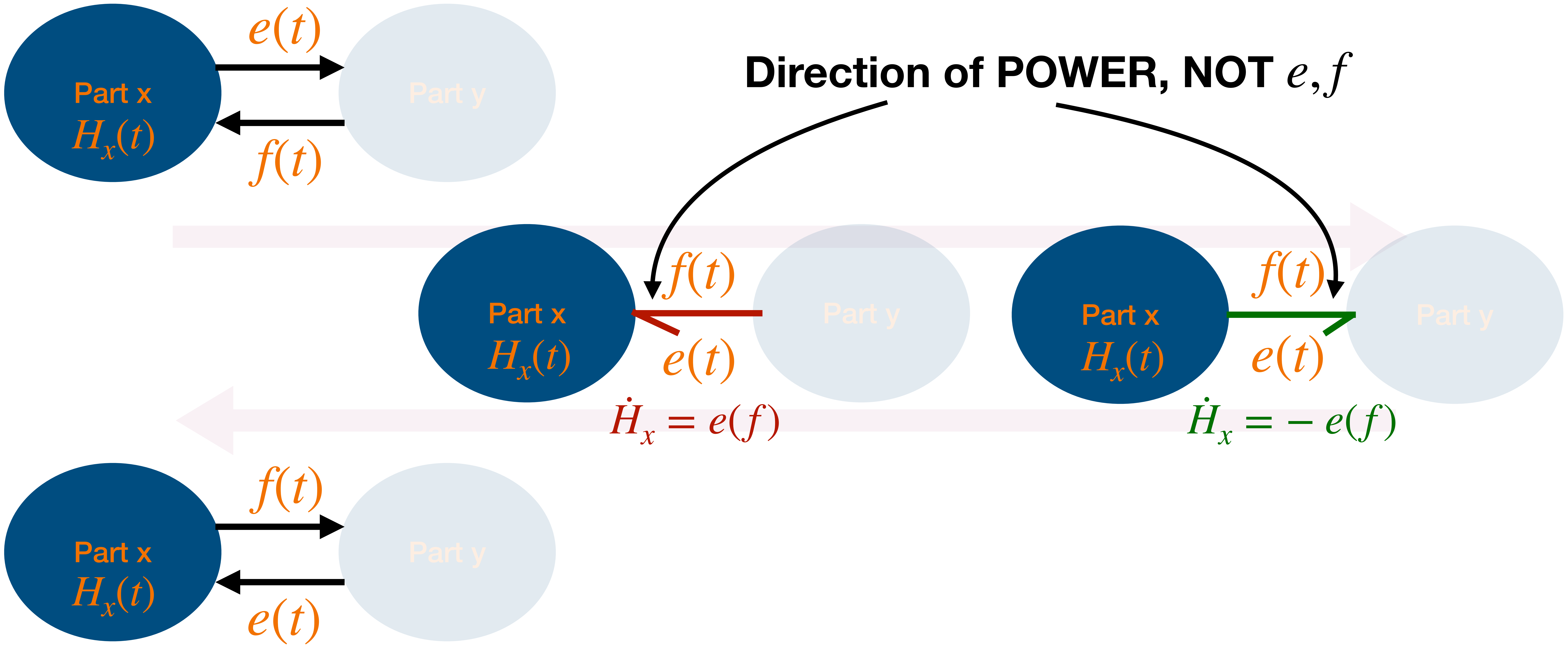
Power Port

POWER BOND: e called effort, f called flow



Power Port

Introducing a new notation: POWER BOND



Power Port

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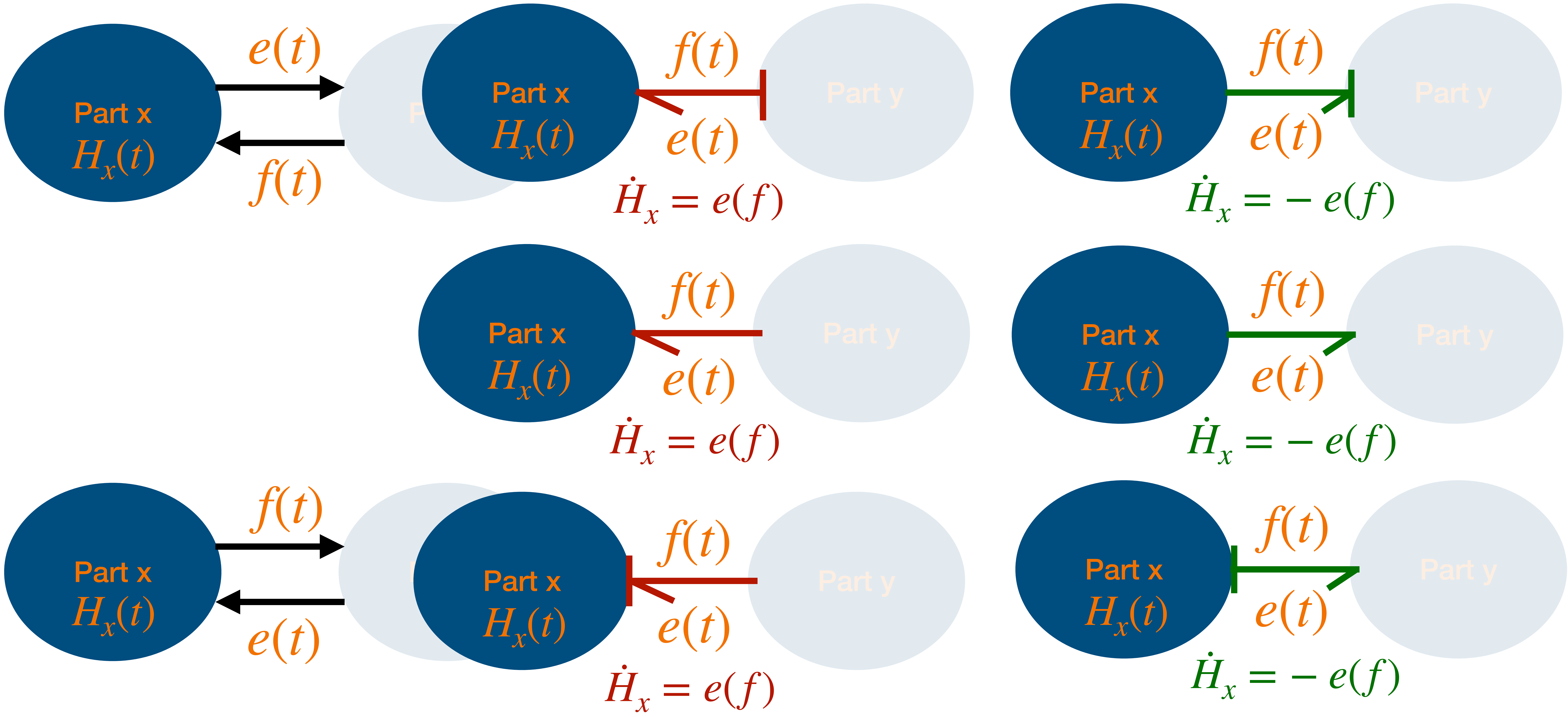
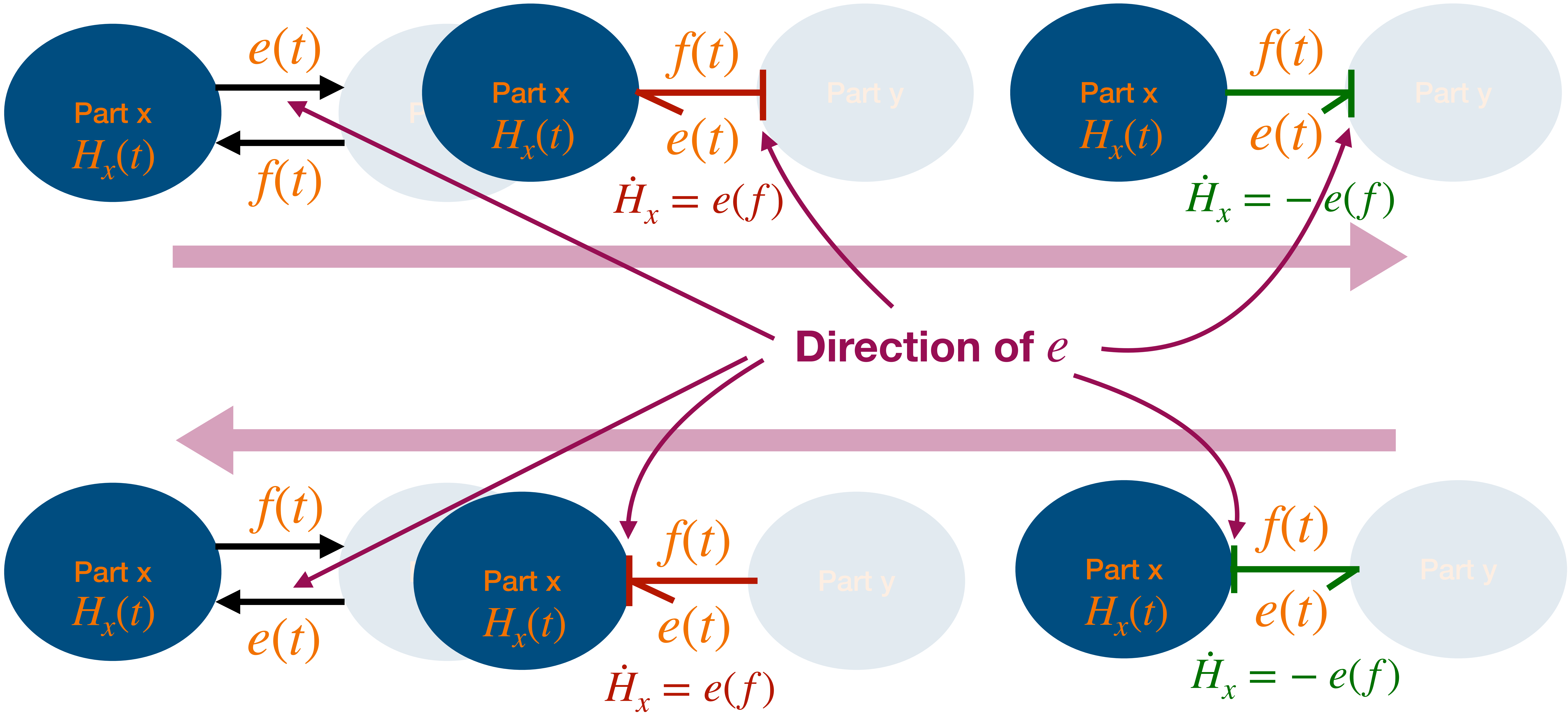


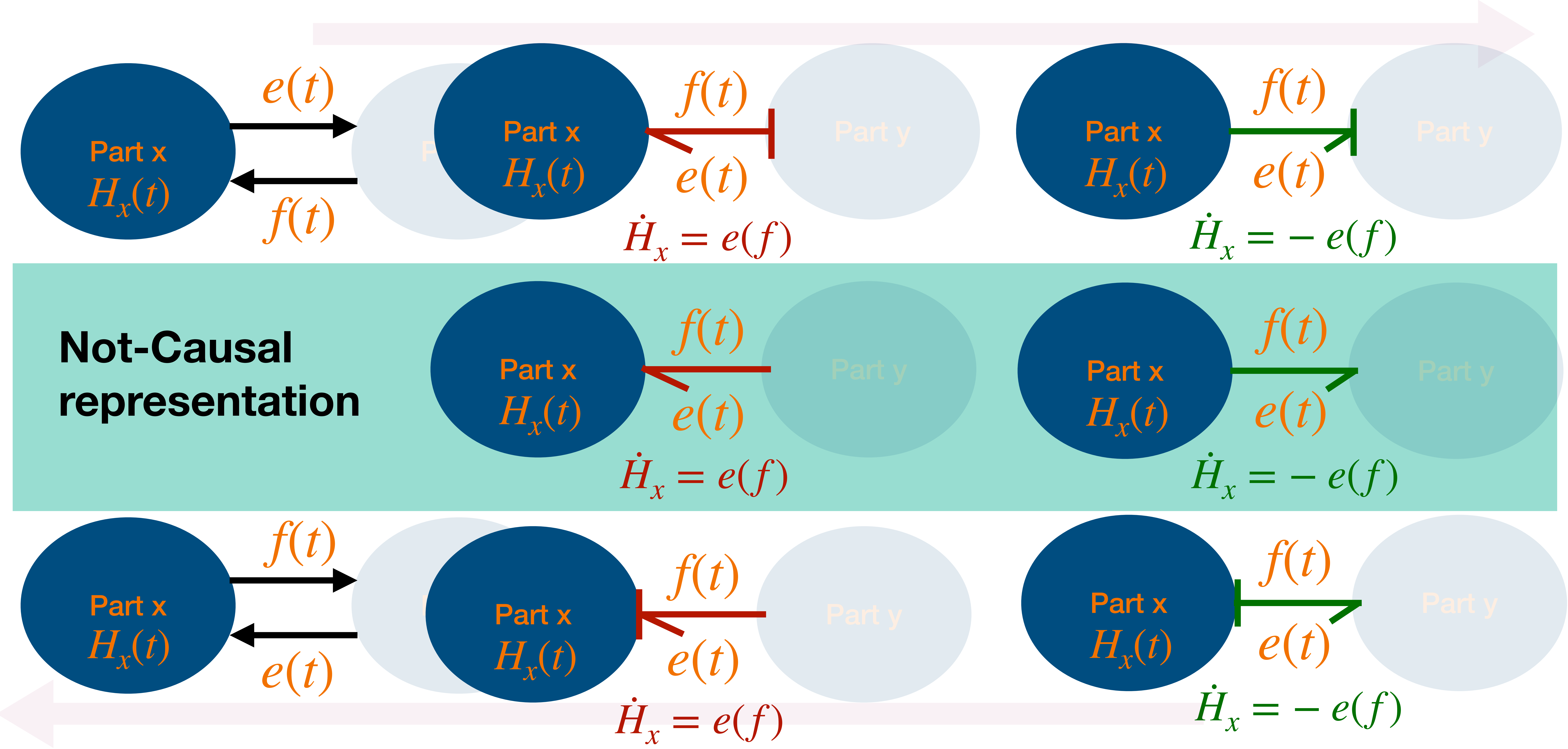
Power Port

Causal Stroke: direction of e



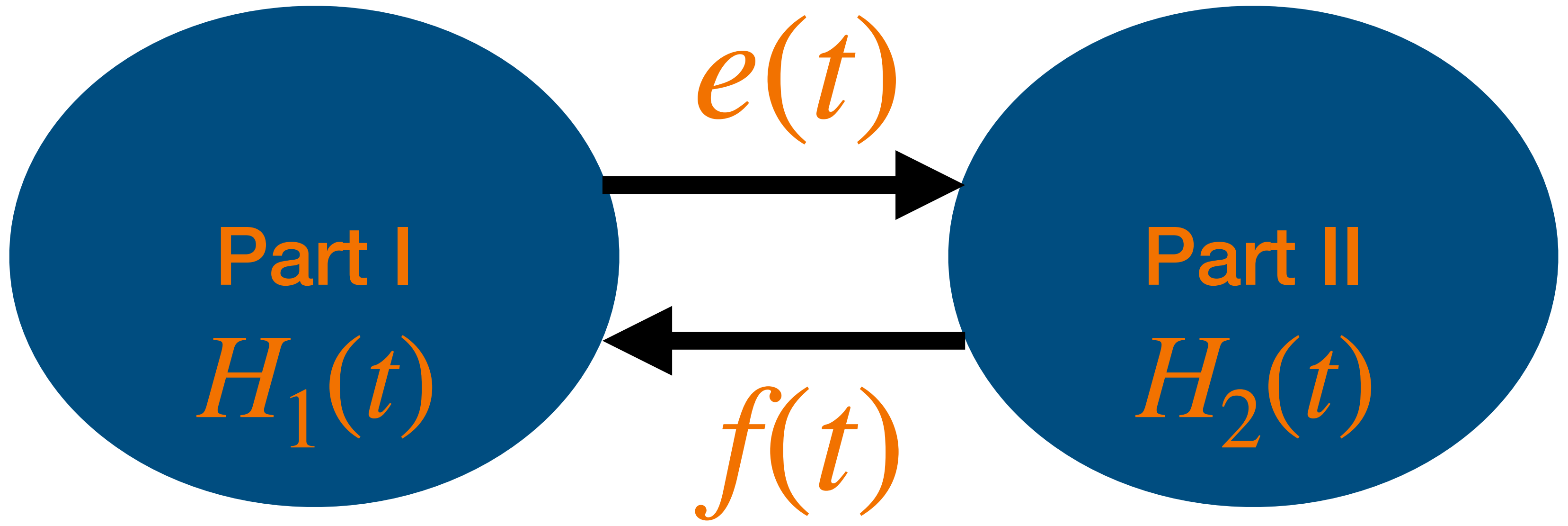
Power Port

All Possibilities: Direction of power IS INDEPENDENT of causal stroke



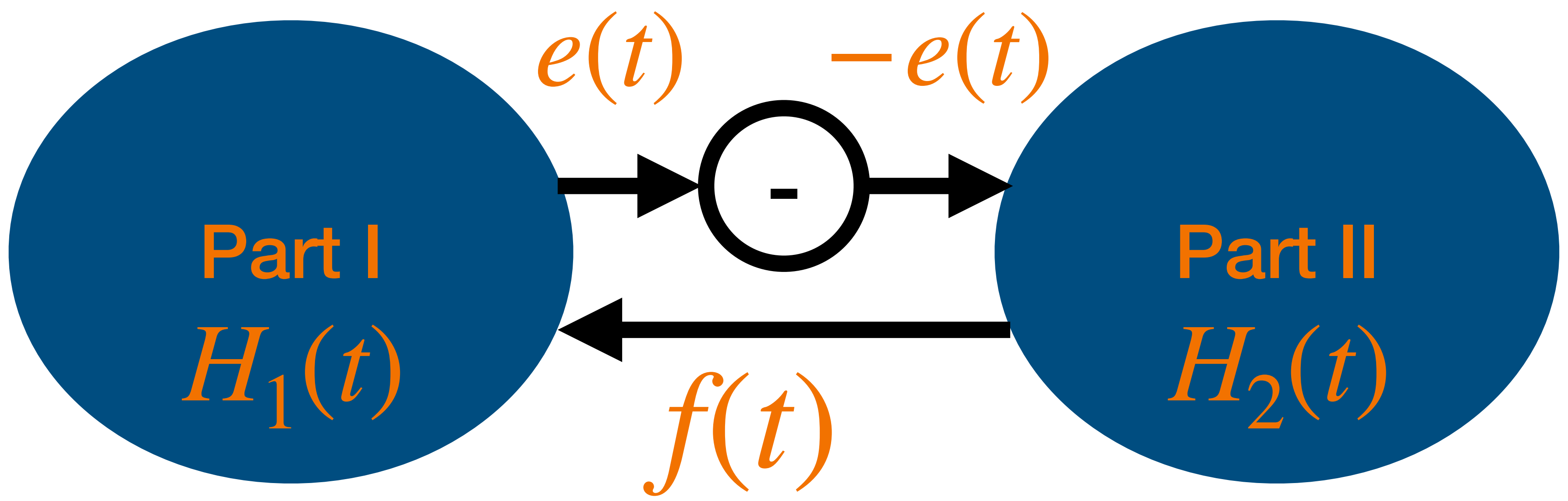
Solving the sign change

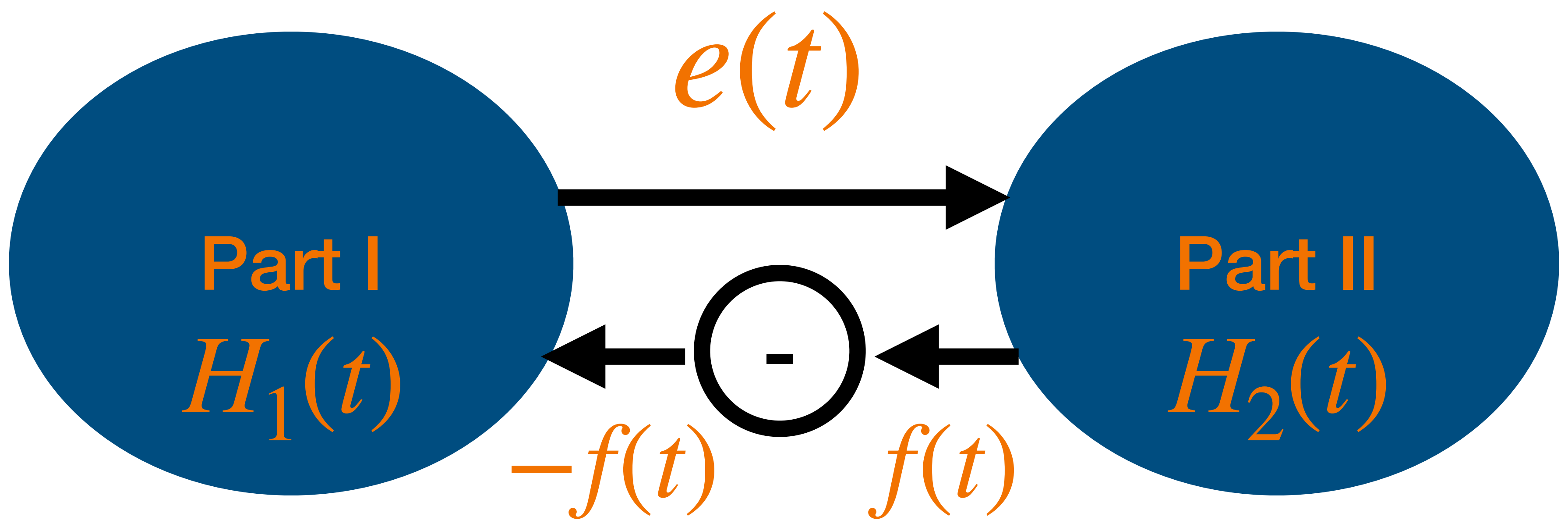
One of the two should flip sign to solve power direction



$$H_1(t) + H_2(t) = H_T \Rightarrow \boxed{\dot{H}_1(t) = -\dot{H}_2(t)}$$

Power Transfer!





We can take a “-“ because they are (co)vectors!!

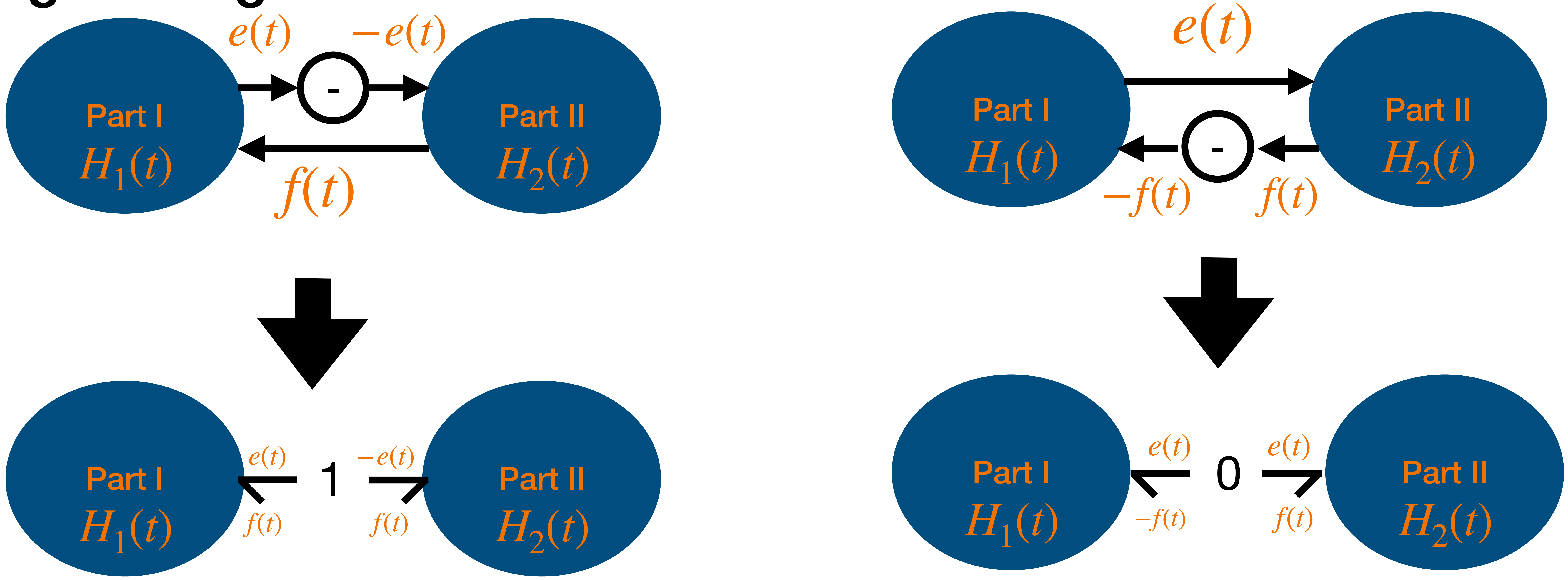
$$\dot{H}_1 = e(f), \dot{H}_2 = (-e)(f) = -e(f) \Rightarrow \dot{H}_1 = -\dot{H}_2$$

$$\dot{H}_1 = e(-f) = -e(f), \dot{H}_2 = e(f) = \Rightarrow \dot{H}_1 = -\dot{H}_2$$

as required. Sign problem solved!

Junctions notation for power bonds

Sign change



1: all bonds attached to it have **same flow** and **efforts sum algebraically**

0: all bonds attached to it have **same effort** and **flows sum algebraically**

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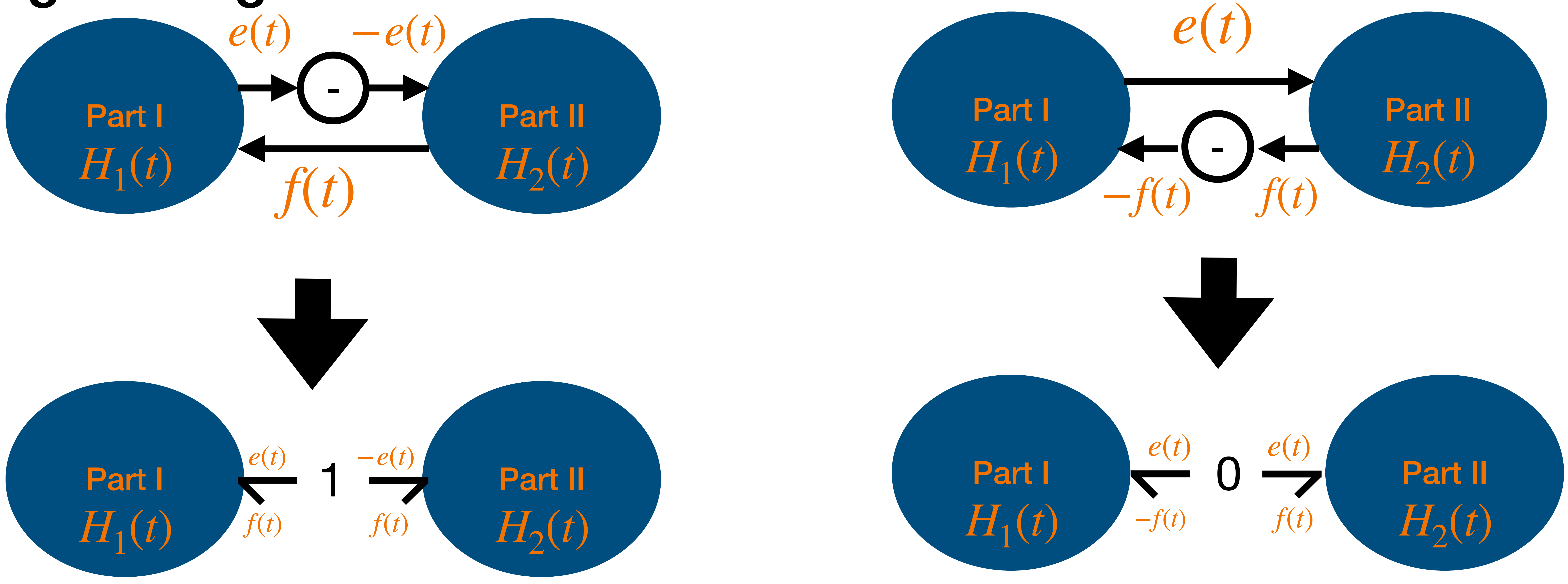
Junctions

Generalising Kirchhoff laws

Junctions notation for power bonds

Sign change



1: all bonds attached to it have **same flow** and **efforts sum algebraically**

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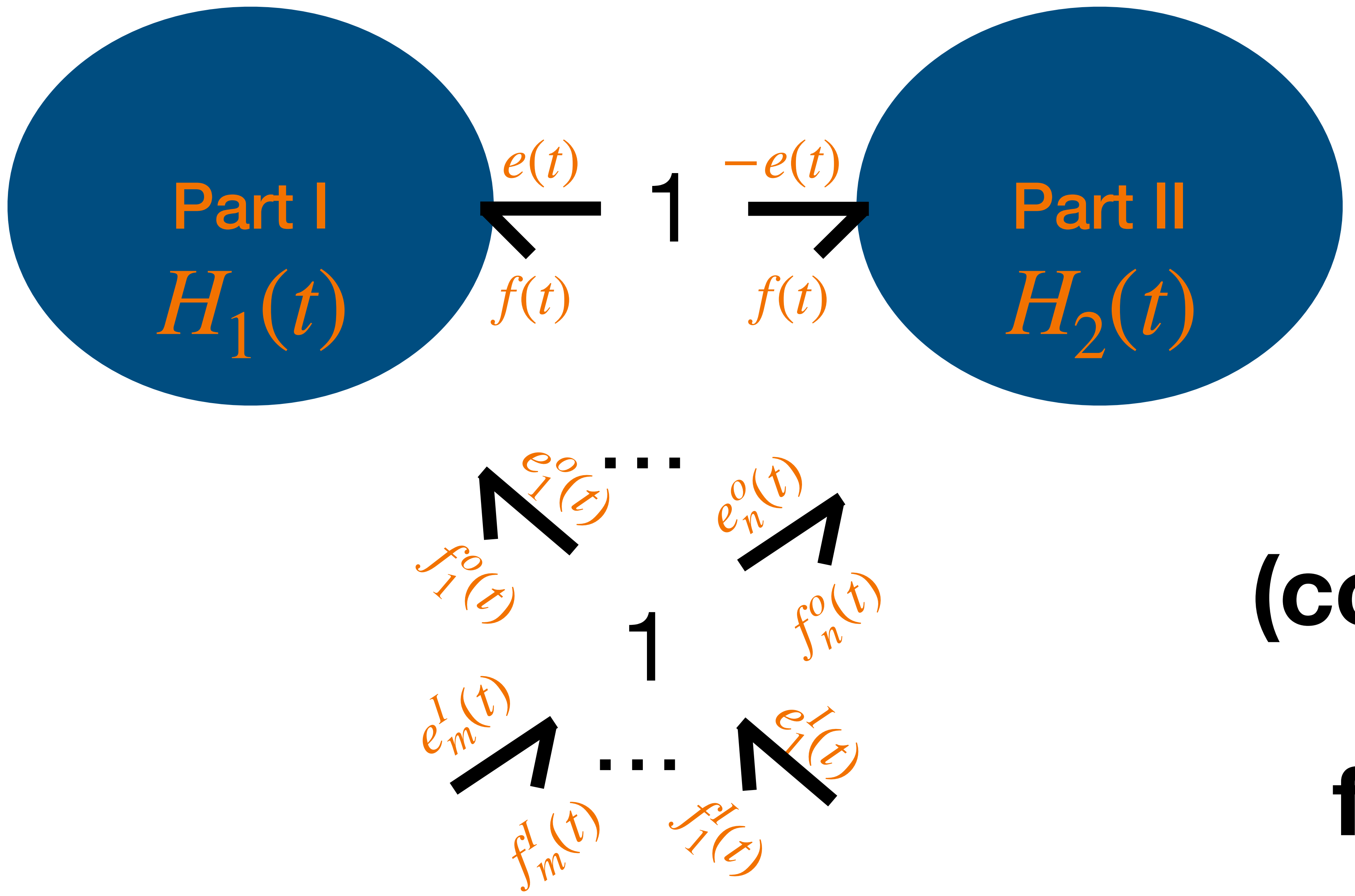
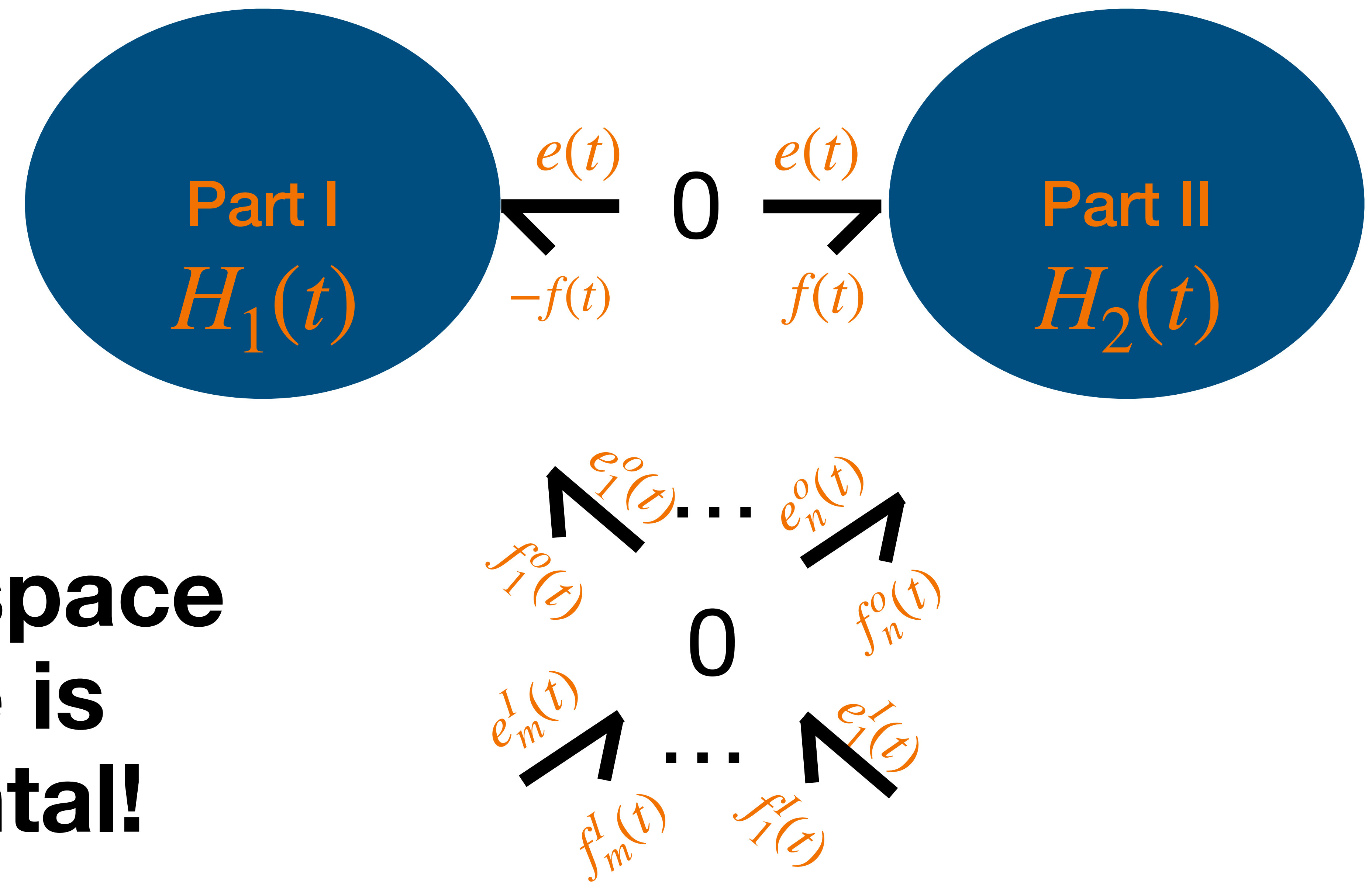
Junctions and generalised Kirchhoff laws

All based on Power preservation



(co)vector space structure is fundamental!



Flow junction

$$f_1^I = \ldots = f_m^I = f_1^o = \ldots = f_n^o$$

$$\sum_{i=1}^{m} e_i^I = \sum_{i=1}^{n} e_i^o$$

Effort junction

$$e_1^I = \ldots = e_m^I = e_1^o = \ldots = e_n^o$$

$$\sum_{i=1}^{m} f_i^I = \sum_{i=1}^{n} f_i^o$$

Power continuity

$$\sum_{i=1}^{m} P_i^I = \sum_{i=1}^{n} P_i^o$$

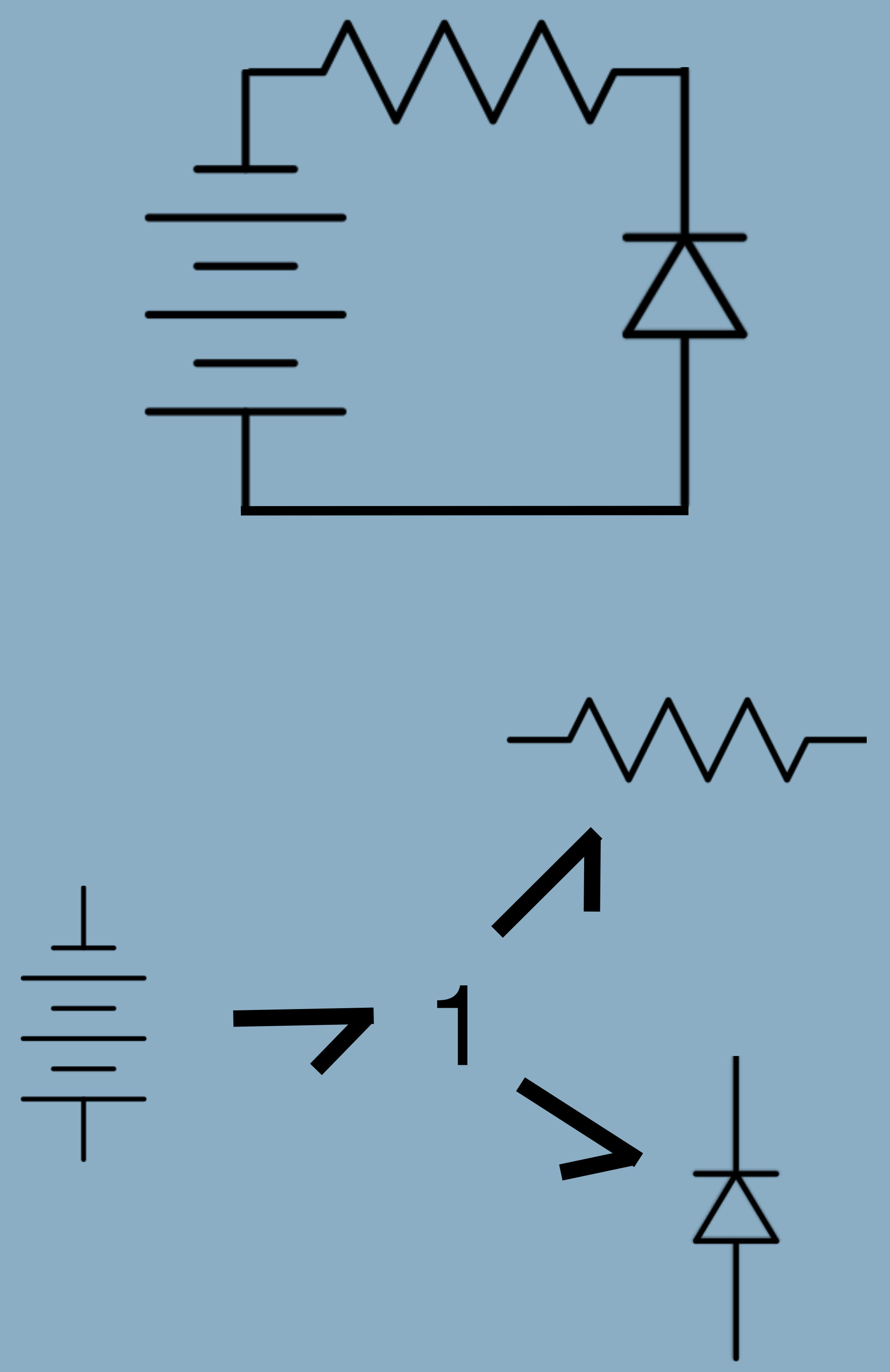
Differences

Prove the following

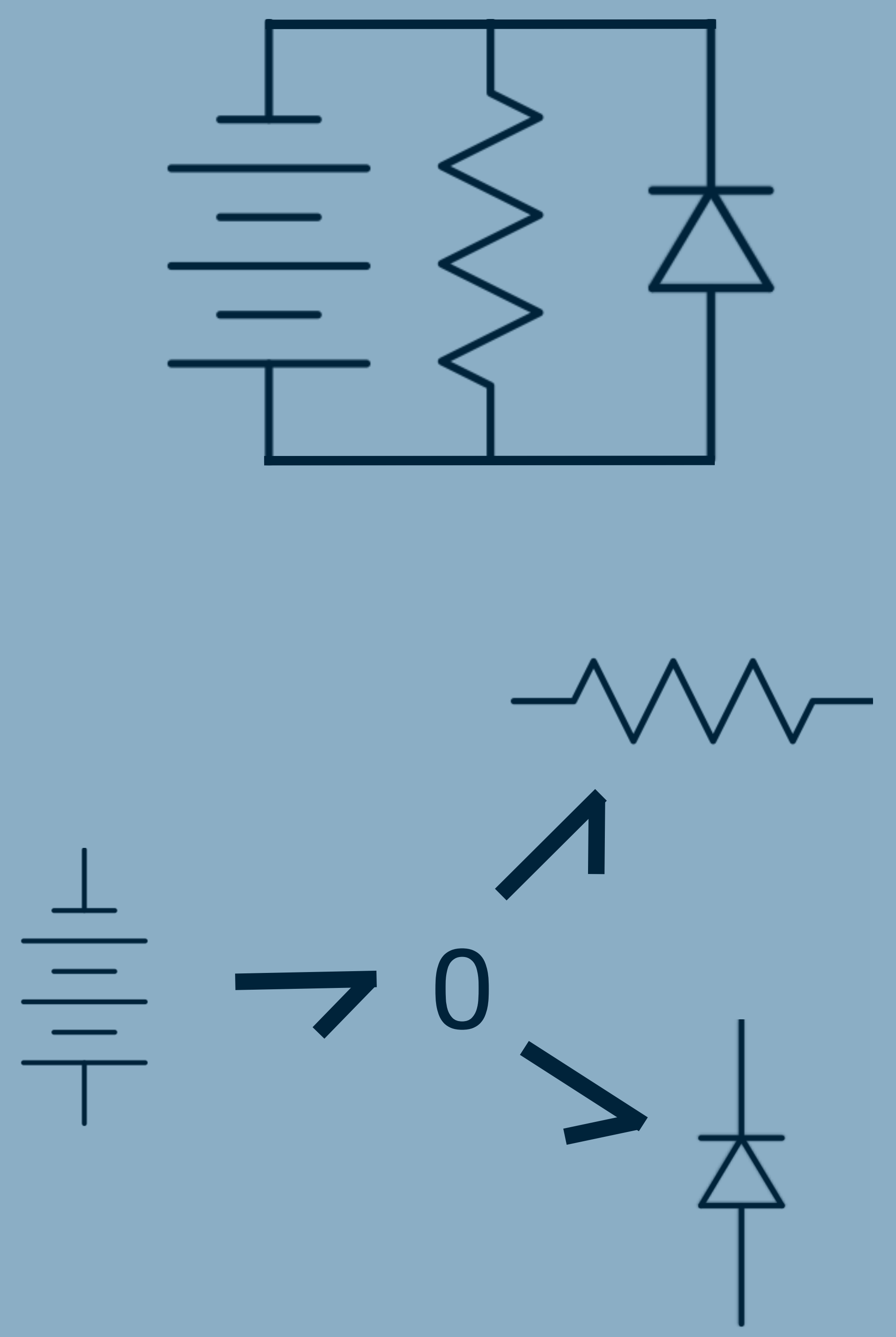
$$
\begin{tikzcd}
f_1 \; 1 \arrow[r, harpoon'] & 0 \arrow[r, harpoon'] \arrow[d, harpoon] & 1 \; f_2 \\
& 1 \; f_1 - f_2 &
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}
e_1 \; 0 \arrow[r, harpoon'] & 1 \arrow[r, harpoon'] \arrow[d, harpoon] & 0 \; e_2 \\
& 0 \; e_1 - e_2 &
\end{tikzcd}
$$

Example

Electrical example: effort=voltage, flow=current



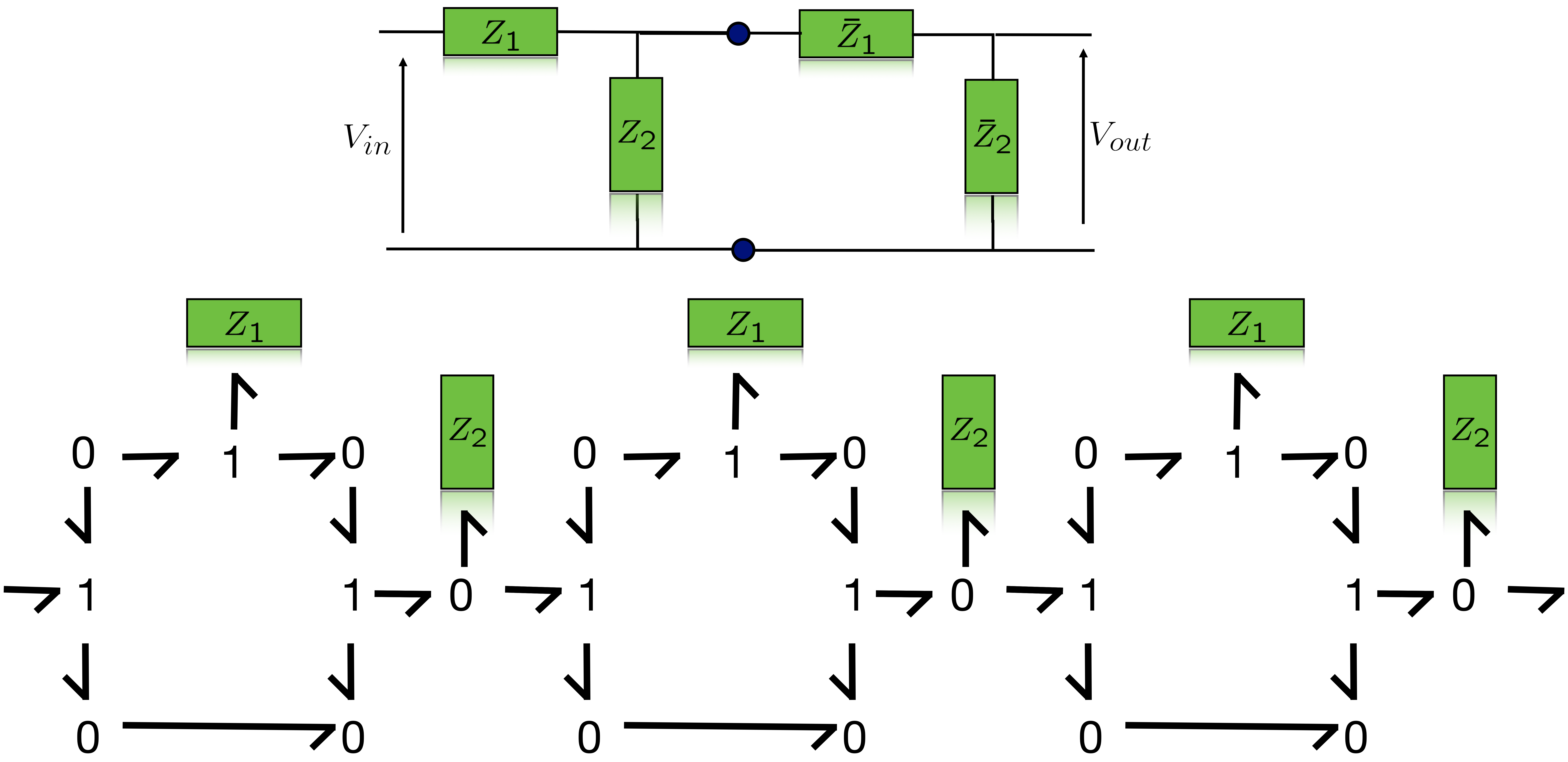
Same current=flow (series)



Same voltage=effort (parallel)

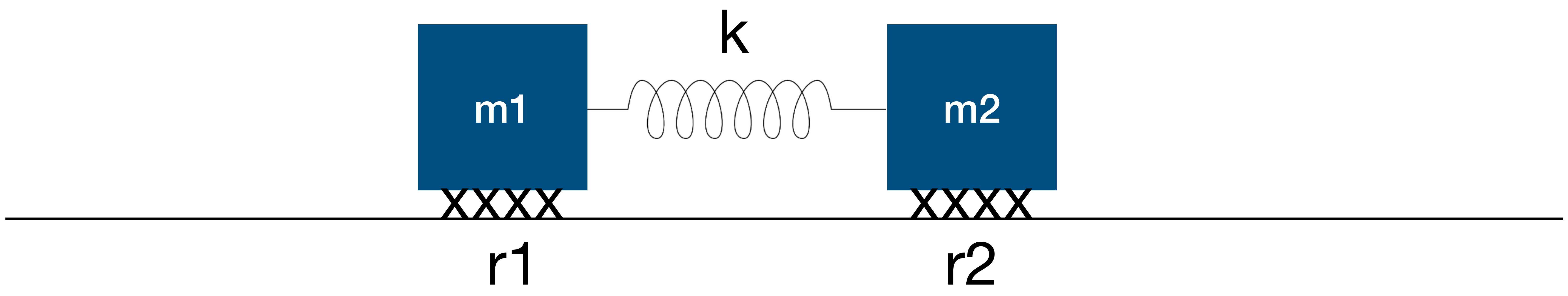
Example

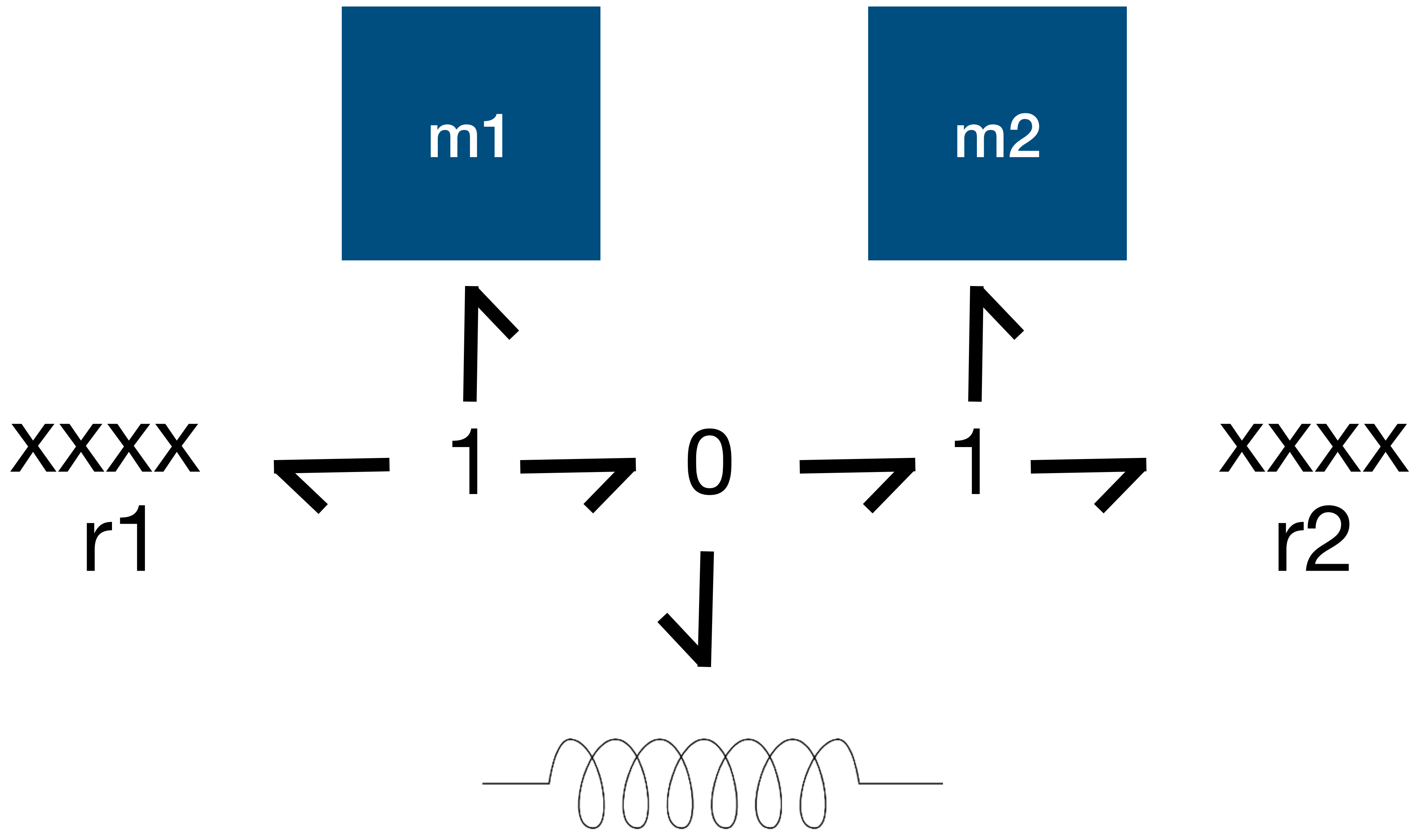
Electrical example: effort=voltage, flow=current



Example

Mechanical **1D** example (in the course generalise to **6D**!)





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Atomic Elements

Storage, Transformers and Gyrators, Sources and Irreversible Transformation

Atomic Elements

Decomposing until the essence of its parts

- We have now introduced how to decompose a system in a physical way and introduced what is needed to do so: **power ports, effort and flows**.
- We have also introduced 1 and 0 junctions to do that and saw that they are actually representing much more: **Generalised Kirchhoff laws**
- If we would keep on decomposing a system in this energetically consistent way, it makes sense to then arrive to a **classification of possible atomic (not further decomposable) elements** which represent **specific behaviours relating to energy** of the physical system under consideration

Atomic Elements

Decomposing until the essence of its parts

What are the simplest **conceptual elements** of which a Physical Model can be composed of?

- **Storage**: elements which purely store energy (also including some transducers!!)
- **Transformation**: power continuous elements which transform efforts and flows
- **Dissipation (of free energy)/Irreversible Transduction**: elements which represents ideal irreversible energy transformation (to heat)
- **Ideal Sources**: elements which represent ideal energy sources

Demo
Conceptual Element v.s. *Components*

Resistance

Physical Resistor

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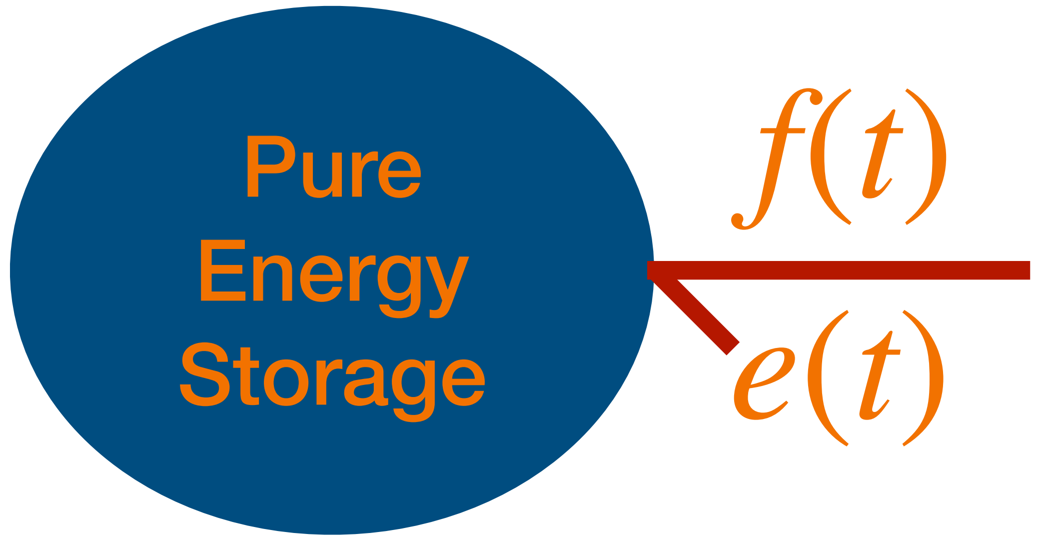
Energy Storage

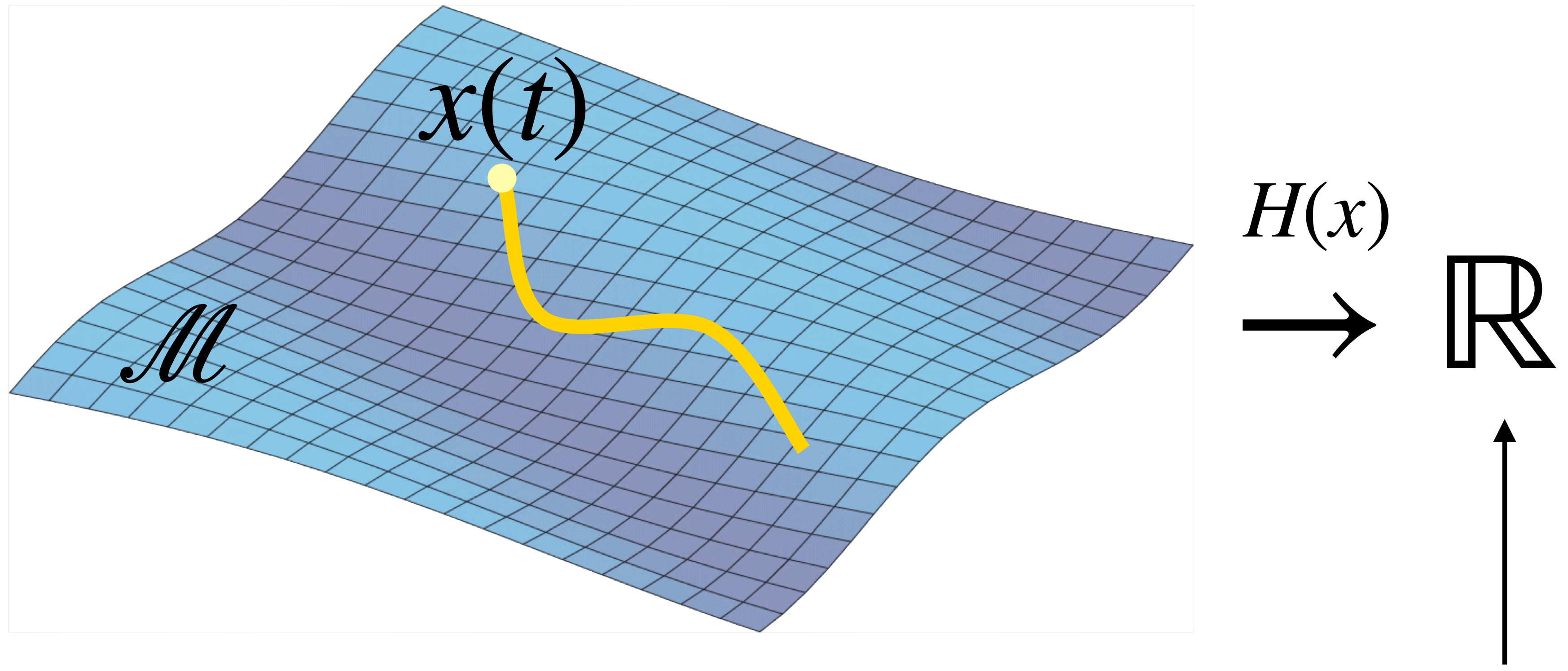
How can we generally model Energy Storage in all physical systems?

Energy Storage

A **pure** energy storage is characterised by 2 things:

1. A **physical state** x
2. An **Energy function** $H(x)$ of the state x called the Hamiltonian $H : \mathscr{M} \to \mathbb{R}; x \mapsto H(x)$





Energy Value

State Space $\mathscr{M}$ is a Manifold, **NOT a vector space** !!

How does it relate to the port: $(f(t), e(t))$?

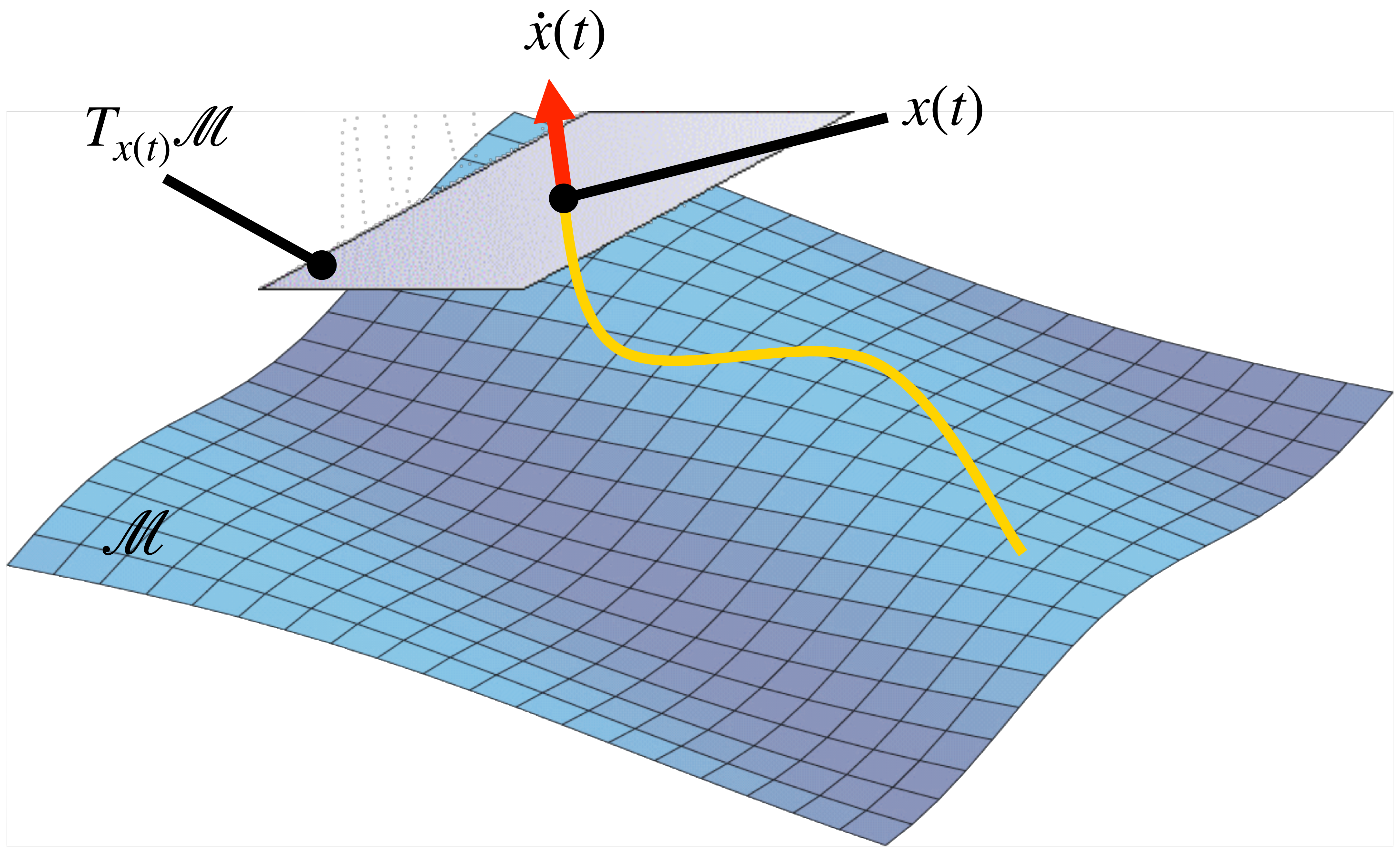
Relation to port Variables

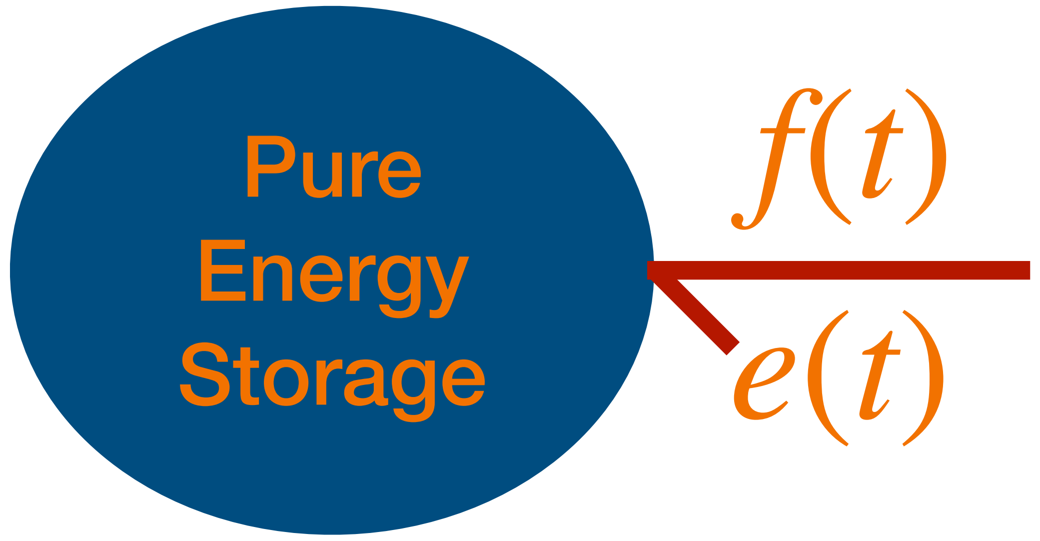
$$H : \mathscr{M} \to \mathbb{R}$$
$$x : T \to \mathscr{M}$$

Chain Rule

$$\dot{H}(t) = \underbrace{\frac{\partial H}{\partial x}}_{dH} \dot{x}$$

Intuitive Definition





$f(t) = \dot{x}(t)$ **Vectors**

$e(t) = dH(x(t))$ **Co-vectors**

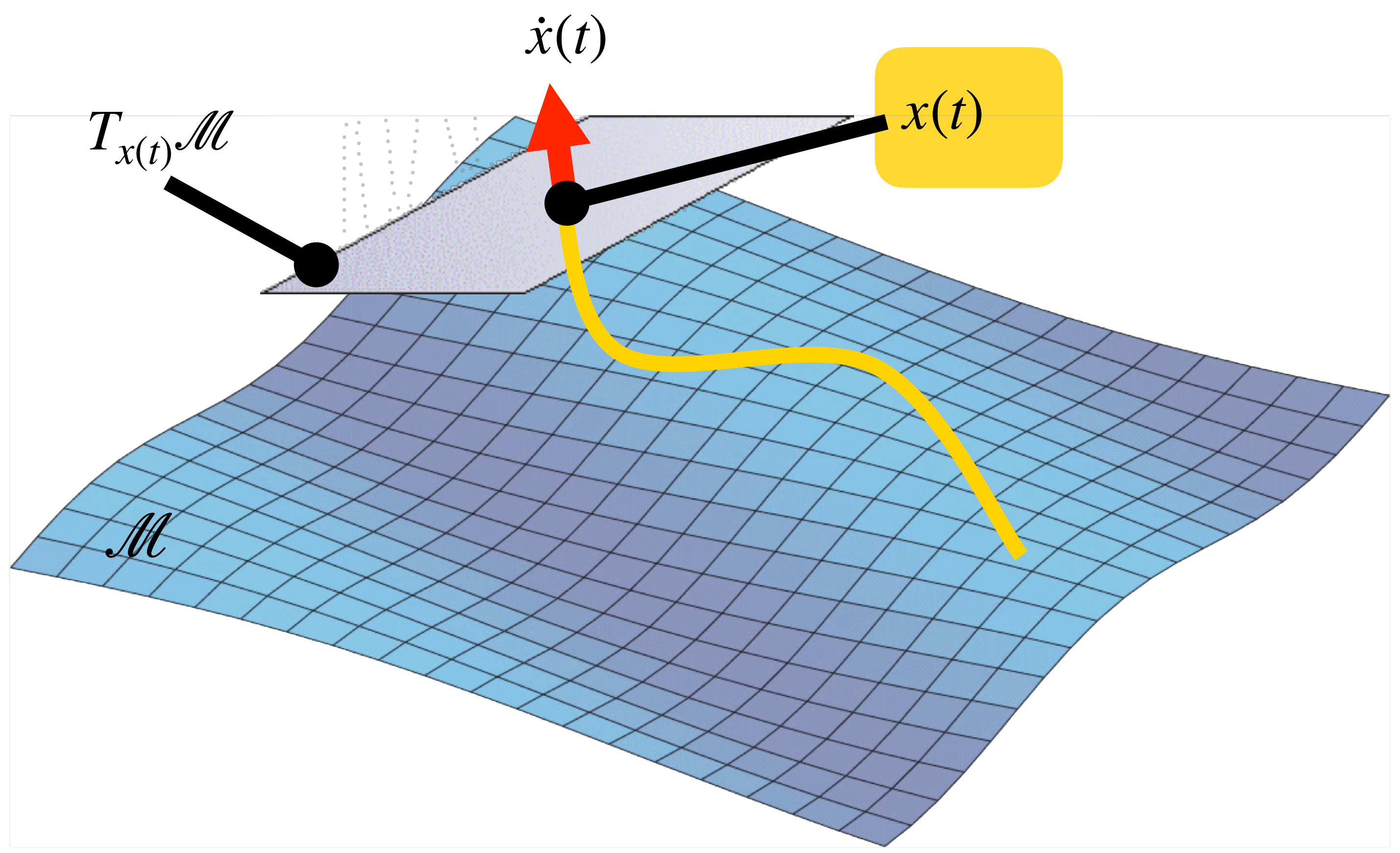
$$\dot{x}(t) \in T_{x(t)}\mathscr{M} \rightarrow (x(t), \dot{x}(t)) \in T\mathscr{M}$$

$$dH(x(t)) \in T^*_{x(t)}\mathscr{M} \rightarrow (x(t), dH(x(t))) \in T^*\mathscr{M}$$

$$< \dot{x}(t), dH(x(t)) > = \dot{H}(t) = P_{\text{in}}(t) = e(f)(t)$$

Relation to port Variables

Intuitive Definition



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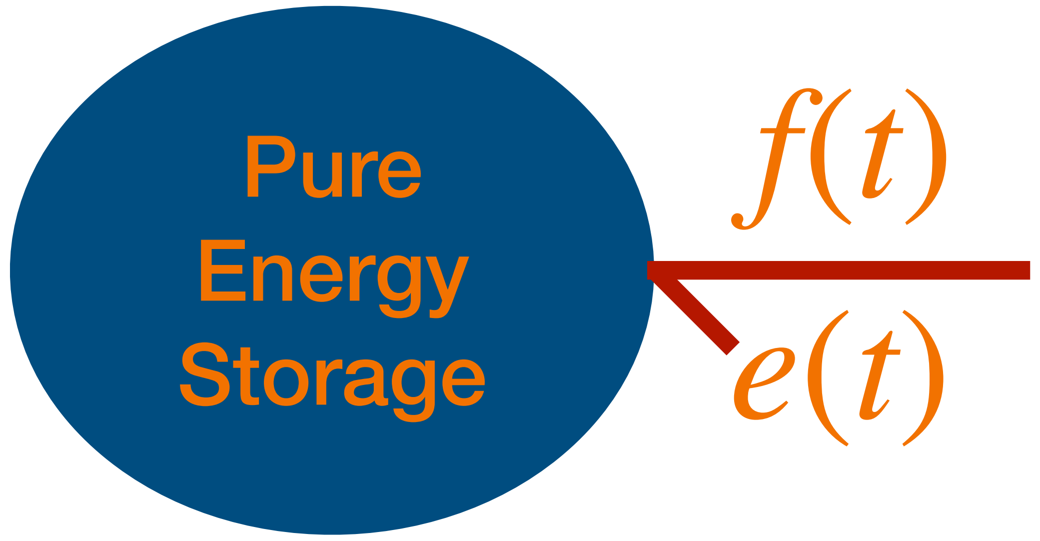


$x(t)$

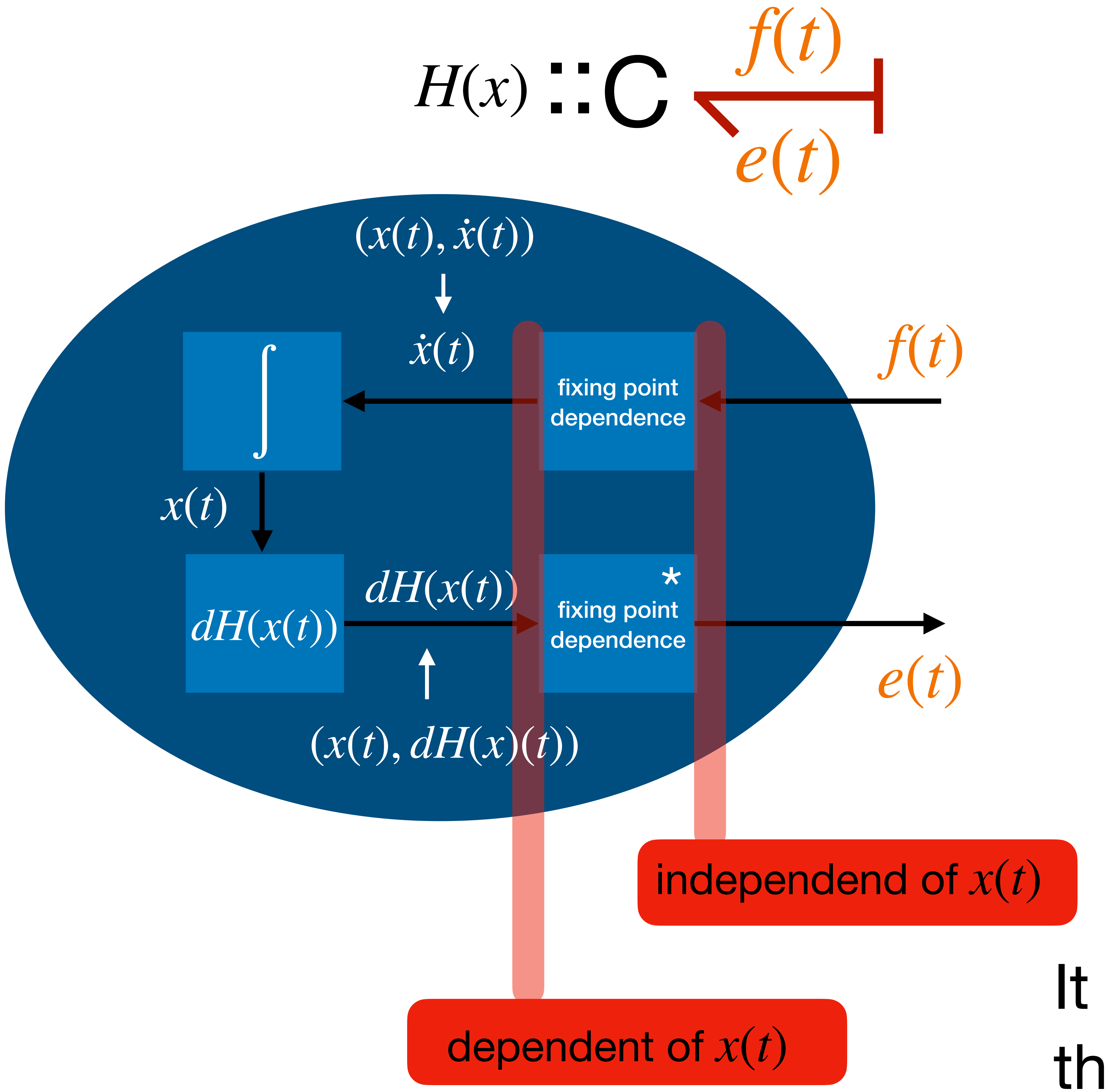
$$\dot{x}(t) \in T_{x(t)}\mathscr{M} \rightarrow (x(t), \dot{x}(t)) \in T\mathscr{M}$$

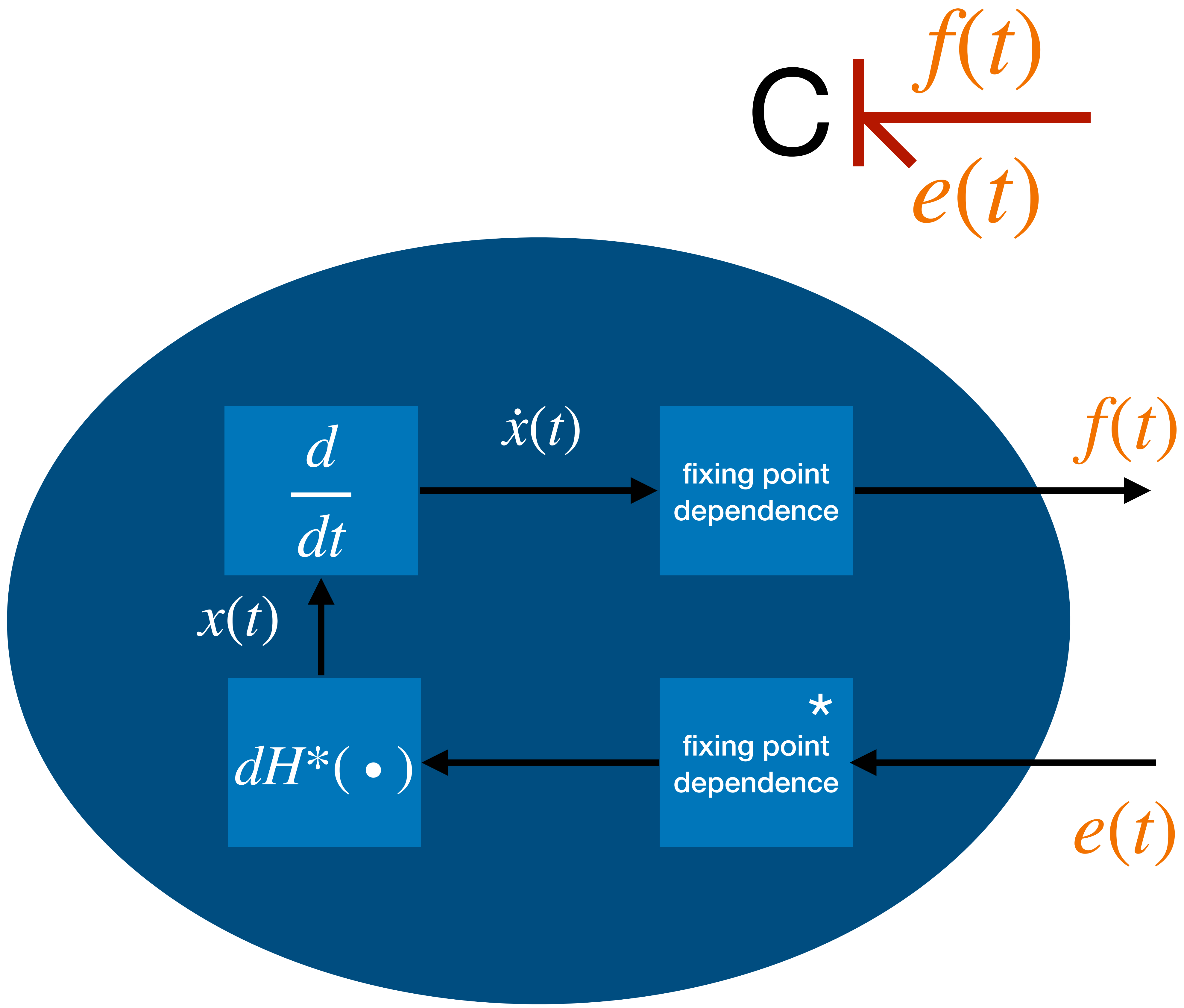
$$dH(x(t)) \in T^*_{x(t)}\mathscr{M} \rightarrow (x(t), dH(x(t))) \in T^*\mathscr{M}$$

$$< \dot{x}(t), dH(x(t)) > = \dot{H}(t) = P_{\text{in}}(t) = e(f)(t)$$

Geometrically there is an issue which needs to be solved more precisely: $f(t)$ does not have "knowledge" of where the state $x(t)$ is. All works out because we use **1 coordinate patch for everything!** *(More later)*

Causal Description





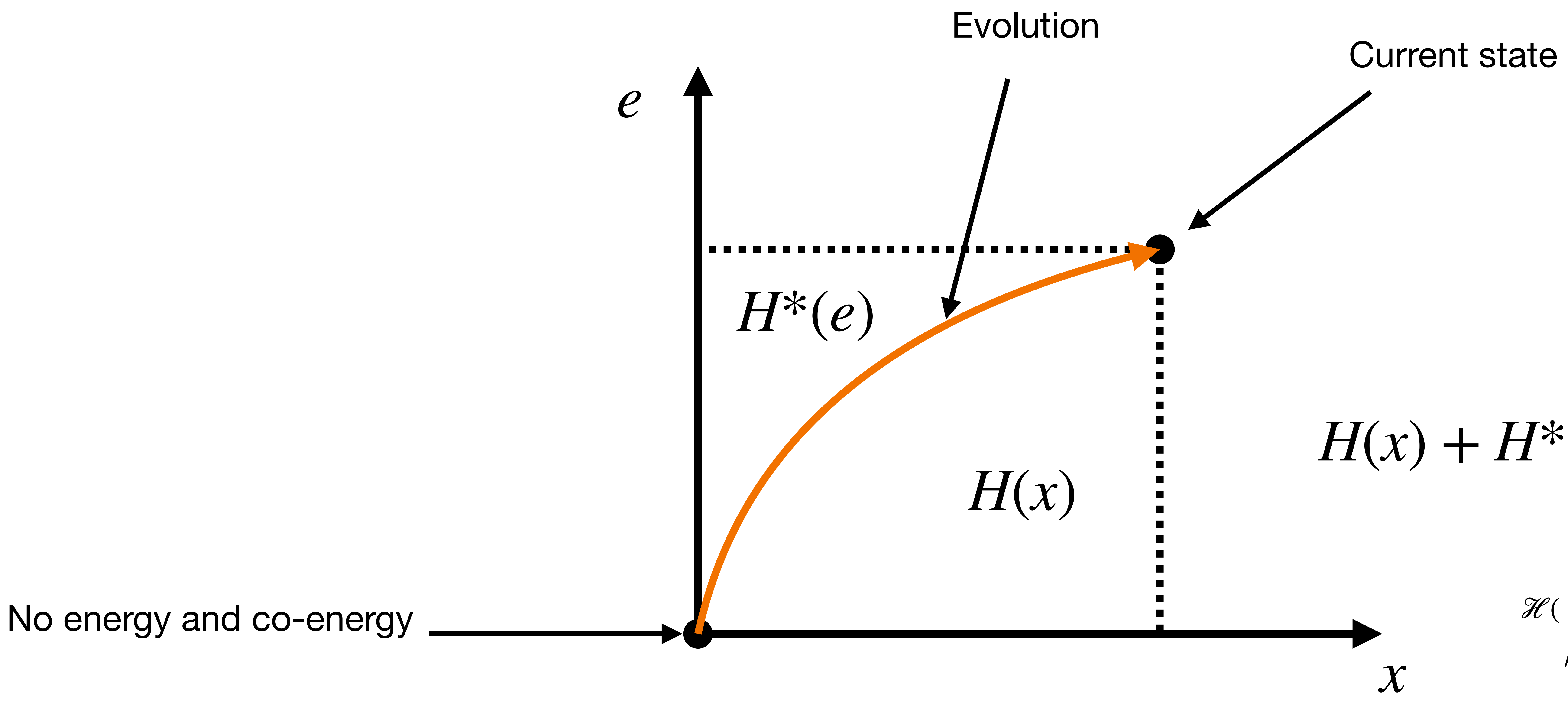
H^* is NOT function of x but of $e(t)$!

It is called **co-energy** and is mathematically the Legendre transform of H

Energy versus co-Energy

Energy and Co-Energy are two **different functions** related by what is called the **Legendre transform** and in nonlinear storages have different values for the same state!

Intuitive **NON PRECISE SCALAR** idea:



$$H(x) + H^*(e) = ex$$

$$\mathscr{H}(\underbrace{x}_{k\text{–form}}) + \mathscr{H}^*(\underbrace{e}_{(n-k)\text{–form}}) = e \wedge x$$

Energy versus co-Energy: The Legendre Transform

$$H(x) \Rightarrow e := \frac{\partial H}{\partial x}$$

$$H^*(e) := \text{sup}_x[ex - H(x)]$$ (Legendre Transform)

$$= [ex - H(x)]_{x=\left(\frac{\partial H}{\partial x}\right)^{-1}(p)}$$

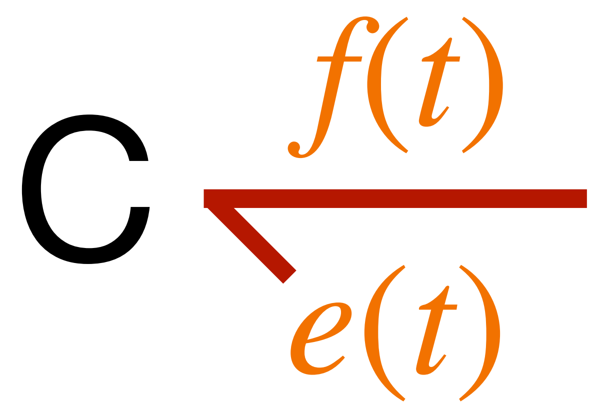
$$H(x) + H^*(e) = ex$$

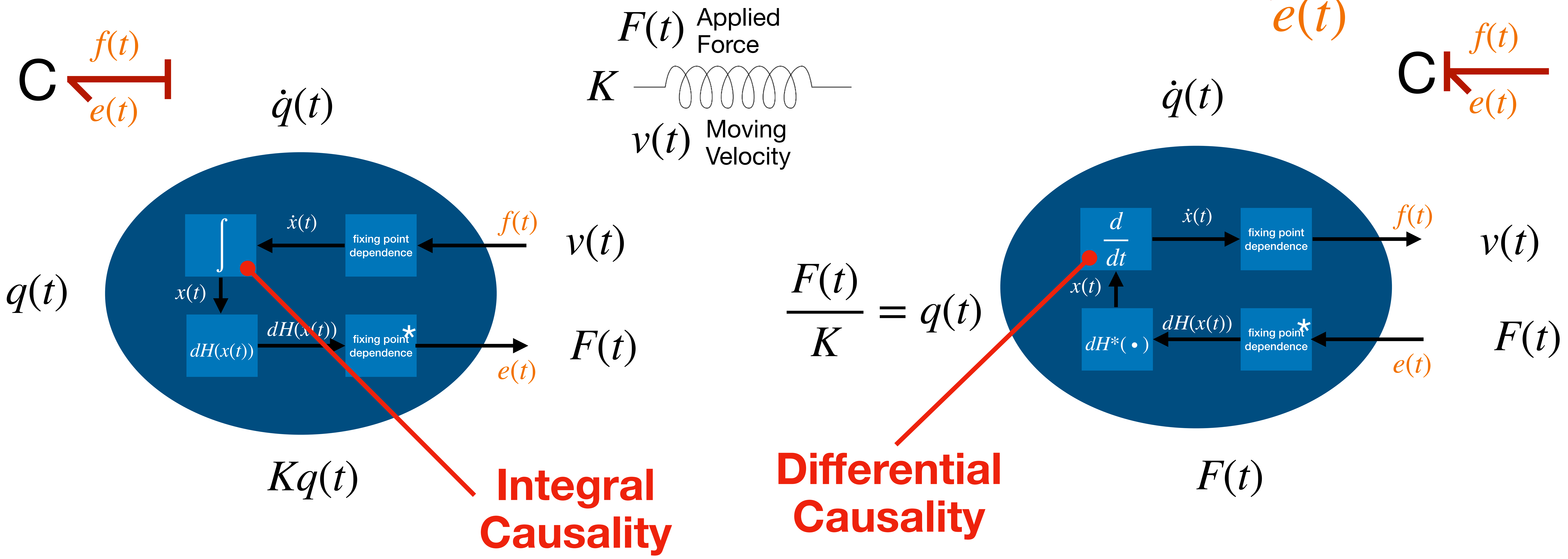
C in **differential** causality

$$\frac{\partial H}{\partial x}\dot{x} + \frac{\partial H^*}{\partial e}\dot{e} = e\dot{x} + x\dot{e}$$

C in **integral** causality

Example: Potential (co)-energy





Potential Energy

State: deformation q

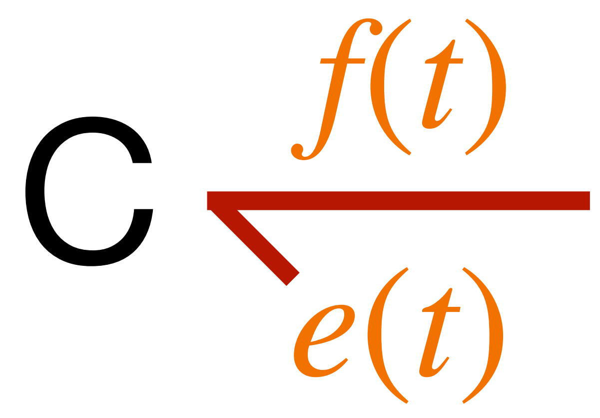
Energy: $H(q) = \frac{1}{2}Kq^2$

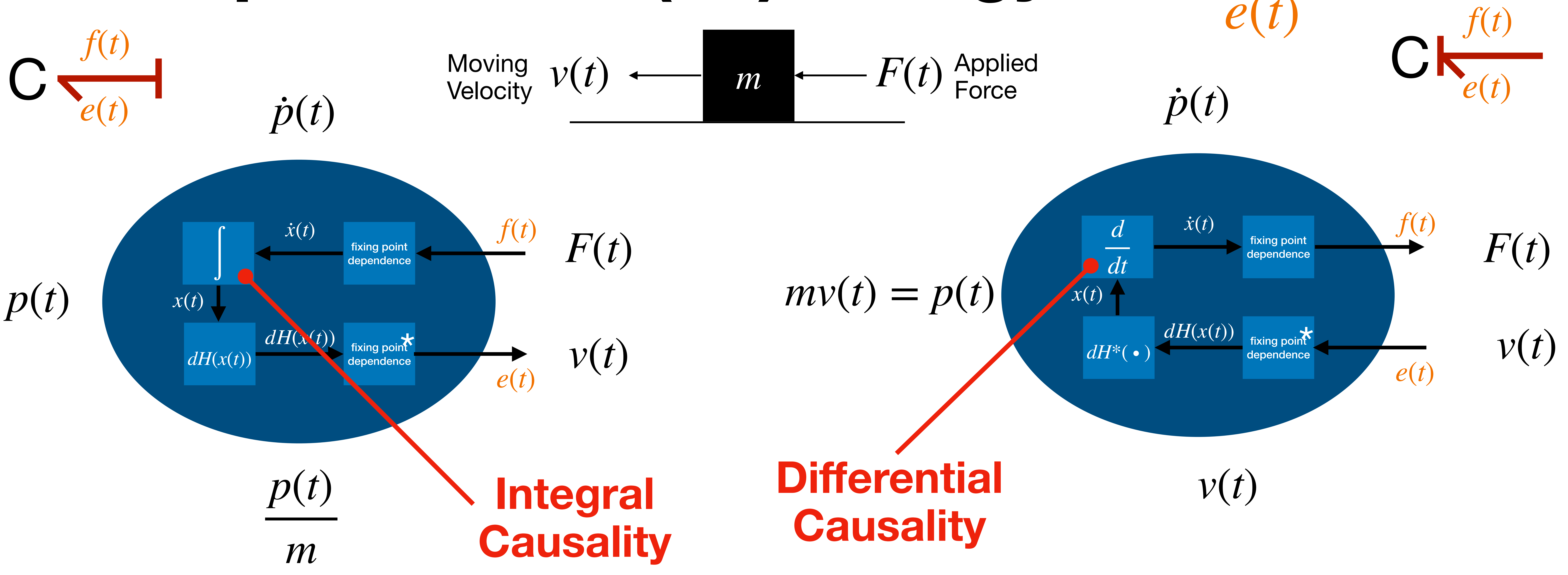
Kinetic Co-Energy

State: force F

Co-Energy: $H^*(F) = \frac{1}{2K}F^2$

Example: Kinetic (co)-energy





Kinetic Energy

State: momenta p

Energy: $H(p) = \frac{1}{2m}p^2$

Kinetic Co-Energy

State: velocity v

Co-Energy: $H^*(v) = \frac{1}{2}mv^2$

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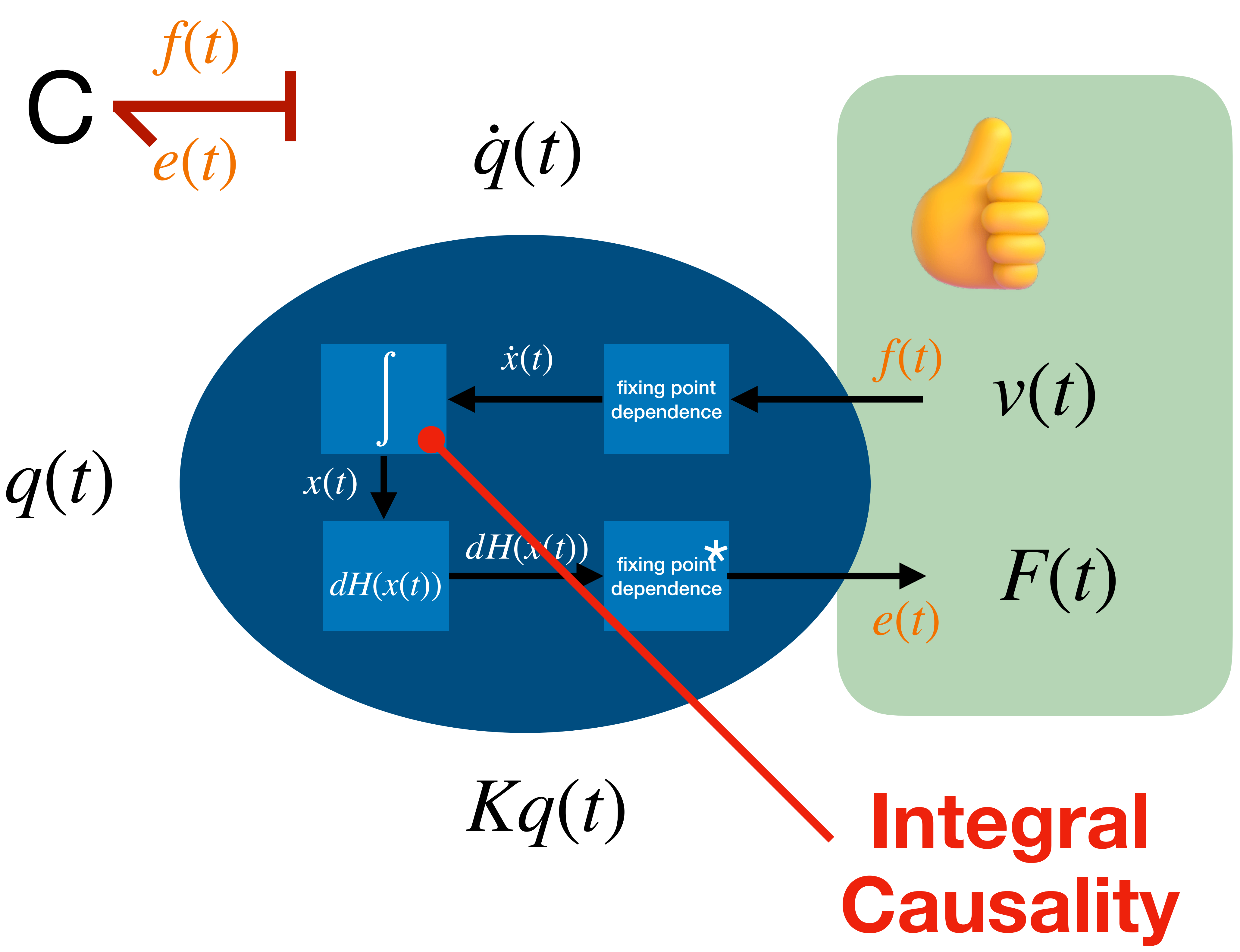
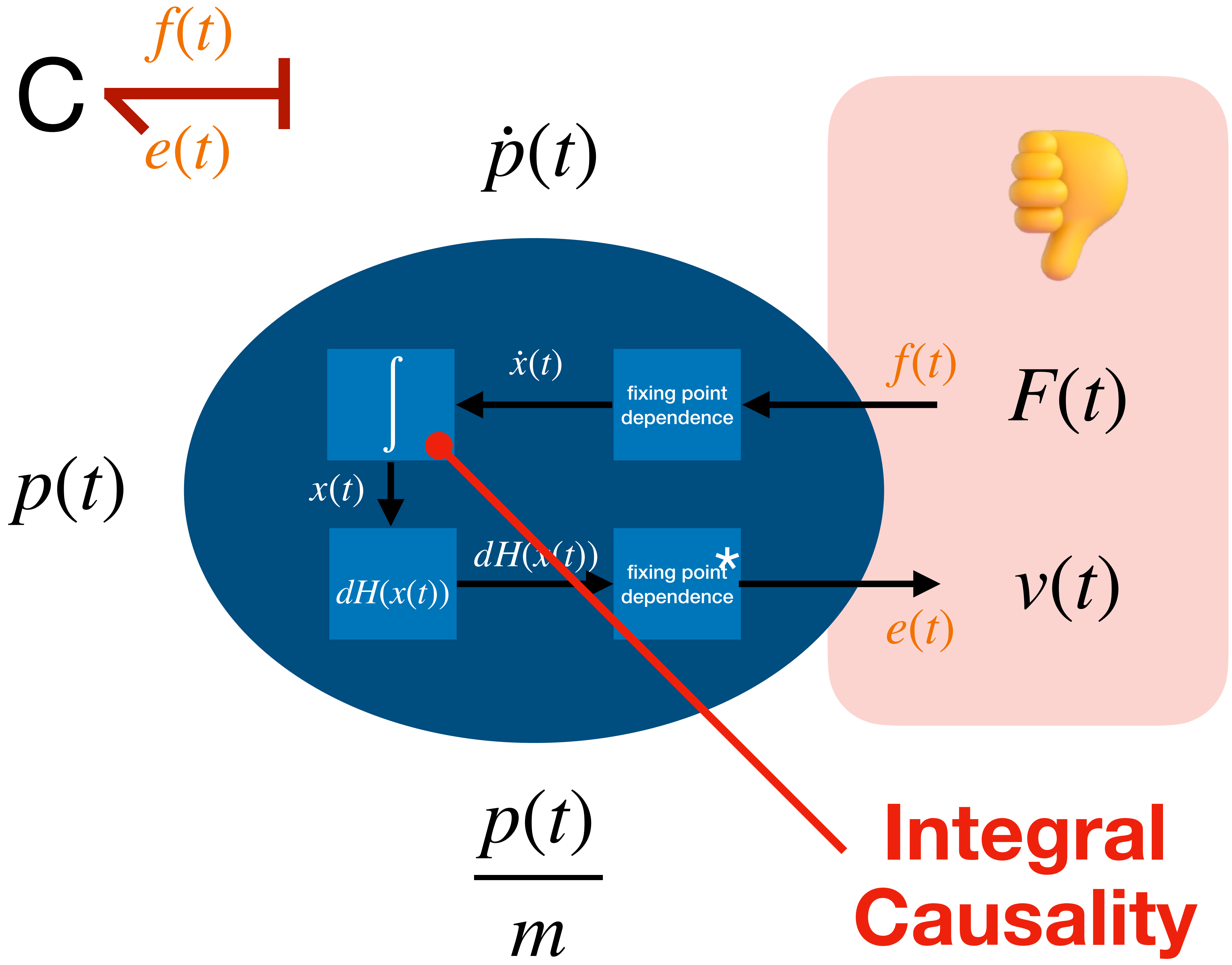
In Bond Graphs $f(t) = v(t)$ and $e(t) = F(t)$





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Kinetic Energy

State: momenta p

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In Bond Graphs $f(t) = v(t)$ and $e(t) = F(t)$

Classification is made on the basis of Thermodynamical reasons:

Physical states are Extensive Thermodynamical Quantities

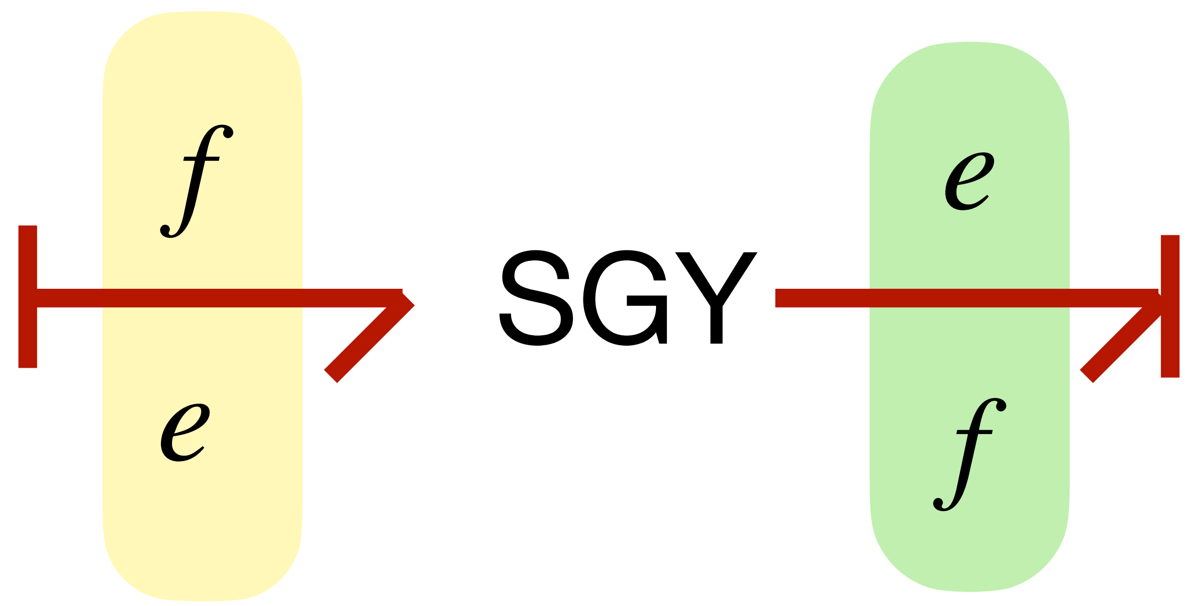
Domain	Sub-domain	Flow	Effort	Physical State is Flow Integral (C)	Physical State is Effort Integral (I)
Mechanical	Potential	Velocity v	Force F	Displacement q	X
Mechanical	Kinetic	Velocity v	Force F	X	Momenta p
Electro-magnetic	Electric	Current i	Voltage u	Electric Charge Q	X
Electro-magnetic	Magnetic	Current i	Voltage u	X	Magnetic Flux ϕ
Thermal	Potential	Entropy Flux $\dot{S}$	Temperature T	Entropy S	X

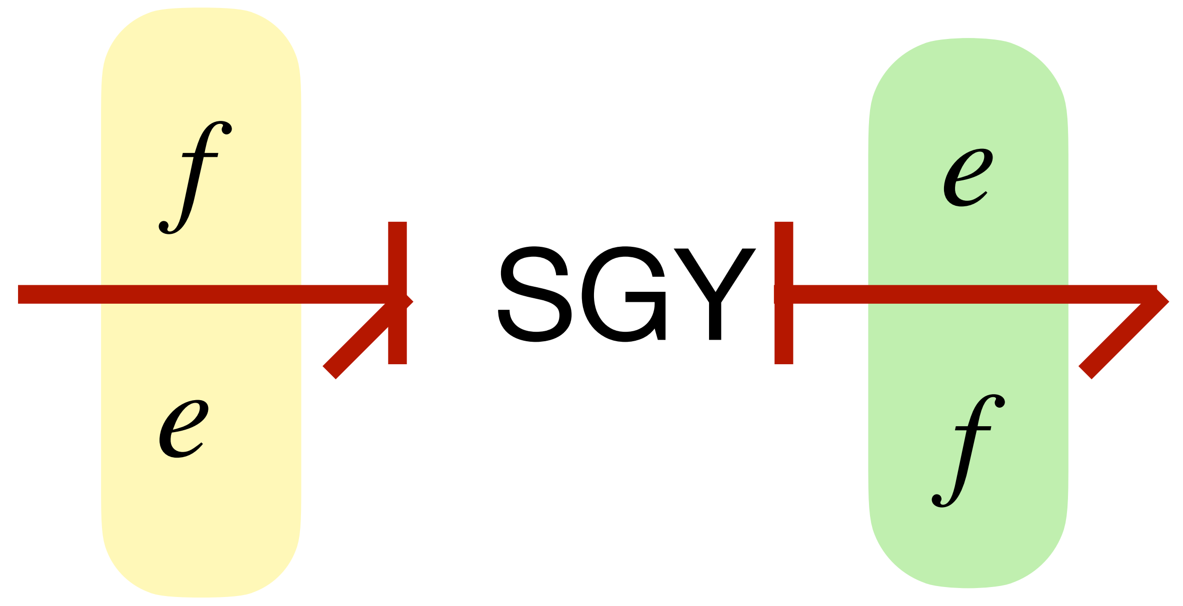
Physical Asymmetry

Irreversible Energy Transduction!

Symplectic Gyrator

A very fundamental element which every domain has except the thermal one





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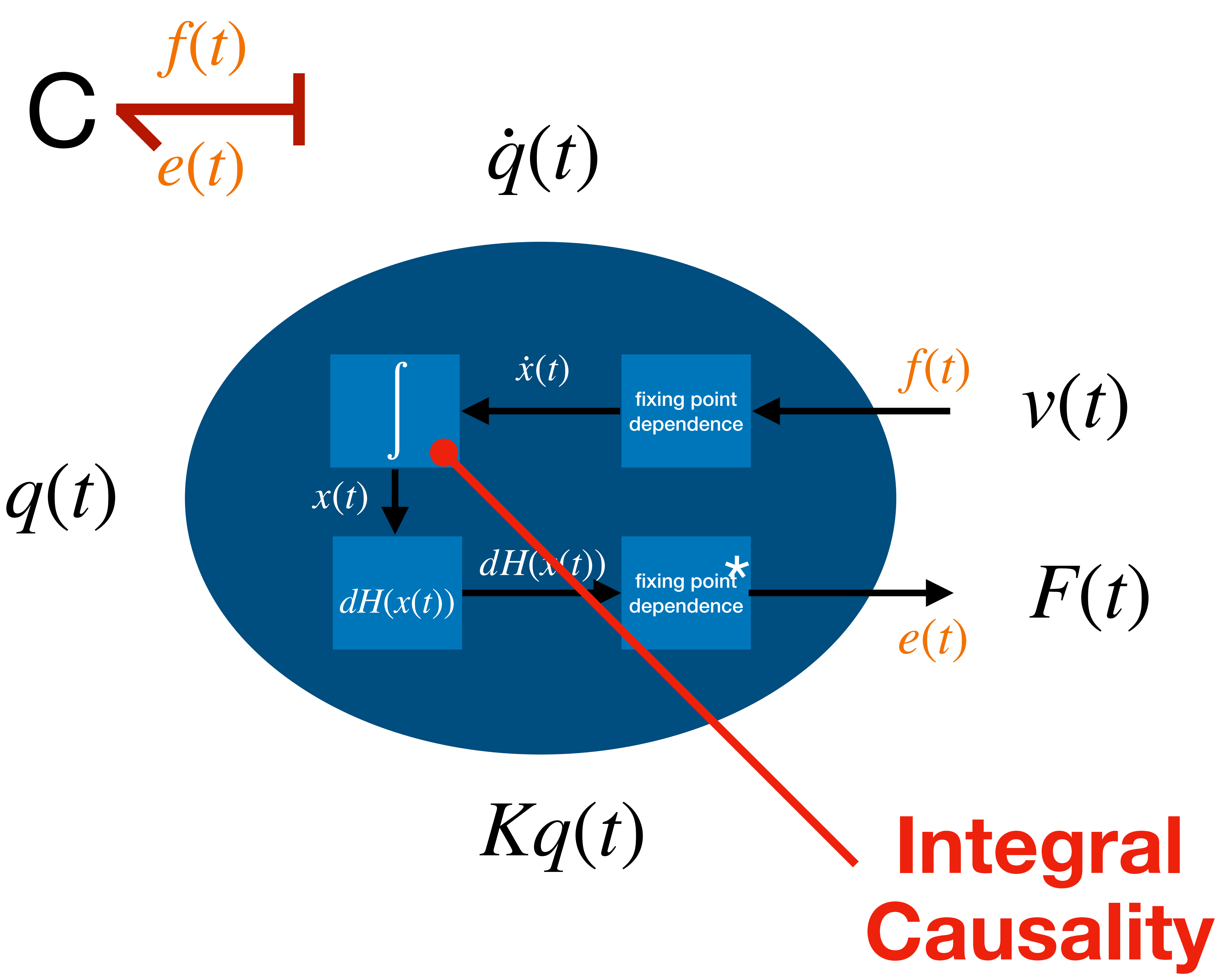

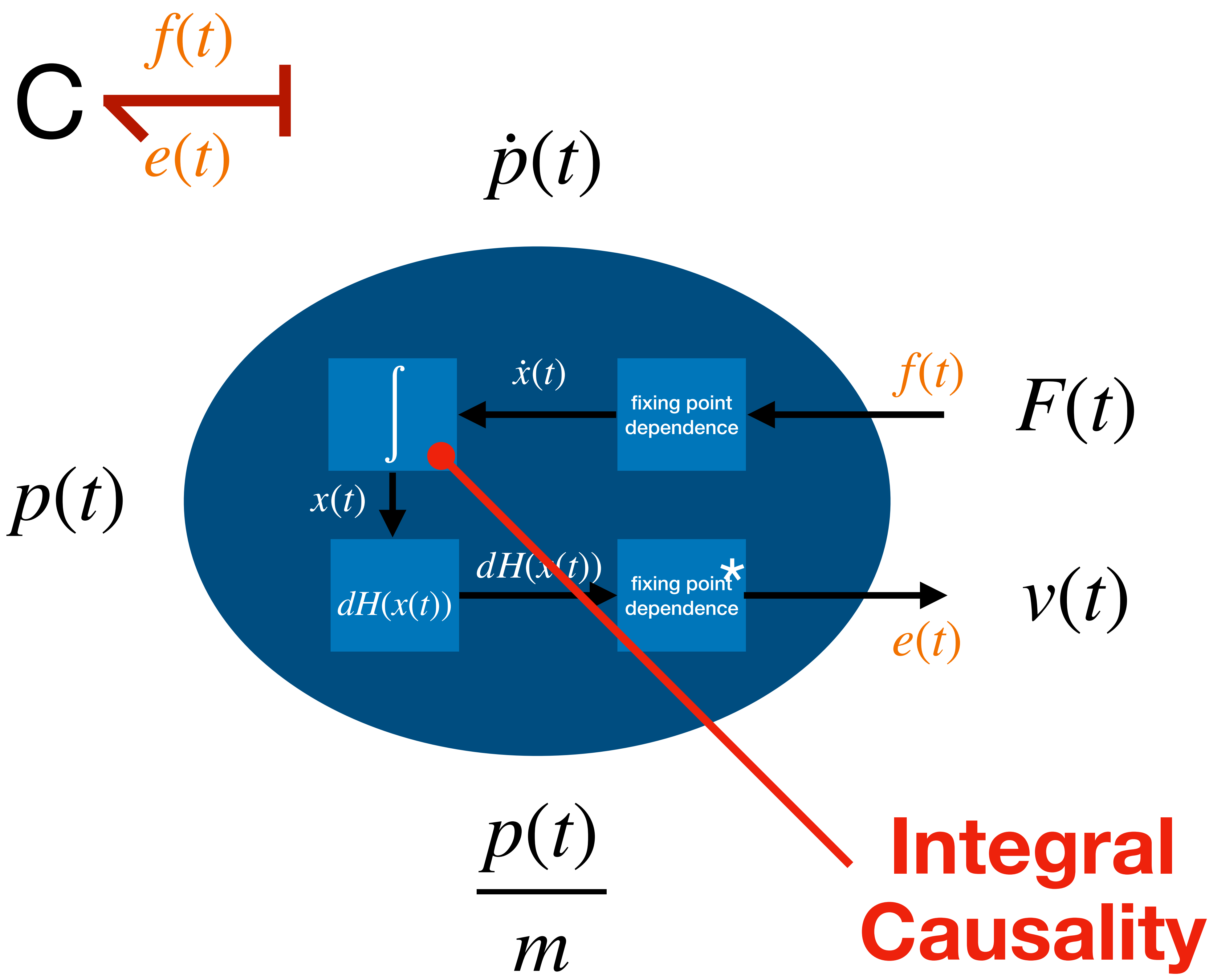

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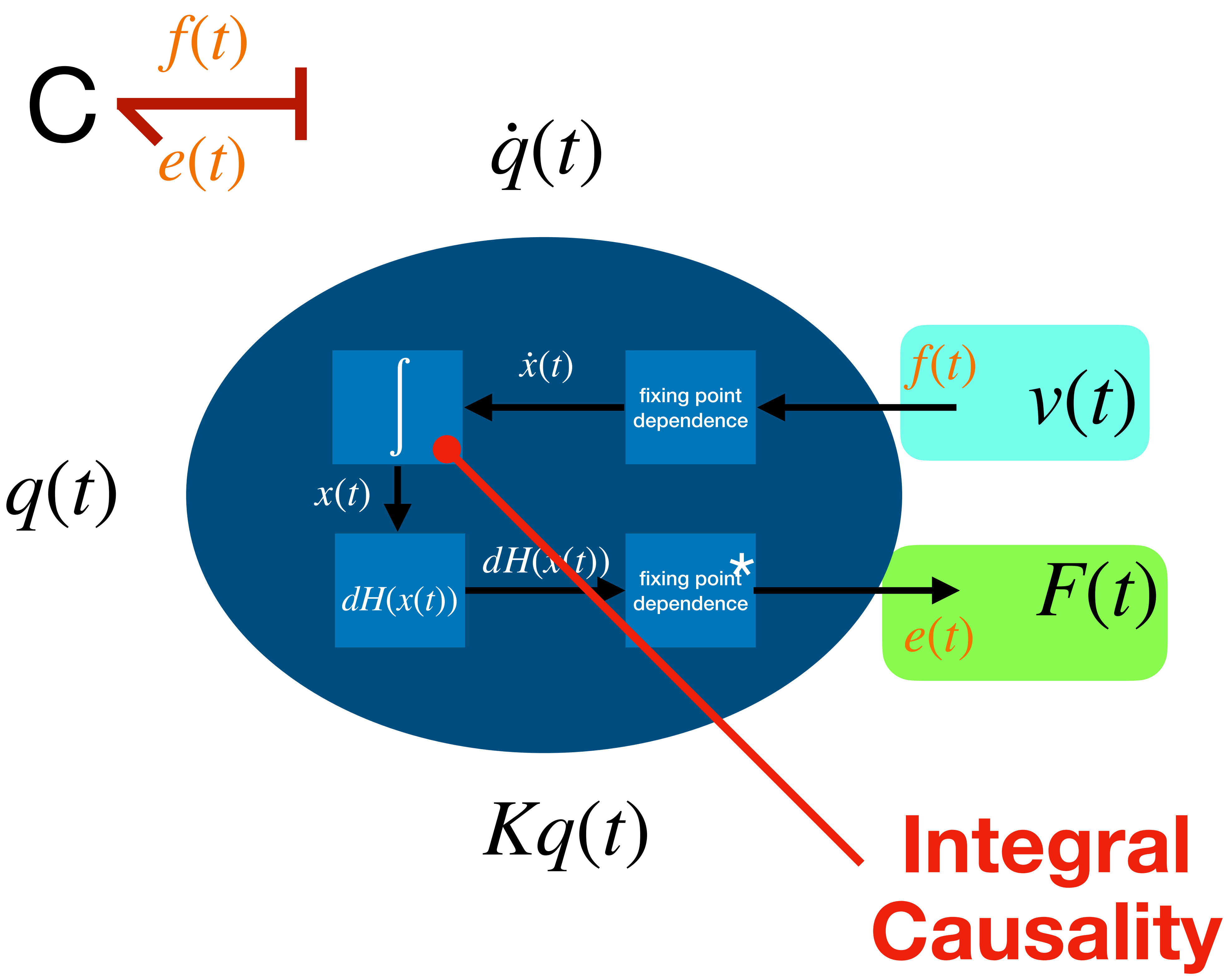
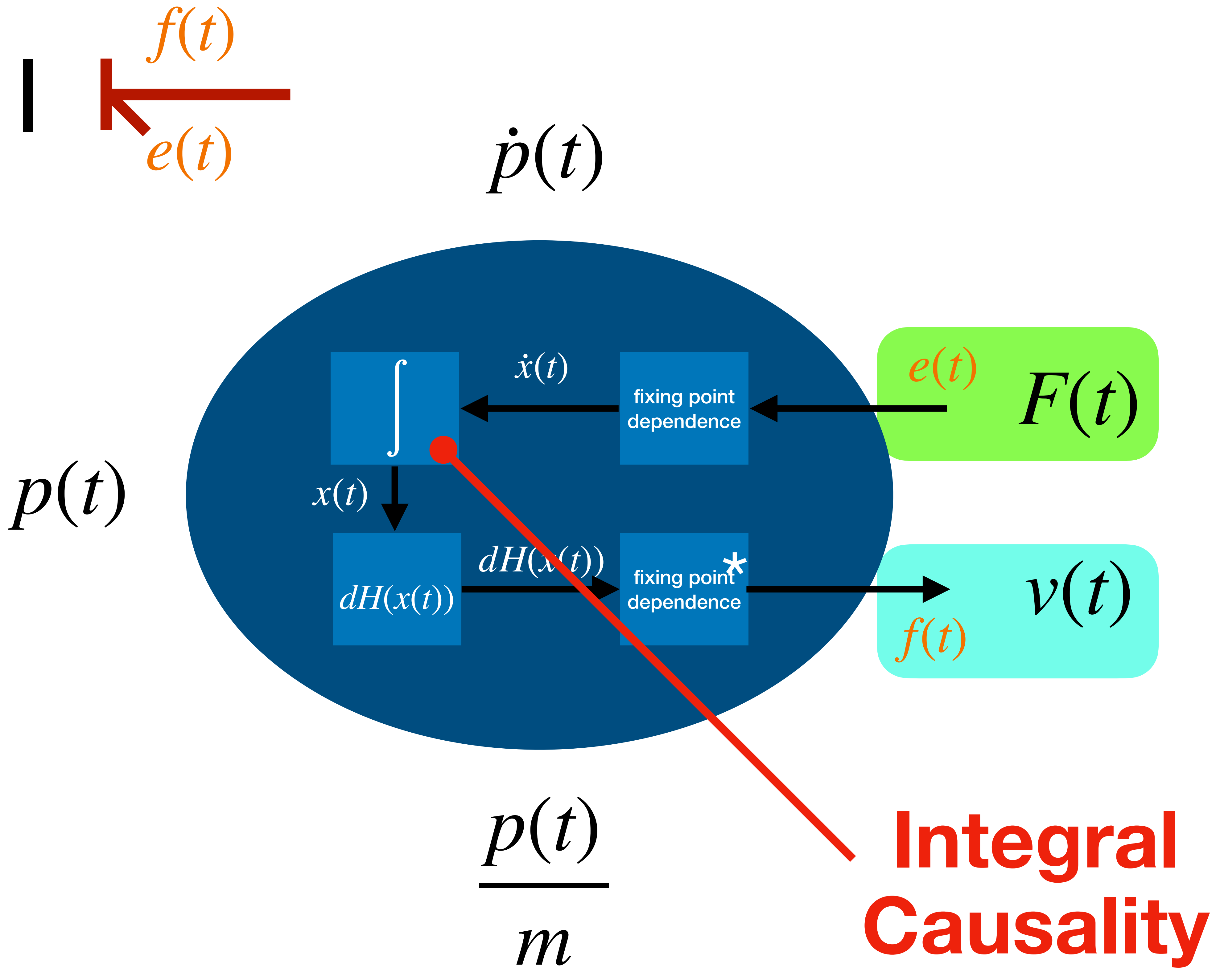
Kinetic Energy

State: momenta p

Energy: $H(p) = \frac{1}{2m}p^2$

In Bond Graphs $f(t) = v(t)$ and $e(t) = F(t)$





Potential Energy

State: deformation q

Energy: $H(q) = \frac{1}{2}Kq^2$

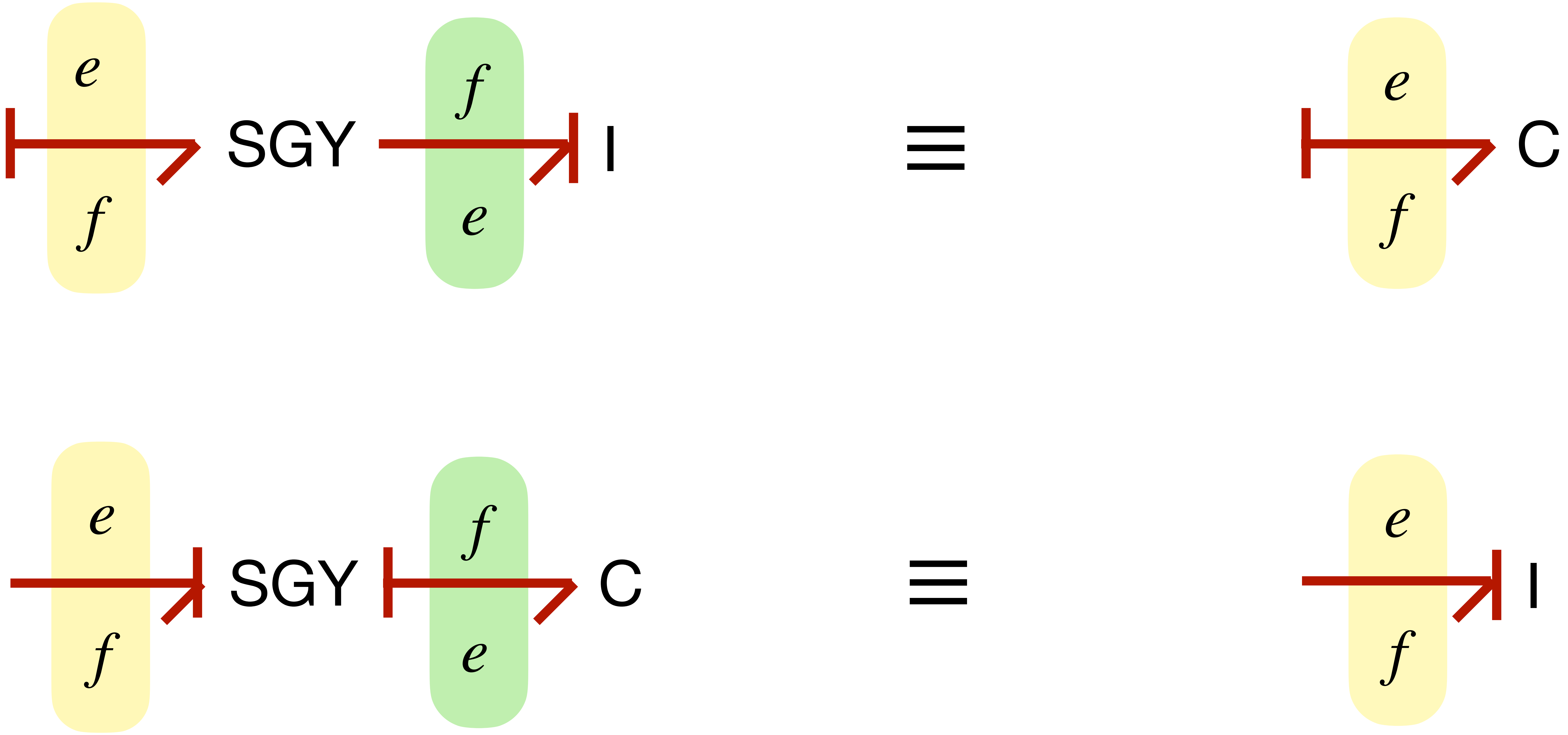
Kinetic Energy

State: momenta p

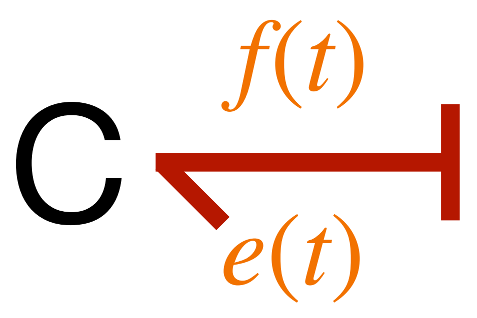
Energy: $H(p) = \frac{1}{2m}p^2$

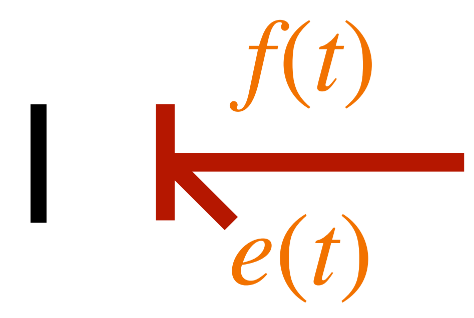
Equivalence in Dual elements

Not exiting in thermal domain!



In Bond Graphs $f(t) = v(t)$ and $e(t) = F(t)$





Power bond ALWAYS oriented TOWARD the storage because by construction $\dot{H} = e(f)$

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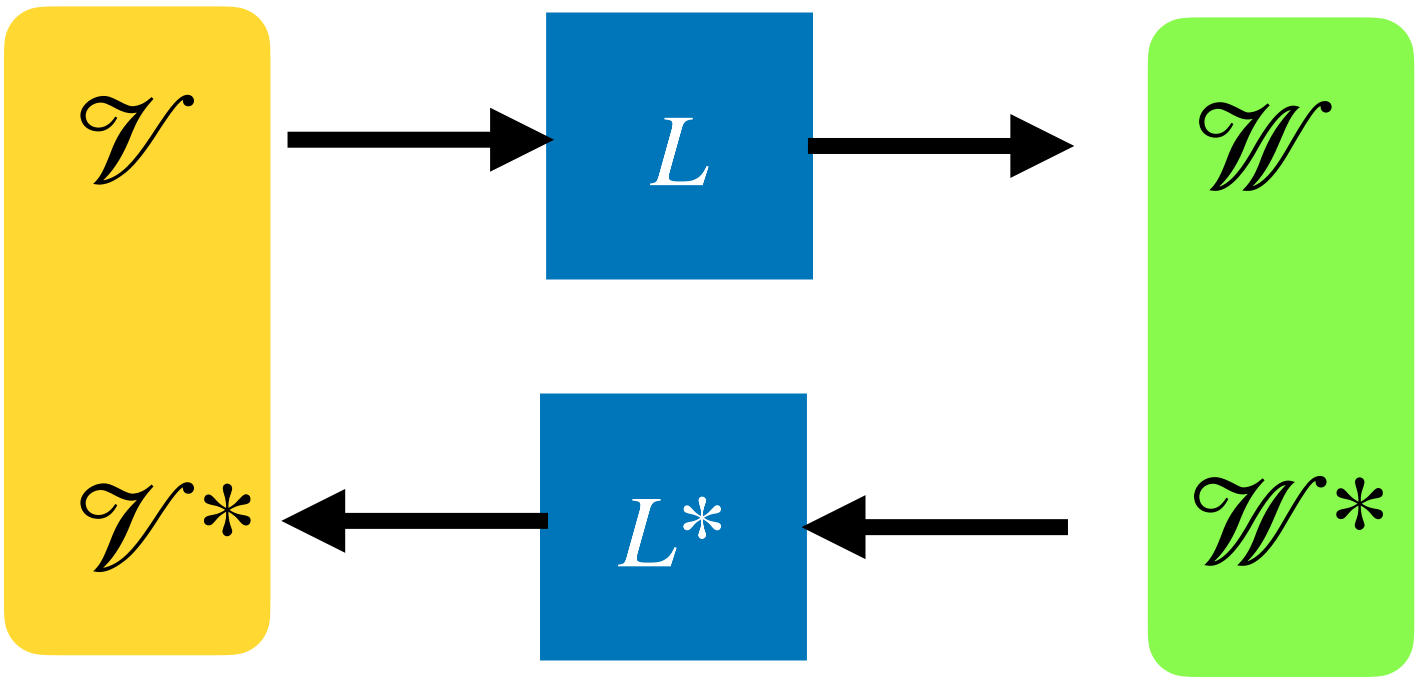
Transformations & Network Structure

How can we model any power continuous "transformation" like levers, ideal gears … ?

Recall: Linear Maps and their dual/adjoint

Given a Linear map L from a vector space $\mathscr{V}$ to a vector space $\mathscr{W}$



It is possible to UNIQUELY define a map L^* of the dual spaces in the opposite direction called **dual or adjoint**

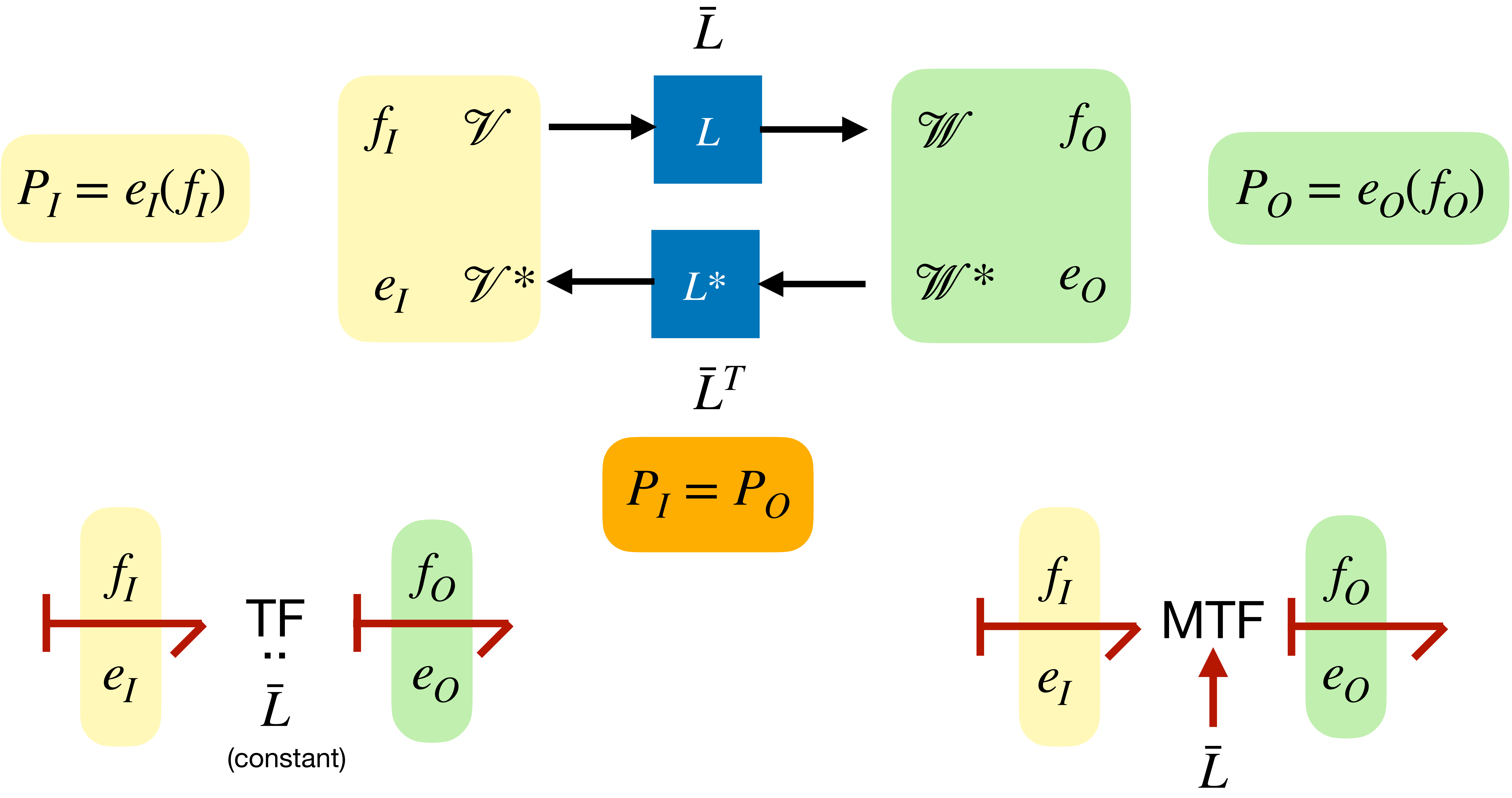
Such that if $v \in \mathscr{V}$, $w \in \mathscr{W}$, $\bar{v} \in \mathscr{V}^*$, $\bar{w} \in \mathscr{W}^*$

$$< v \,|\, L^*(\bar{w}) > \; = \; < L(v) \,|\, \bar{w} >$$

For finite dimensional vector spaces, in coordinates the representation of L^* is a matrix which is the transpose of the matrix representing L

Transformers

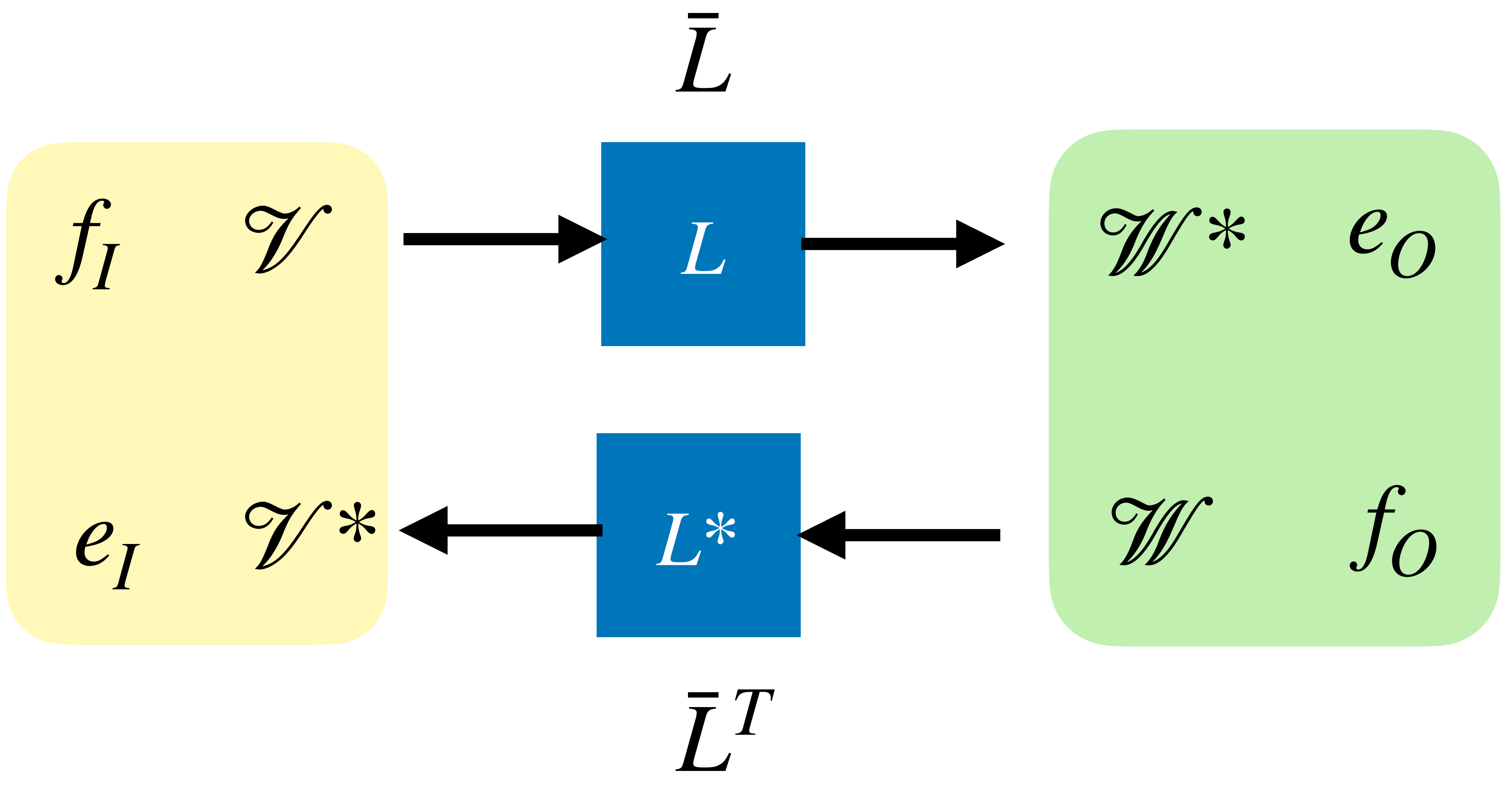
Flow to Flow



Gyrators

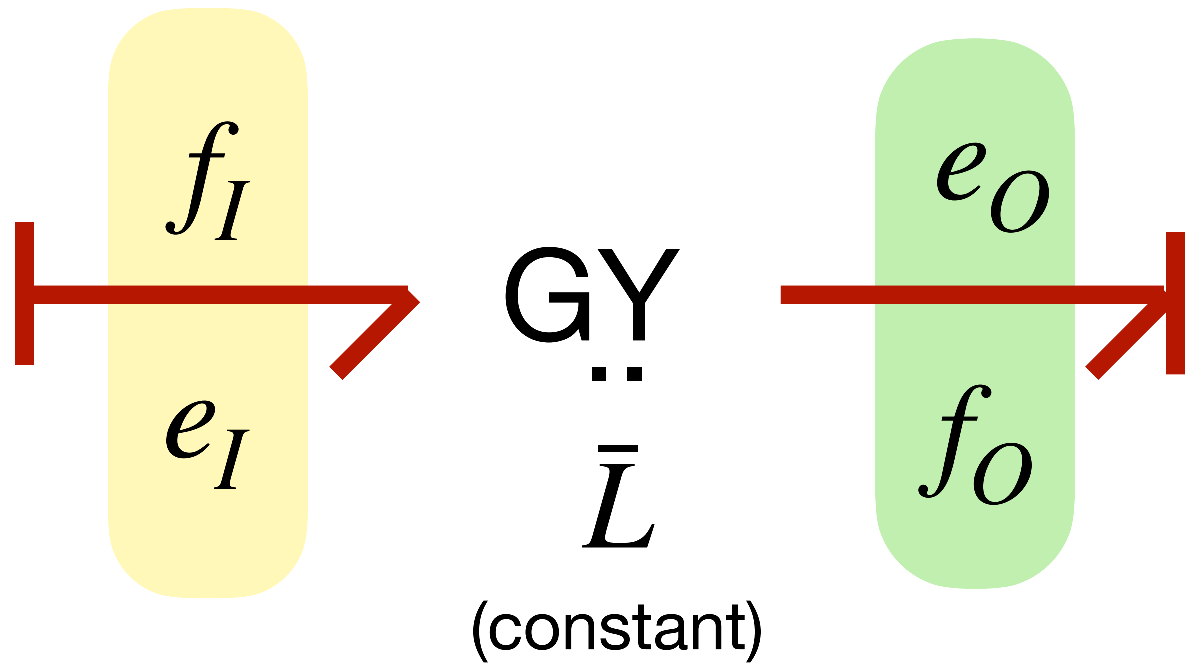
Flow to Effort

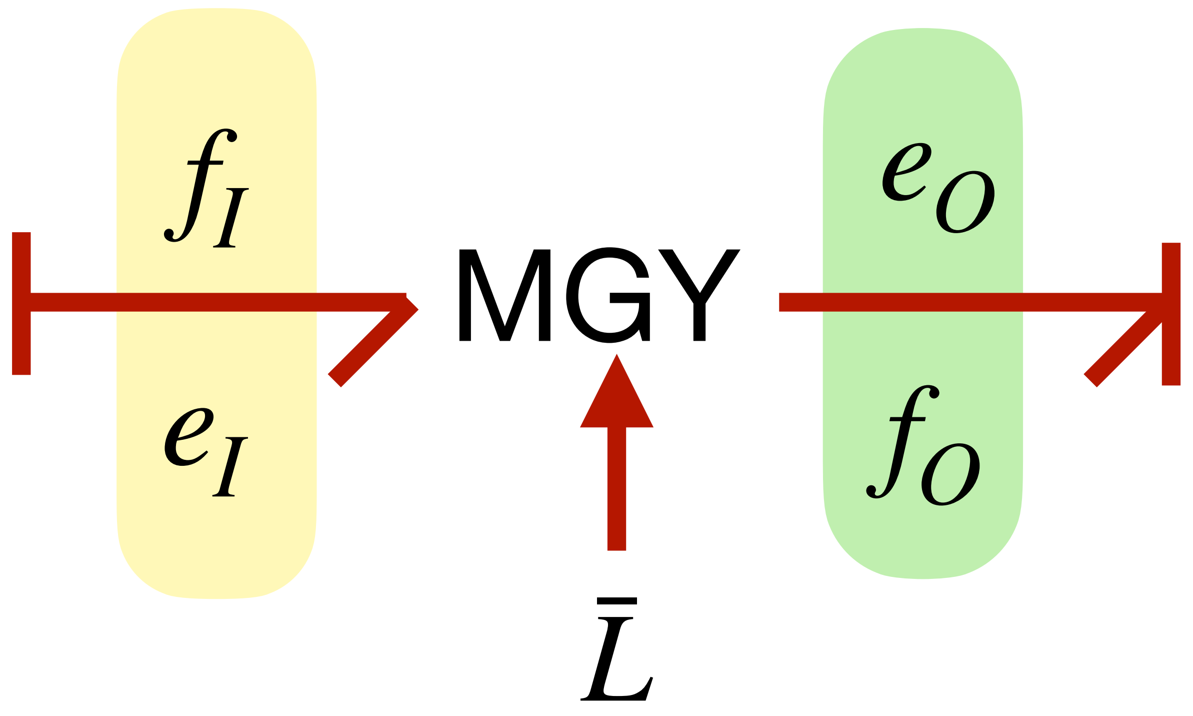
$P_I = e_I(f_I)$



$P_O = e_O(f_O)$

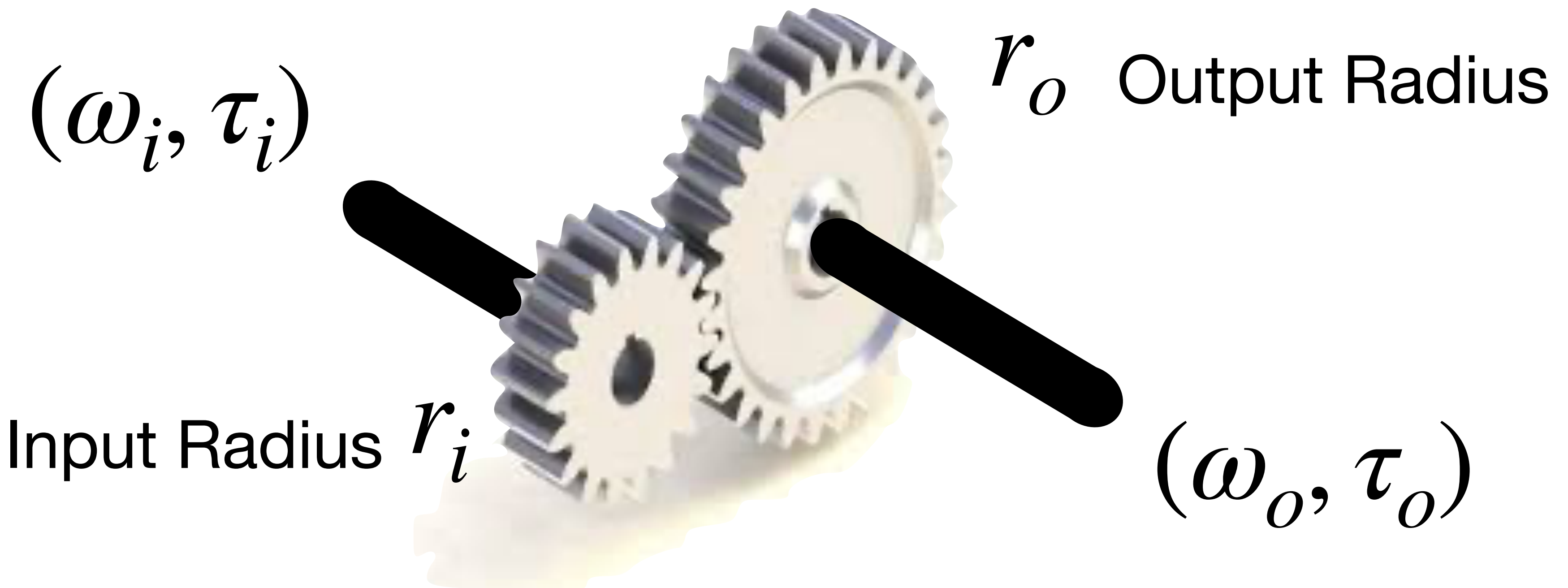
$P_I = P_O$





Example: Gears

Transformer: Flow to Flow



$\omega_i \in \mathbb{R}$ Input Angular velocity (Input flow)

$\tau_i \in \mathbb{R}^* = \mathbb{R}$ Input Torque (Input effort)

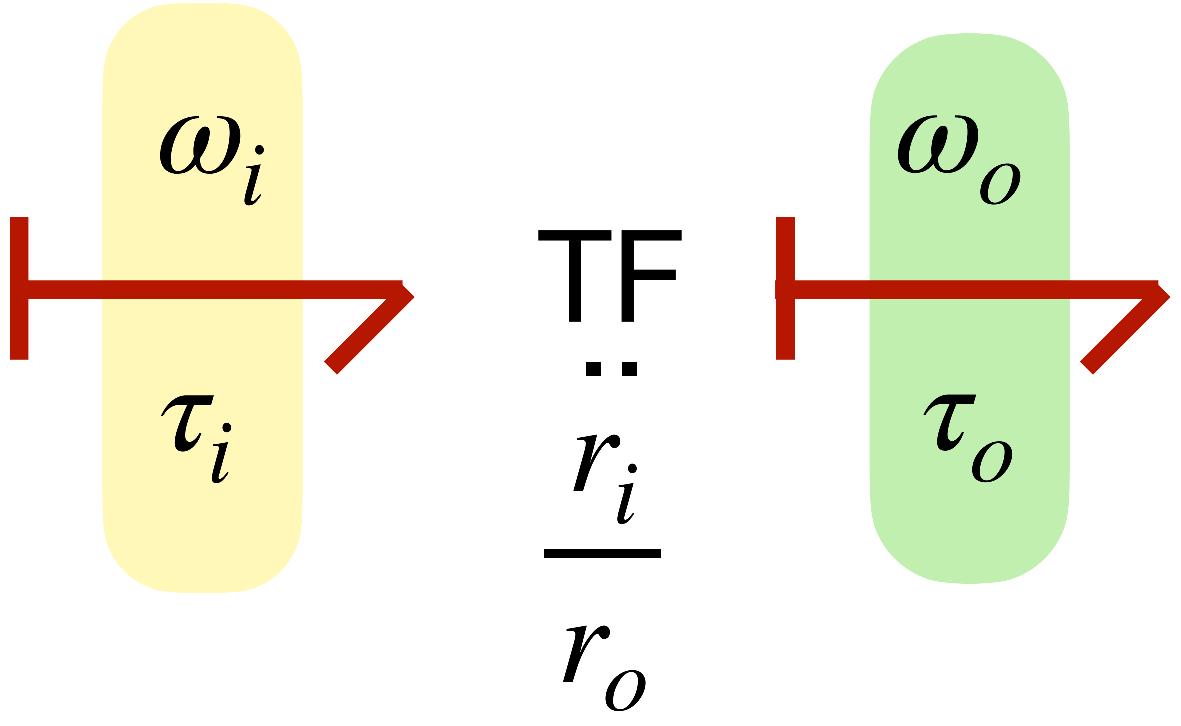
$\omega_o \in \mathbb{R}$ Output Angular velocity (Output Flow)

$\tau_o \in \mathbb{R}^* = \mathbb{R}$ Output Torque (Output effort)

$$P_i = \tau_i \omega_i = \tau_o \omega_o = P_o$$

Power Continuity

$$r_o \omega_o = r_i \omega_i \Rightarrow \omega_o = \underbrace{\frac{r_i}{r_o}}_{\bar{L}} \omega_i \Rightarrow \tau_i = \underbrace{\frac{r_i}{r_o}}_{\bar{L}^T = \bar{L}} \tau_o$$



Example: Electrical DC motor

Gyrator: Flow to Effort

$I \in \mathbb{R}$	Electrical Current (Input flow)
$V \in \mathbb{R}^* = \mathbb{R}$	Voltage (Input effort)
$\omega \in \mathbb{R}$	Output Angular velocity (Output flow)
$\tau \in \mathbb{R}^* = \mathbb{R}$	Output Torque (Output effort)

$$P_i = VI = \tau\omega = P_o$$

Power Continuity

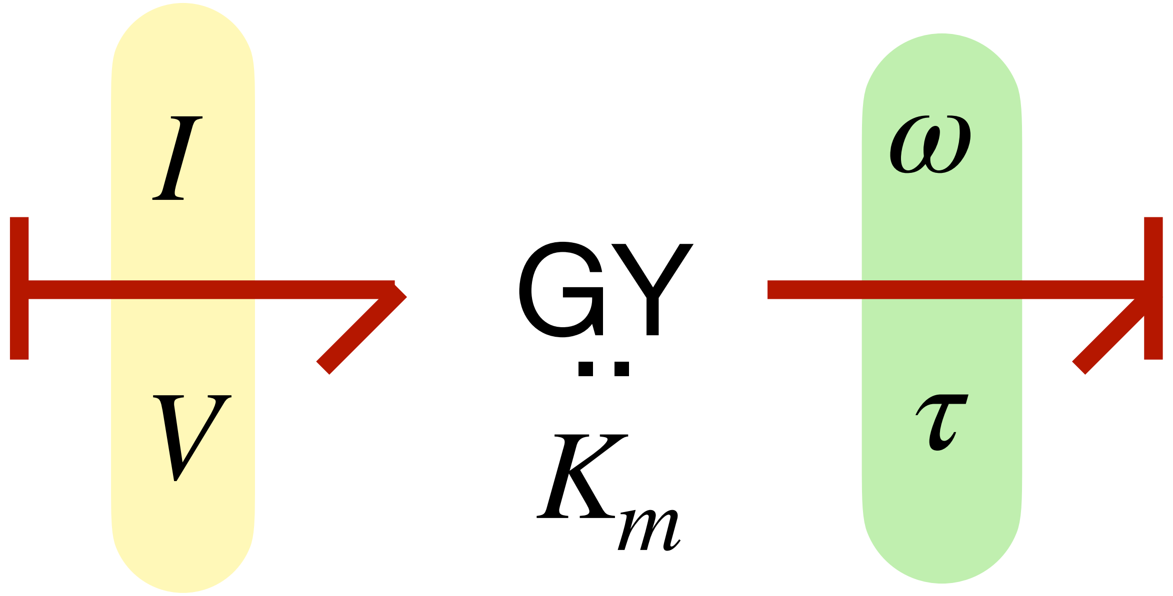
Back emf

$$\tau = K_m I \Rightarrow V = \underbrace{K_m^T}_{K_m = K_e} \omega$$

motor constant

electrical constant

!



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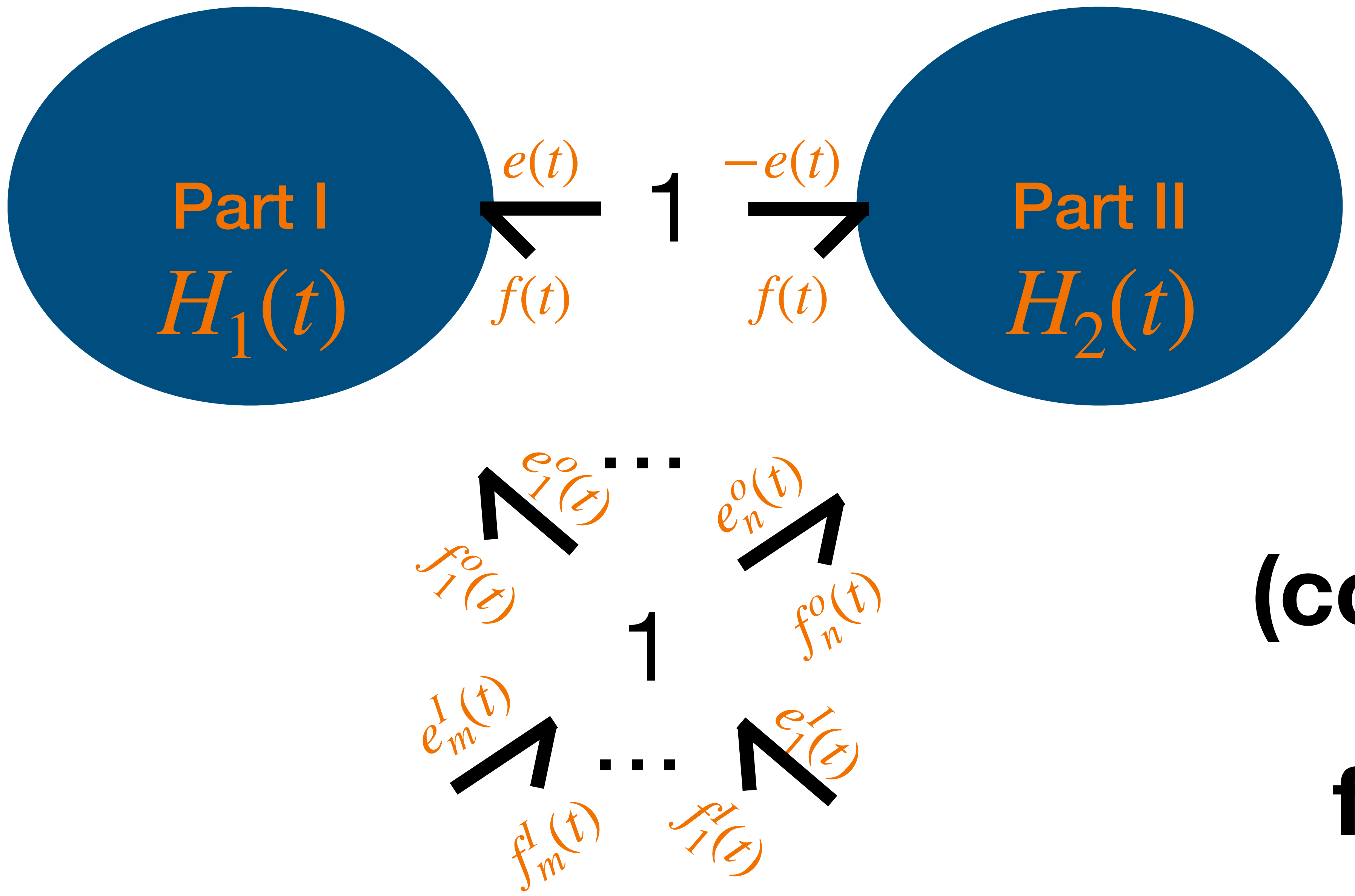
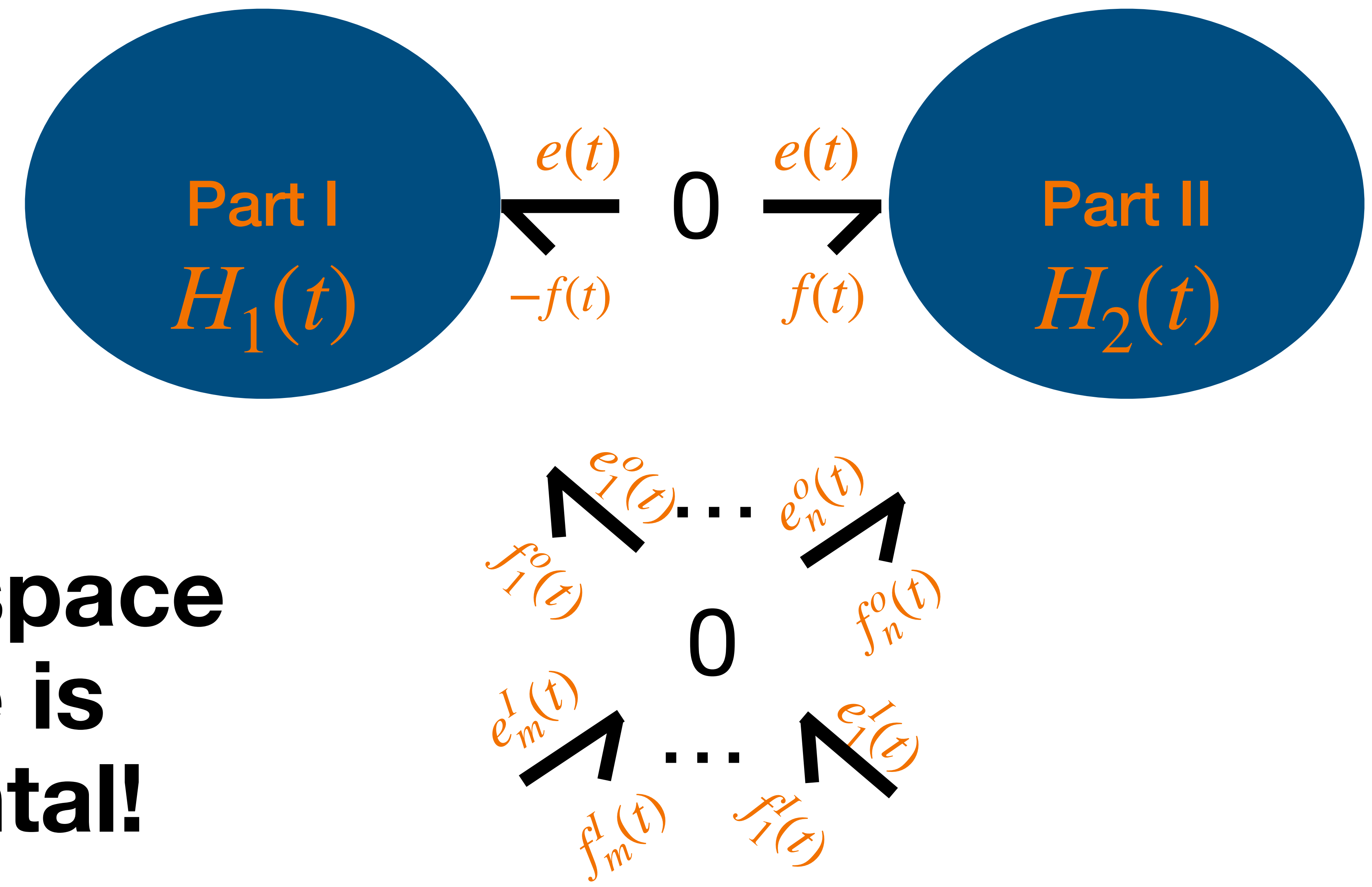
Junctions and generalised Kirchhoff laws

All based on Power preservation



(co)vector space structure is fundamental!



Flow junction

$$f_1^I = \ldots = f_m^I = f_1^o = \ldots = f_n^o$$

$$\sum_{i=1}^{m} e_i^I = \sum_{i=1}^{n} e_i^o$$

Effort junction

$$e_1^I = \ldots = e_m^I = e_1^o = \ldots = e_n^o$$

$$\sum_{i=1}^{m} f_i^I = \sum_{i=1}^{n} f_i^o$$

Power continuity

$$\sum_{i=1}^{m} P_i^I = \sum_{i=1}^{n} P_i^o$$

More General Power Continuos Elements

All based on Power preservation:

Power continuity $\sum_{i=1}^{m} P_i^I = \sum_{i=1}^{n} P_i^o$

But

Flow junction

$$f_1^I = \ldots = f_m^I = f_1^o = \ldots = f_n^o$$

$$\sum_{i=1}^{m} e_i^I = \sum_{i=1}^{n} e_i^o$$

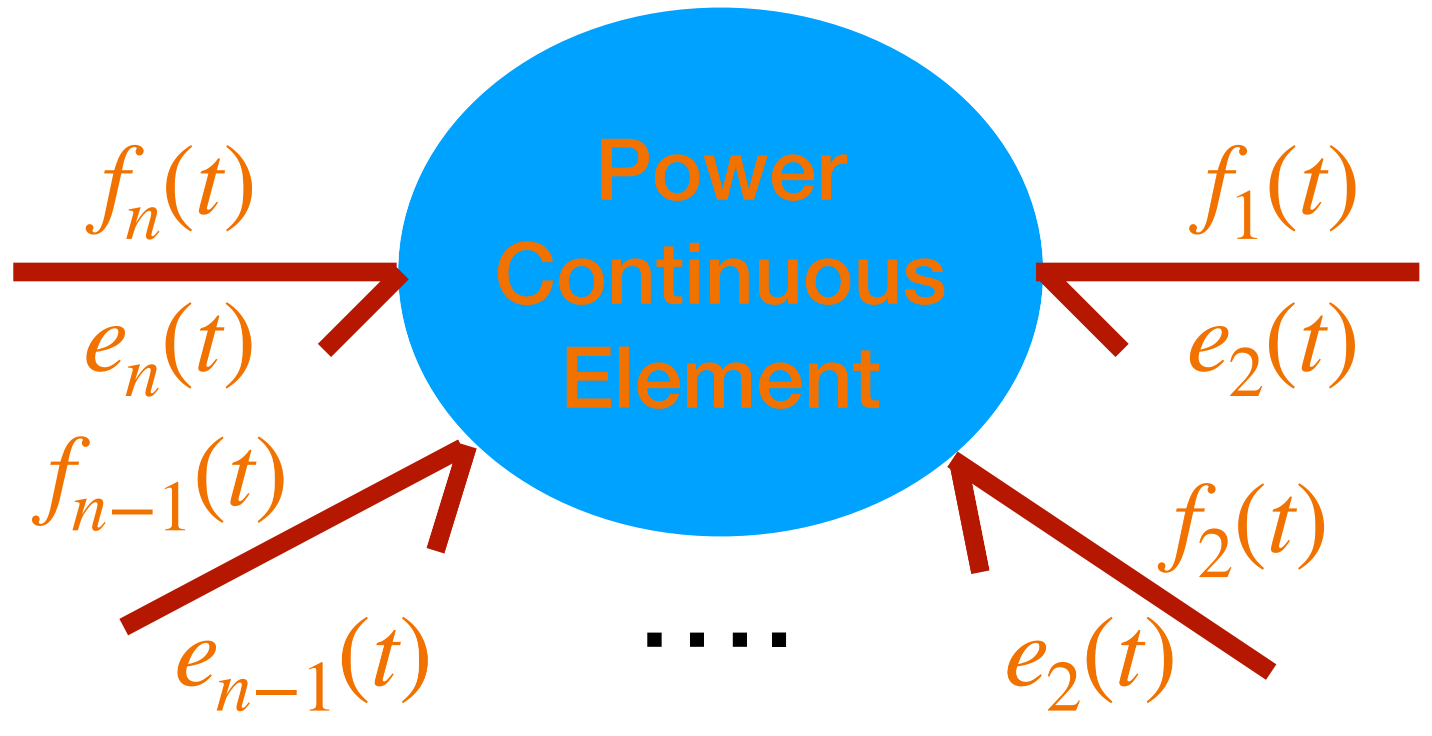
Effort junction

$$e_1^I = \ldots = e_m^I = e_1^o = \ldots = e_n^o$$

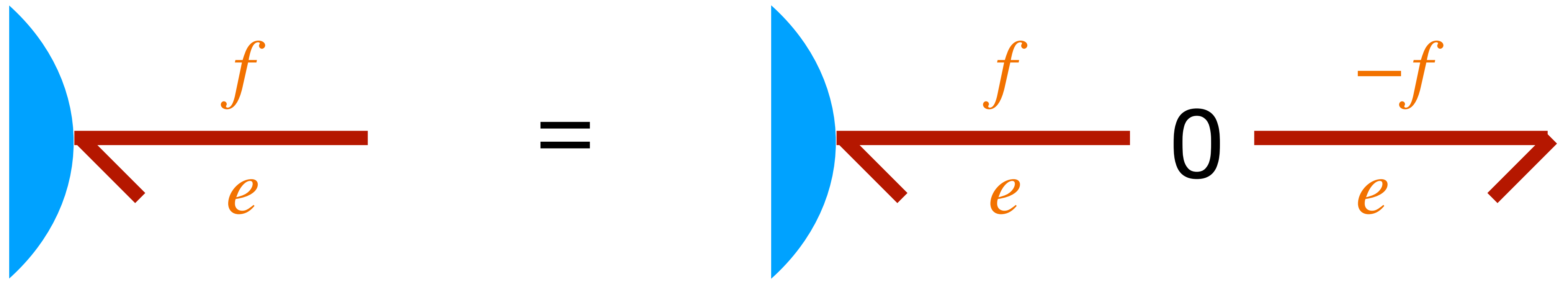
$$\sum_{i=1}^{m} f_i^I = \sum_{i=1}^{n} f_i^o$$

Modify
Relations
of
efforts and flows

Power continuity $\sum_{i=1}^{n} P_i^I = 0$

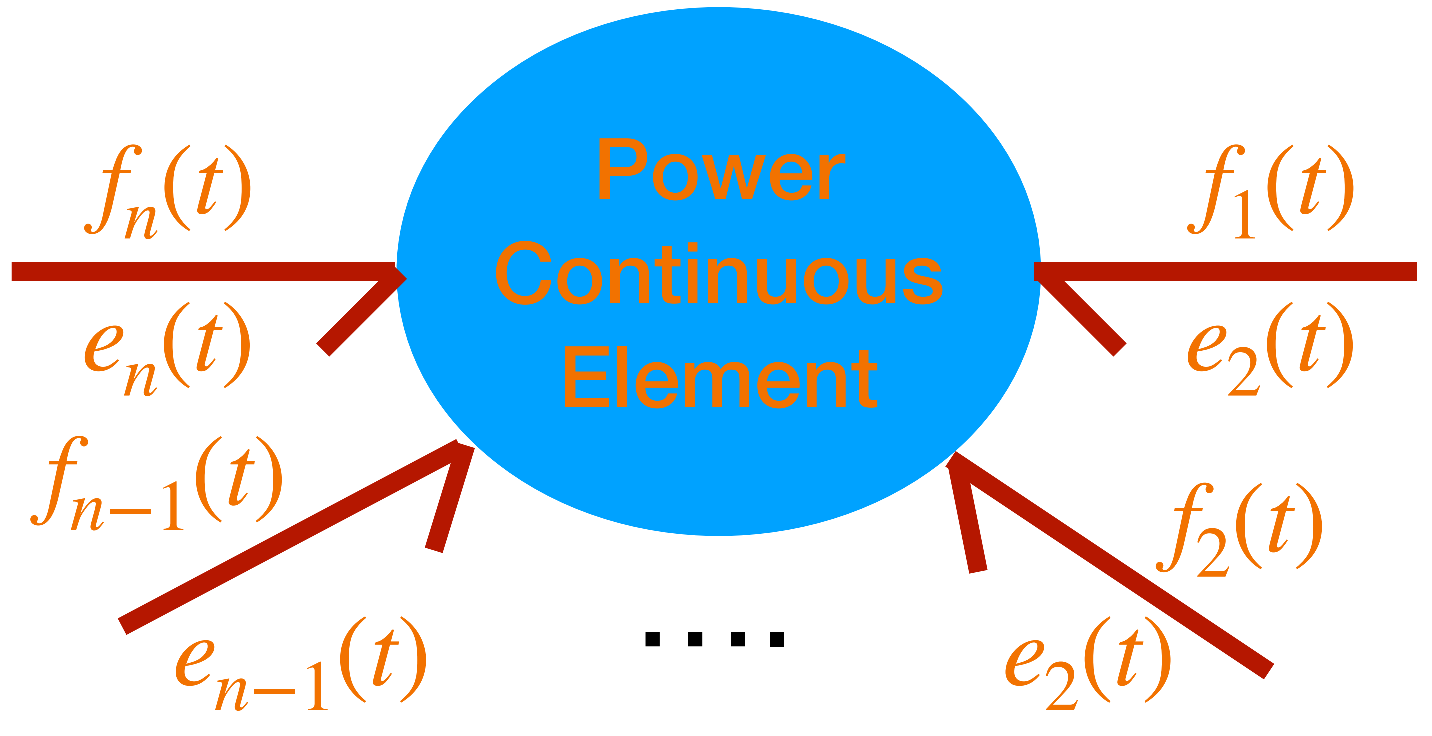


Notice



More General Power Continuos Elements

All based on Power preservation:



Power continuity $\sum_{i=1}^{n} P_i^I = 0$

Let's Put all efforts in a vector and flows in another:

$$e := \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_{n-1} \\ e_n \end{pmatrix} \quad f := \begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_{n-1} \\ f_n \end{pmatrix}$$

Power continuity ➔

$$e^T f = 0$$

$$e_1(f_1) + e_2(f_2) + \ldots + e_n(f_n) = 0$$

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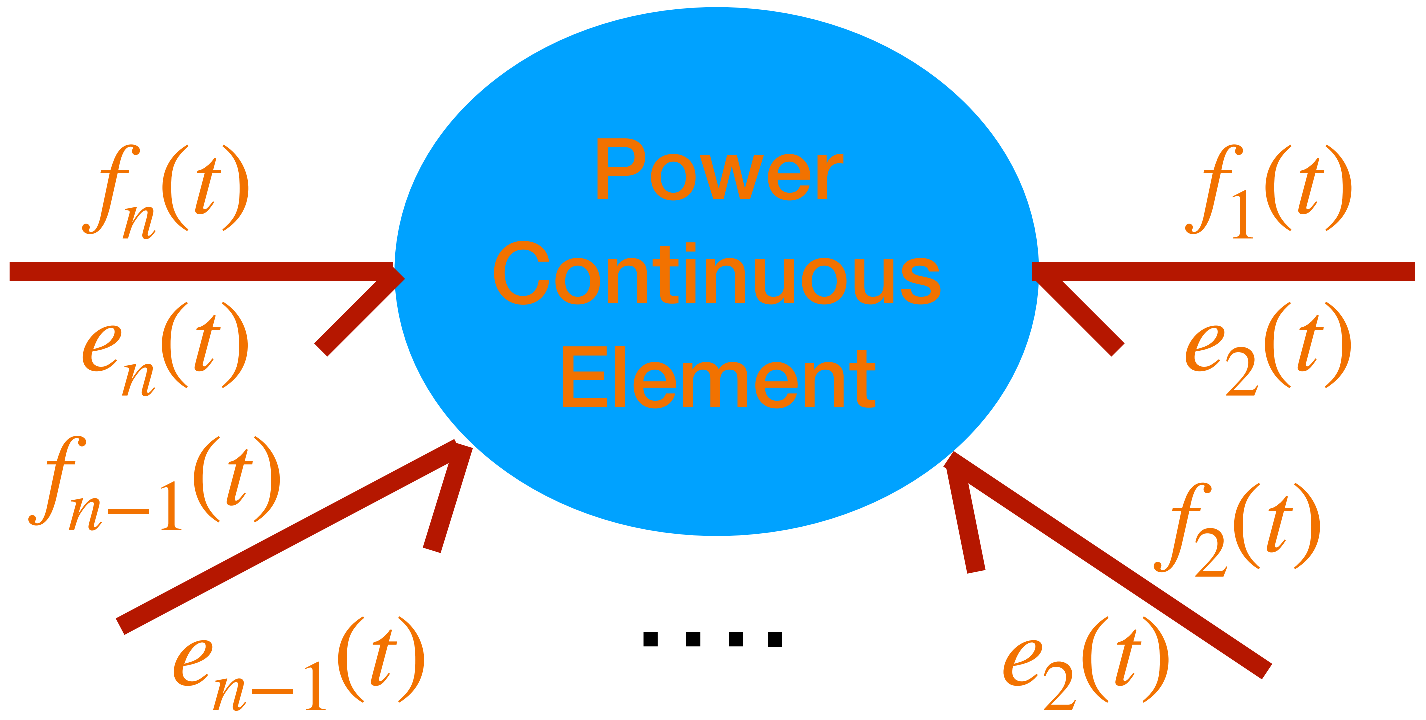
More General Power Continuous Elements

All based on Power preservation:



Power continuity

$$e^T f = f^T e = 0$$

Consider any **general**, **linear**, **possibly time-varying** relation (E, F matrices) of the form $Ee + Ff = 0^n$. What are the conditions to ensure Power Continuity?

$$A := (E\ F), p := \begin{pmatrix} e \\ f \end{pmatrix}$$

$$Ee + Ff = 0 \quad \Leftrightarrow \quad (E\ F)\begin{pmatrix} e \\ f \end{pmatrix} = 0 \quad \Leftrightarrow \quad Ap = 0$$

Under certain conditions and a fundamental theorem of Linear Algebra

$$Ap = 0 \quad \Leftrightarrow \quad p^\perp = A^T \lambda$$

$$a \perp b \Leftrightarrow a \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} b = 0$$

$$(E\ F)\begin{pmatrix} e \\ f \end{pmatrix} = 0$$

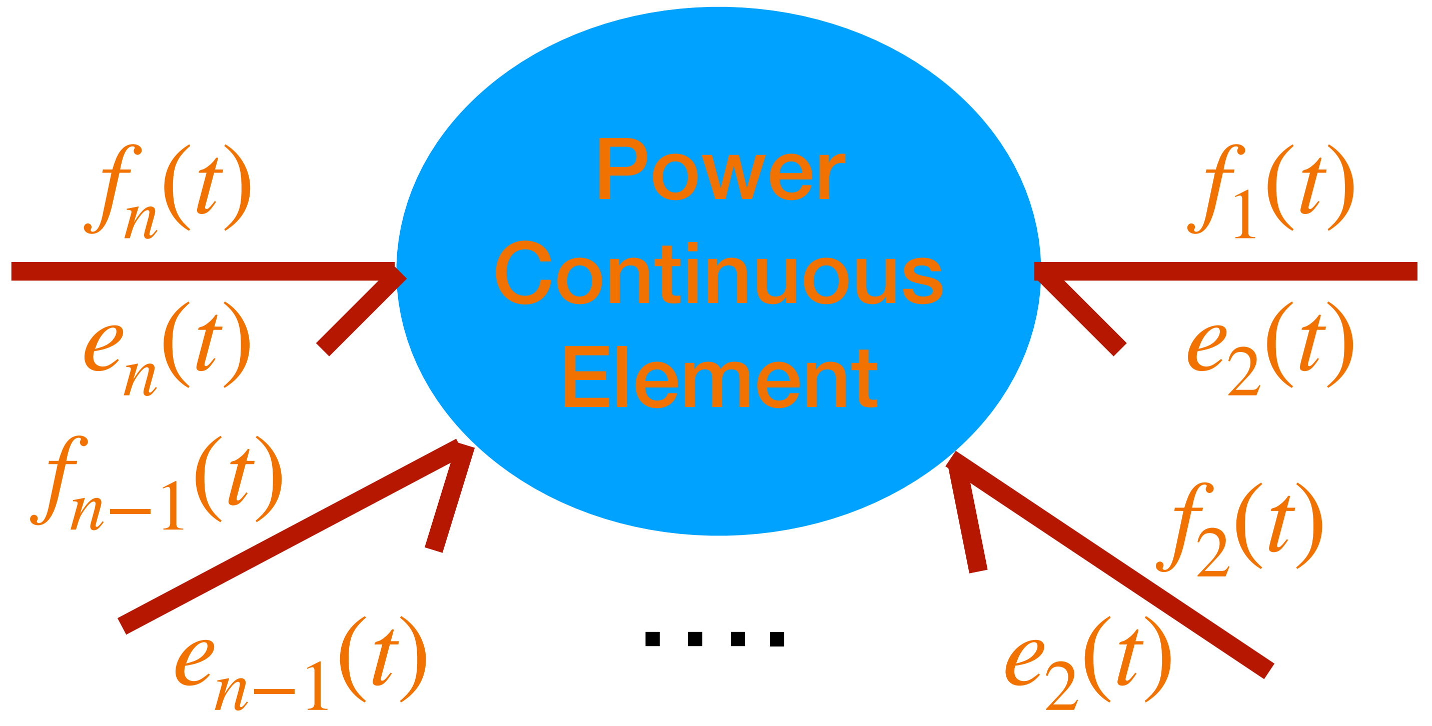
$$\begin{pmatrix} f^T \\ e^T \end{pmatrix} = \begin{pmatrix} F^T \\ E^T \end{pmatrix} \lambda$$

$$EF^T + FE^T = 0$$

Power continuity condition

More General Power Continuos Elements

All based on Power preservation:



Power continuity ➜ $e^T f = f^T e = 0$

$$EF^T + FE^T = 0$$

Power continuity condition

Consider any **general, linear, possibly time-varying** relation (E, F matrices) of the form

$$Ee + Ff = 0$$ **A-causal**

It can be seen that it would be possible to partition the previous equation as:

$$\begin{pmatrix} E_i & E_o \end{pmatrix} \begin{pmatrix} e_i \\ e_o \end{pmatrix} + \begin{pmatrix} F_o & F_i \end{pmatrix} \begin{pmatrix} f_o \\ f_i \end{pmatrix} = 0 \quad \Rightarrow \quad \begin{pmatrix} F_o & E_o \end{pmatrix} \begin{pmatrix} f_o \\ e_o \end{pmatrix} + \begin{pmatrix} E_i & F_i \end{pmatrix} \begin{pmatrix} e_i \\ f_i \end{pmatrix} = 0$$

s.t. $\begin{pmatrix} F_o & E_o \end{pmatrix}$ is invertible ➜

$$\begin{pmatrix} f_o \\ e_o \end{pmatrix} = \underbrace{-\begin{pmatrix} F_o & E_o \end{pmatrix}^{-1} \begin{pmatrix} E_i & F_i \end{pmatrix}}_{J} \begin{pmatrix} e_i \\ f_i \end{pmatrix}$$

Causal

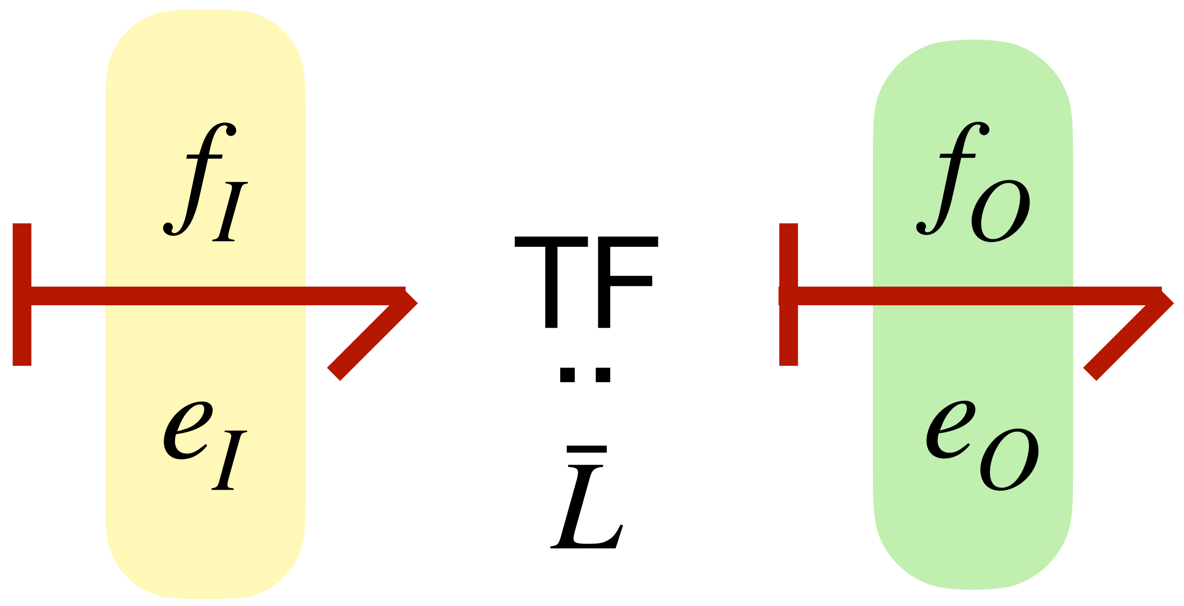
$$e^T f = 0$$

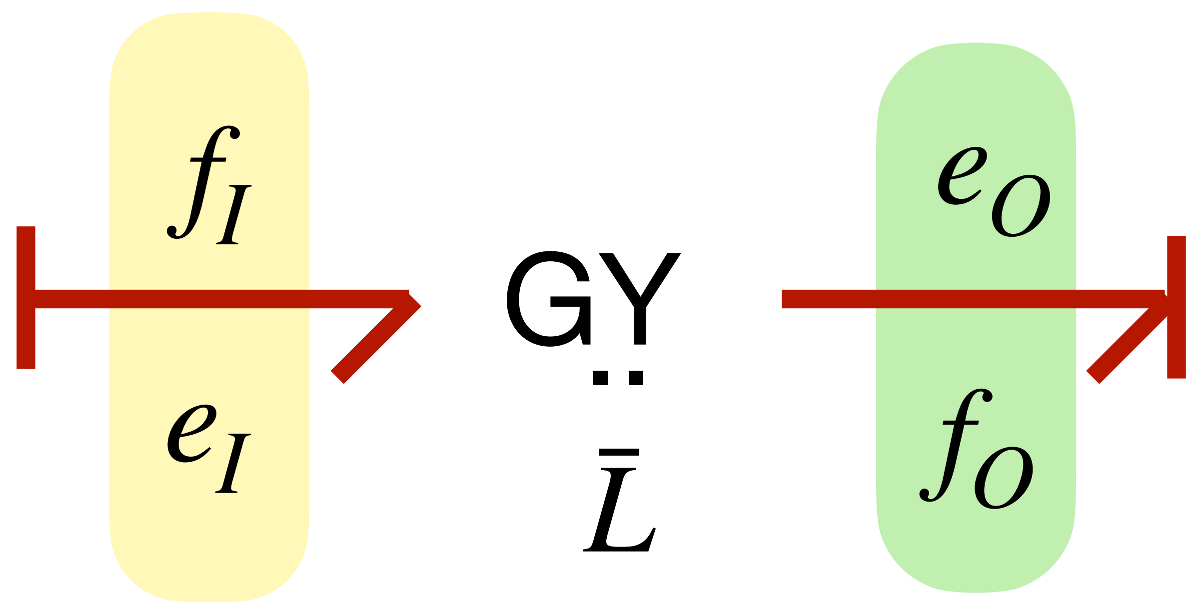
$$\Downarrow$$

$$J = -J^T$$

Exercise

Calculate E, F and J for the following cases





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Energy “Dissipation”

How can we model free energy dissipation of better said irreversible energy transformation to heat ?

Free-Energy Dissipation

$P = e(f) \geq 0 \; \forall e, f$

Energy cannot be dissipated

$\mathscr{V} \ni f$

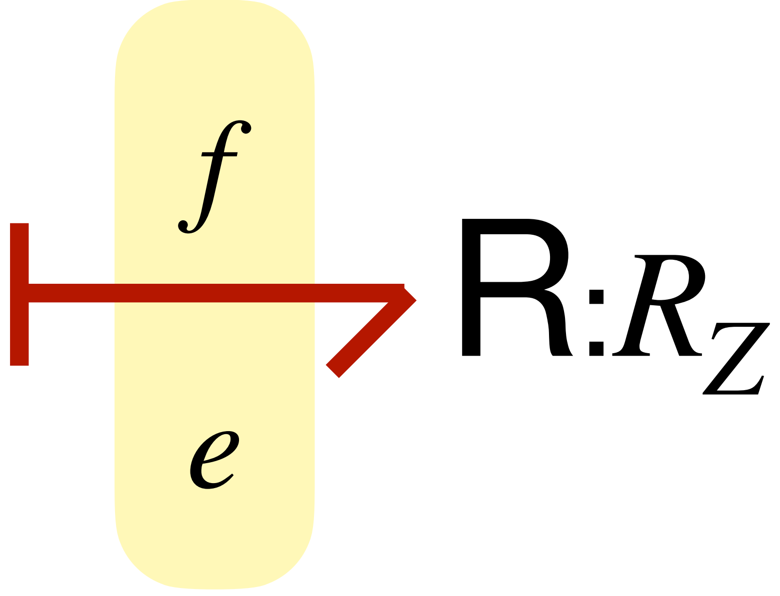
$\mathscr{V}^* \ni e$

R_Z

$e = R_Z(f) \Rightarrow (R_Z(f))(f) \geq 0$

linear case using basis

$\bar{e} = \bar{R}_Z \bar{f} \Rightarrow \bar{f}^T \bar{R}_Z \bar{f} \geq 0$



$\mathscr{V} \ni f$

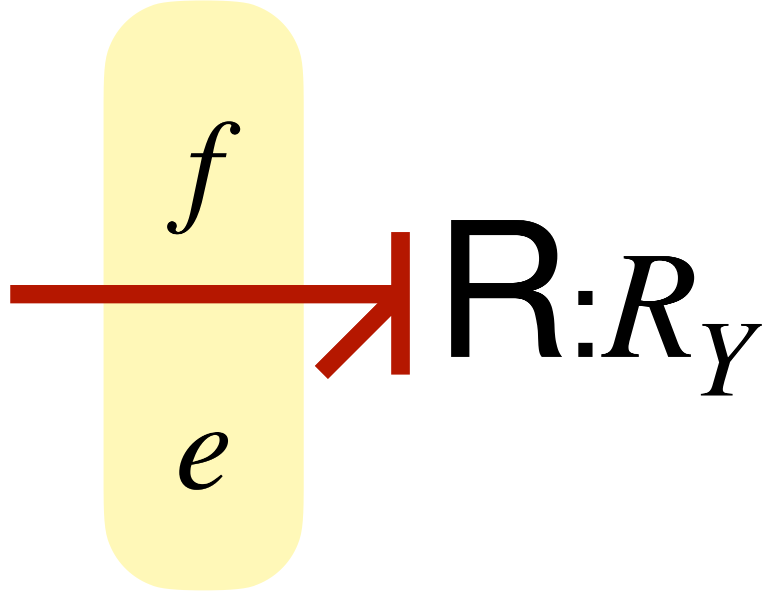
$\mathscr{V}^* \ni e$

R_Y

$f = R_Y(e) \Rightarrow (R_Y(e))(e) \geq 0$

linear case using basis

$\bar{f} = \bar{R}_Y \bar{e} \Rightarrow \bar{e}^T \bar{R}_Y \bar{e} \geq 0$



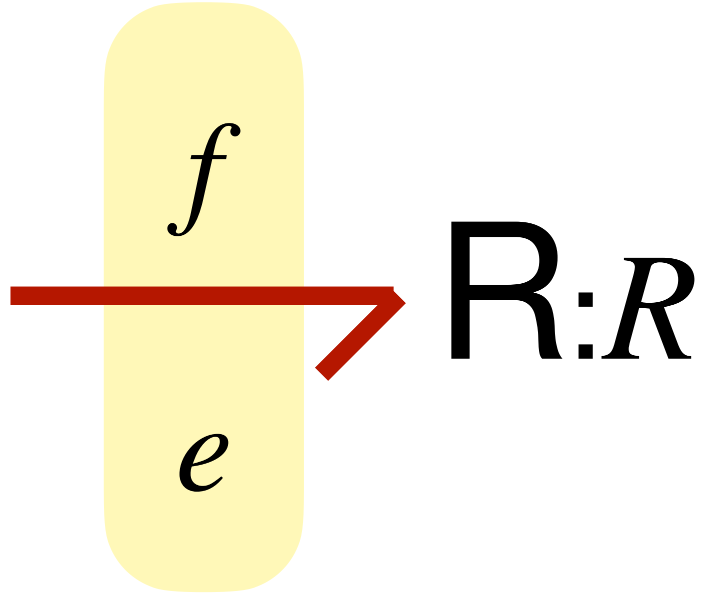
$\mathscr{V} \ni f$

$\mathscr{V}^* \ni e$

$R(f, e) = 0$

Relation

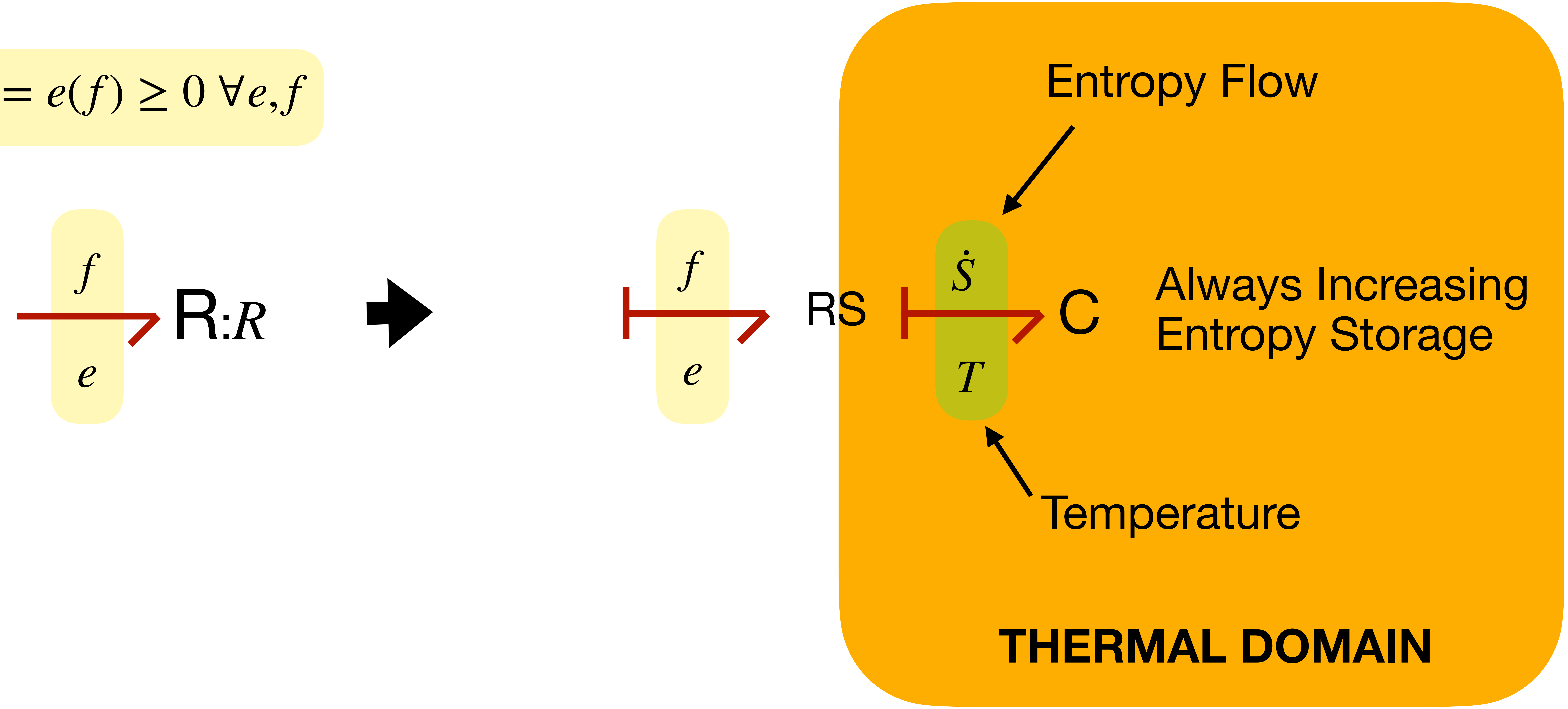
$P = e(f) \geq 0 \; \forall e, f$



Irreversible Energy to Thermal

Adding a port to the thermal domain

$$P = e(f) \geq 0 \; \forall e,f$$



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Ideal Sources

How can we model ideal source which may represent an interconnection of the Physical World to the Digital World for example ?

Ideal Sources

Energy continuity breaks here

$a : \mathrm{S_e} \xrightarrow[e]{f}$

always $e = a$
independent of f

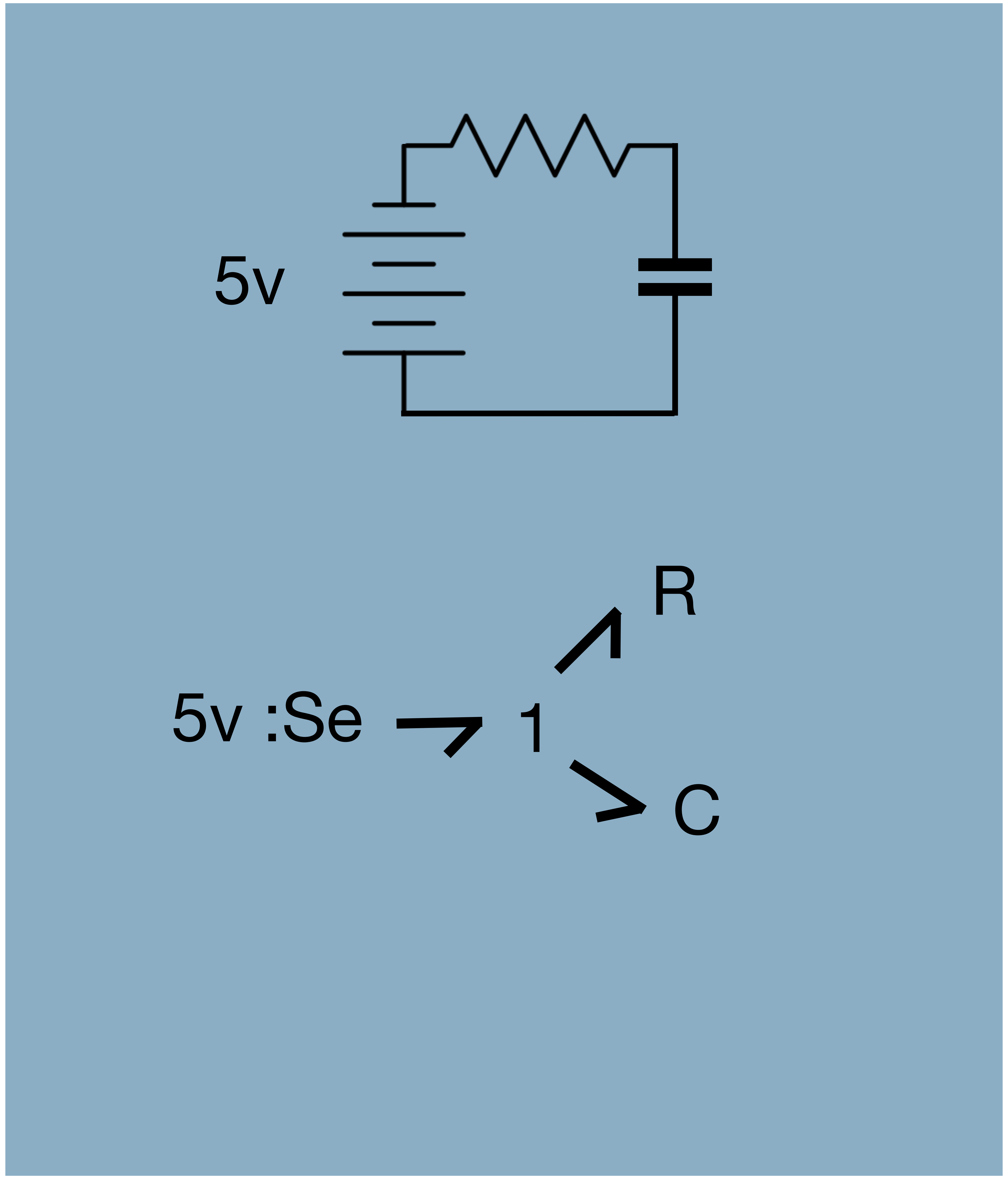
$a(t) \longrightarrow \mathrm{MS_e} \xrightarrow[e]{f}$

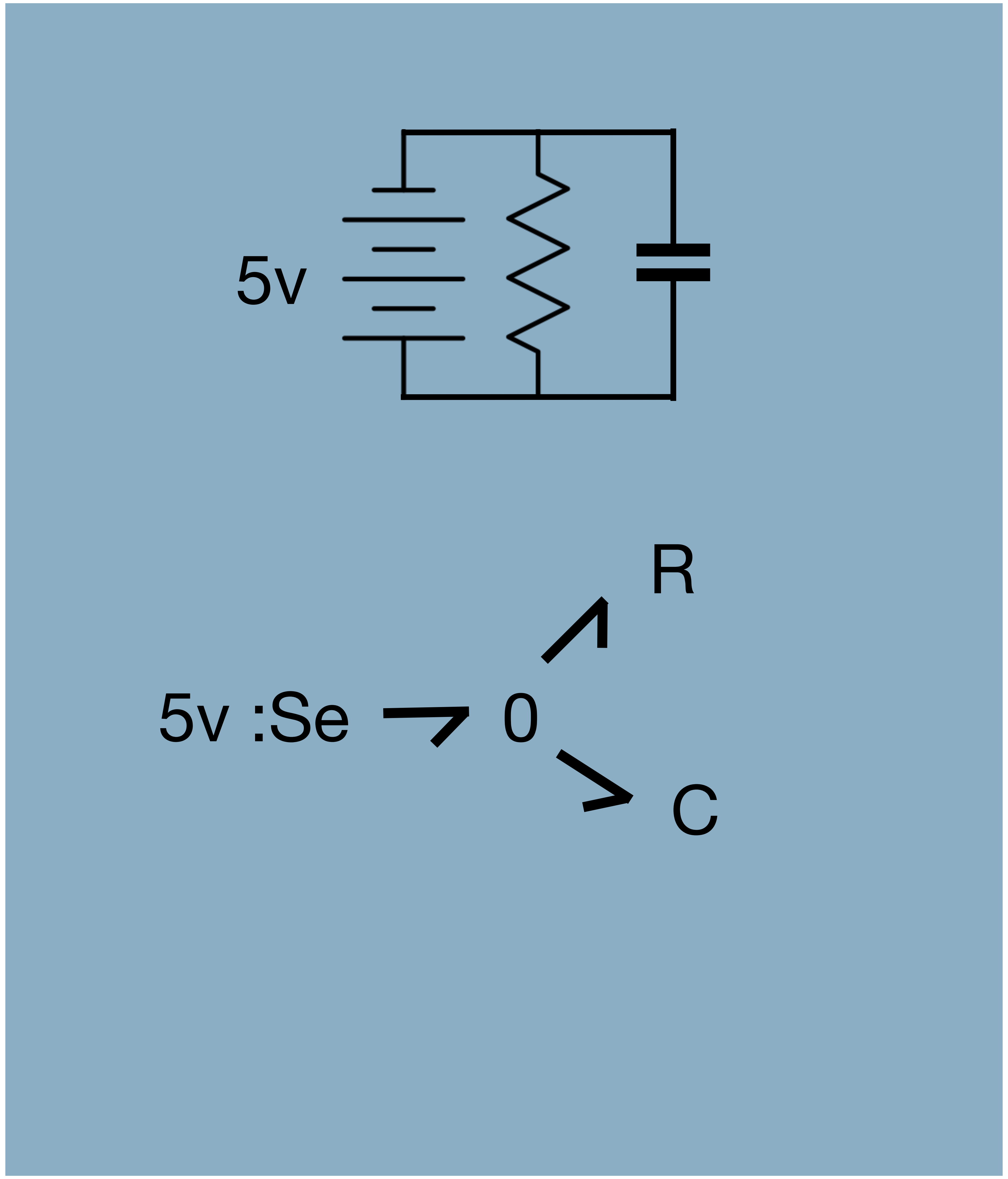
$a : \mathrm{S_f} \xrightarrow[e]{f}$

always $f = a$
independent of e

$a(t) \longrightarrow \mathrm{MS_f} \xrightarrow[e]{f}$

Example





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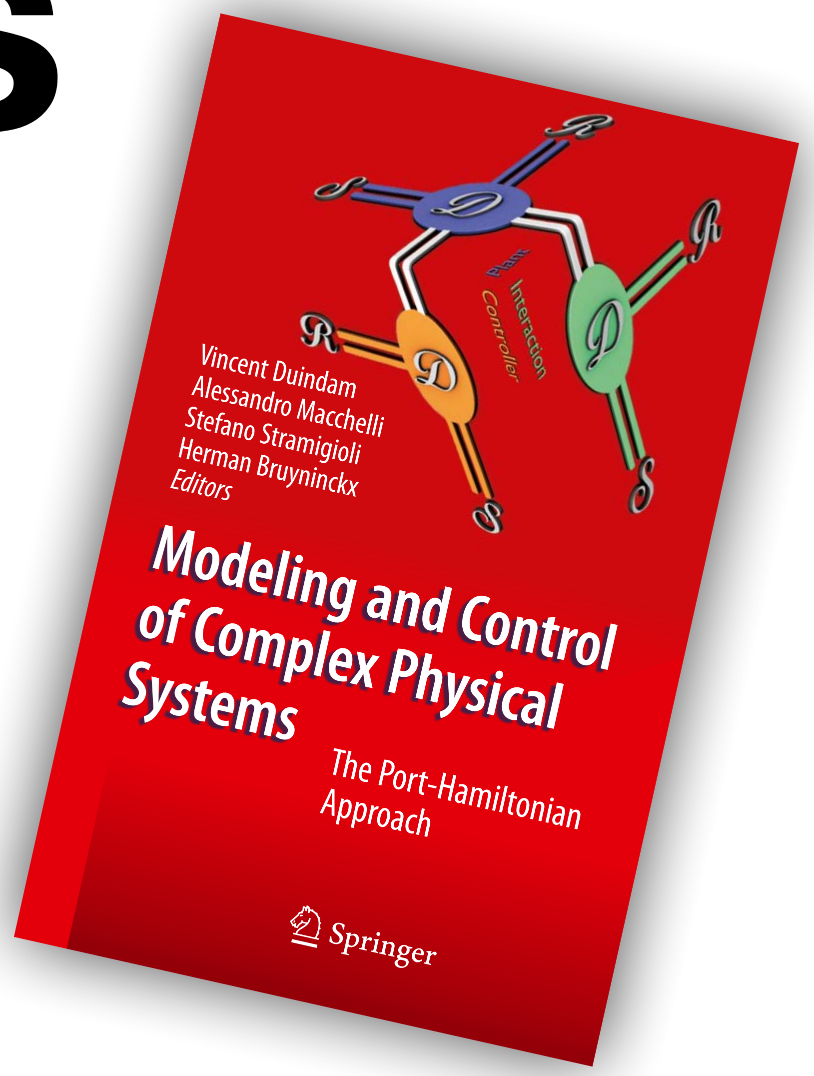
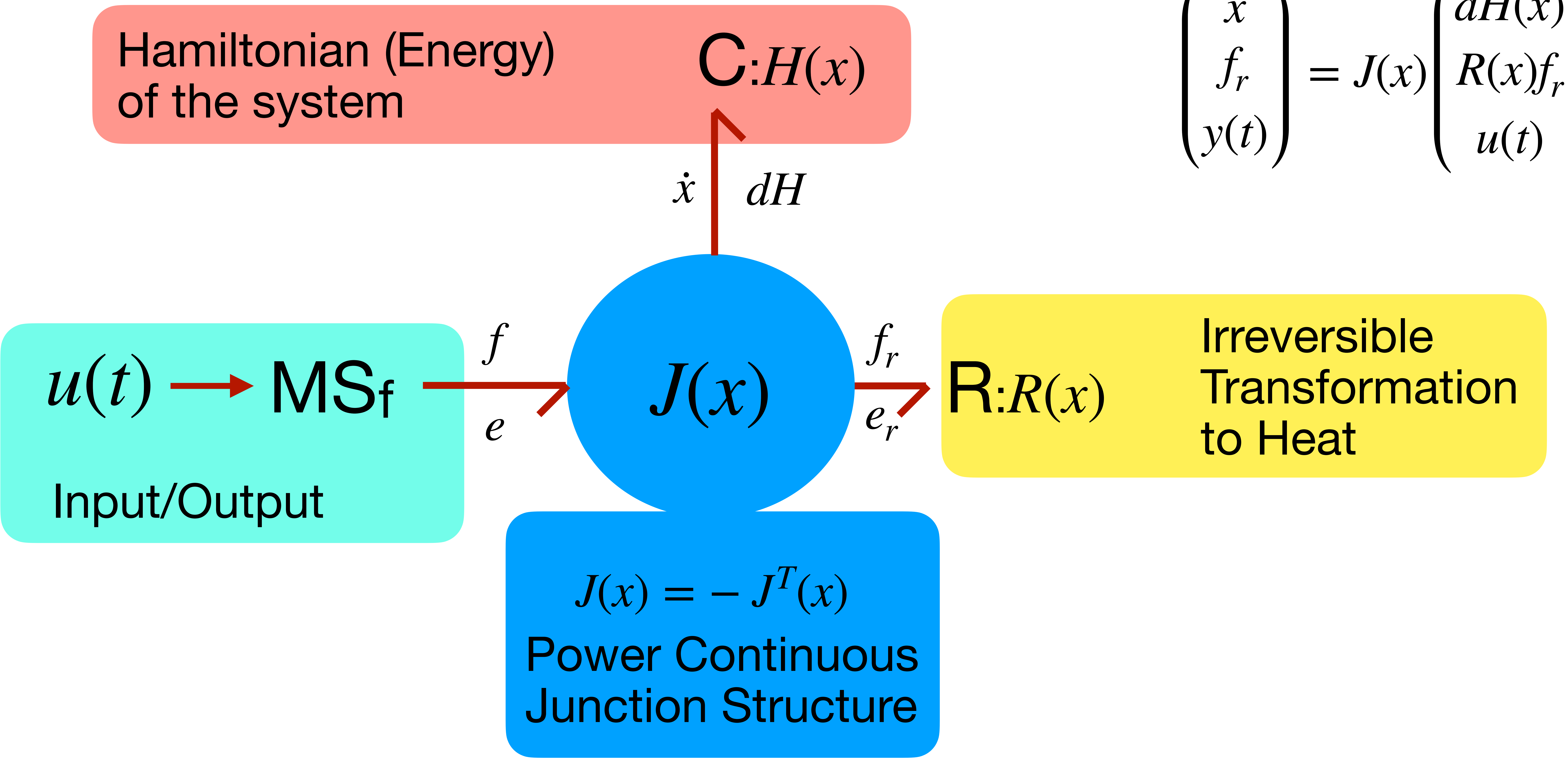
Port Hamiltonian Systems

A VERY brief introduction



General Structure

$$\begin{pmatrix} \dot{x} \\ f_r \\ y(t) \end{pmatrix} = J(x) \begin{pmatrix} dH(x) \\ R(x) f_r \\ u(t) \end{pmatrix}$$



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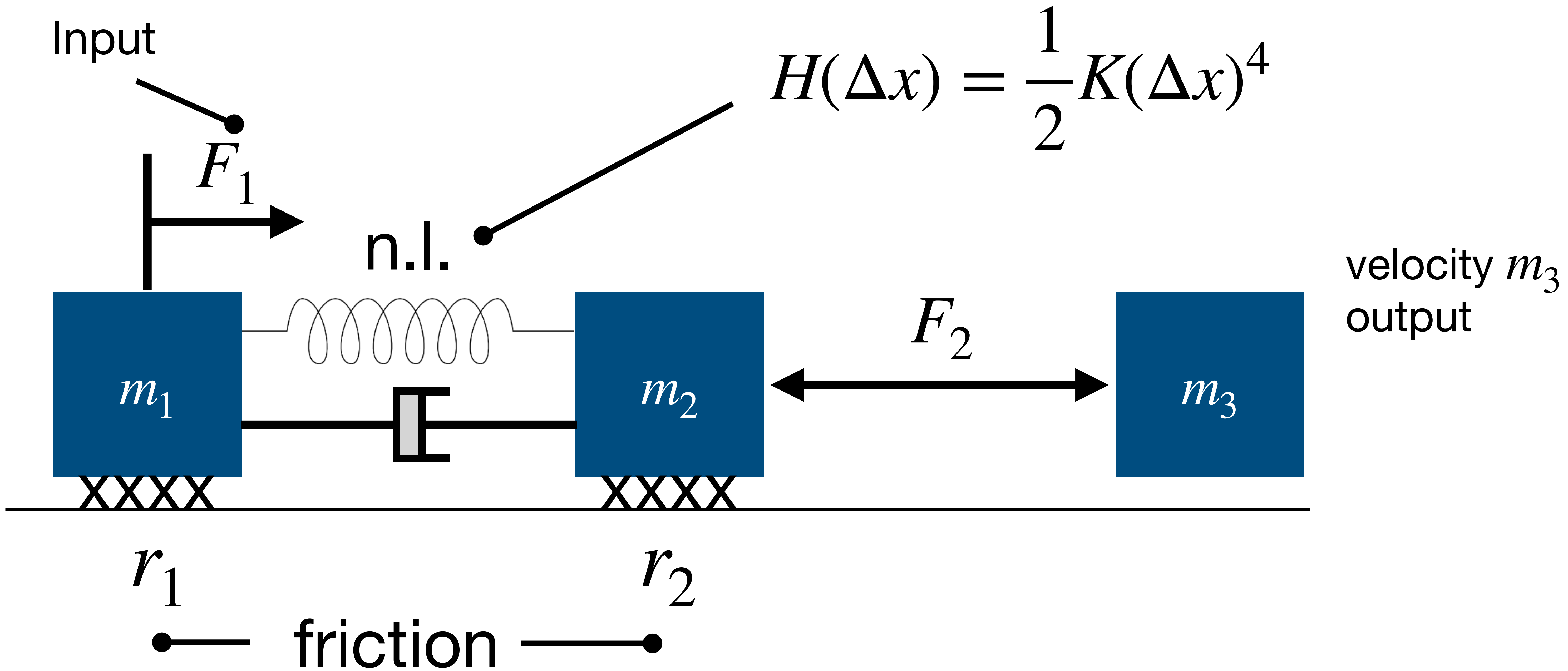
Example

In Mechanical domain

CONCEPTUAL MODEL



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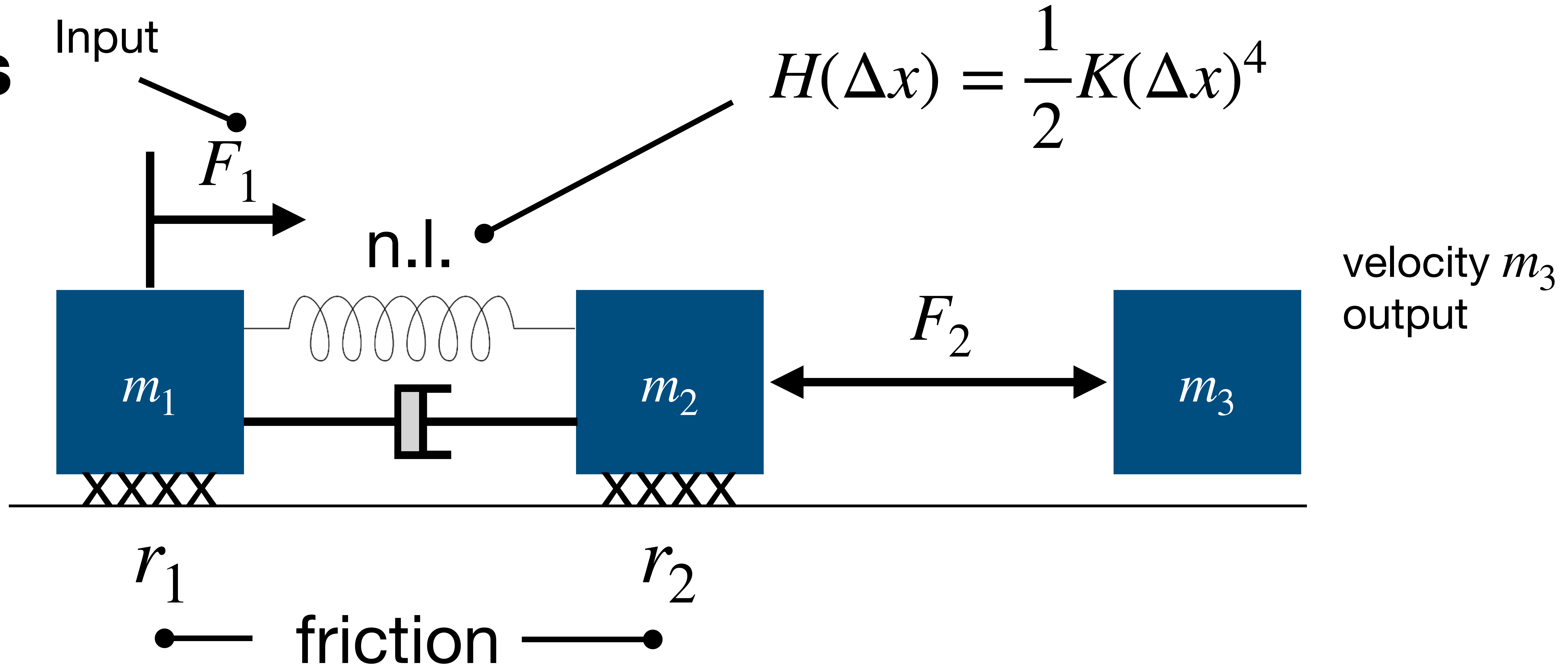
1. Fix velocity references using 1 junction



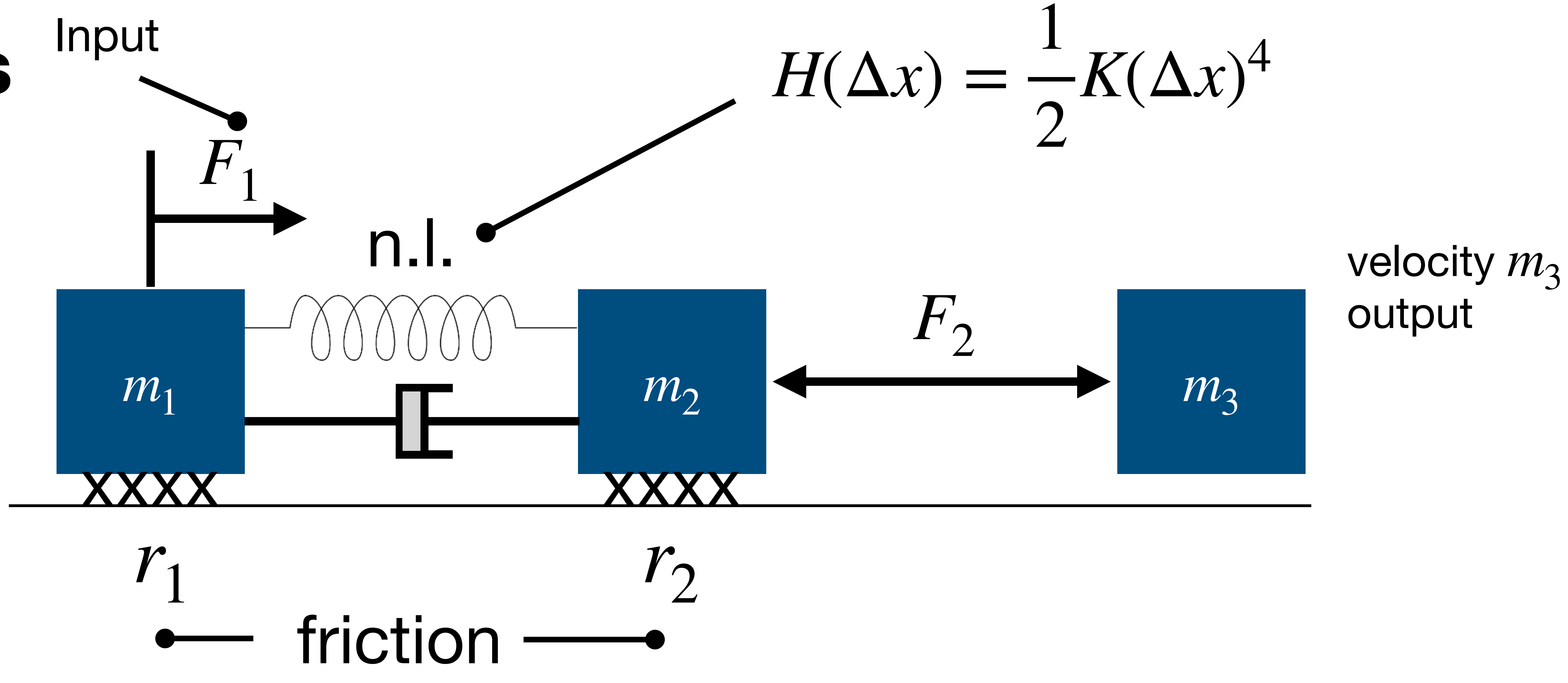
$\underset{0}{\overset{1}{\vdots}}$ $\underset{v_1}{\overset{1}{\vdots}}$ $\underset{v_2}{\overset{1}{\vdots}}$ $\underset{v_3}{\overset{1}{\vdots}}$

Remember: Differences

That is why we have a Vector and Co-Vector Space structure on the bonds!

$$
\begin{tikzcd}
f_1 \ 1 \arrow[r, harpoon] & 0 \arrow[r, harpoon] \arrow[d, harpoon] & 1 \ f_2 \\
& 1 \ f_1 - f_2 &
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}
e_1 \ 0 \arrow[r, harpoon] & 1 \arrow[r, harpoon] \arrow[d, harpoon] & 0 \ e_2 \\
& 0 \ e_1 - e_2 &
\end{tikzcd}
$$

1. Fix velocity references using 1 junction



1	1	1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$
0	v_1	v_2	v_3

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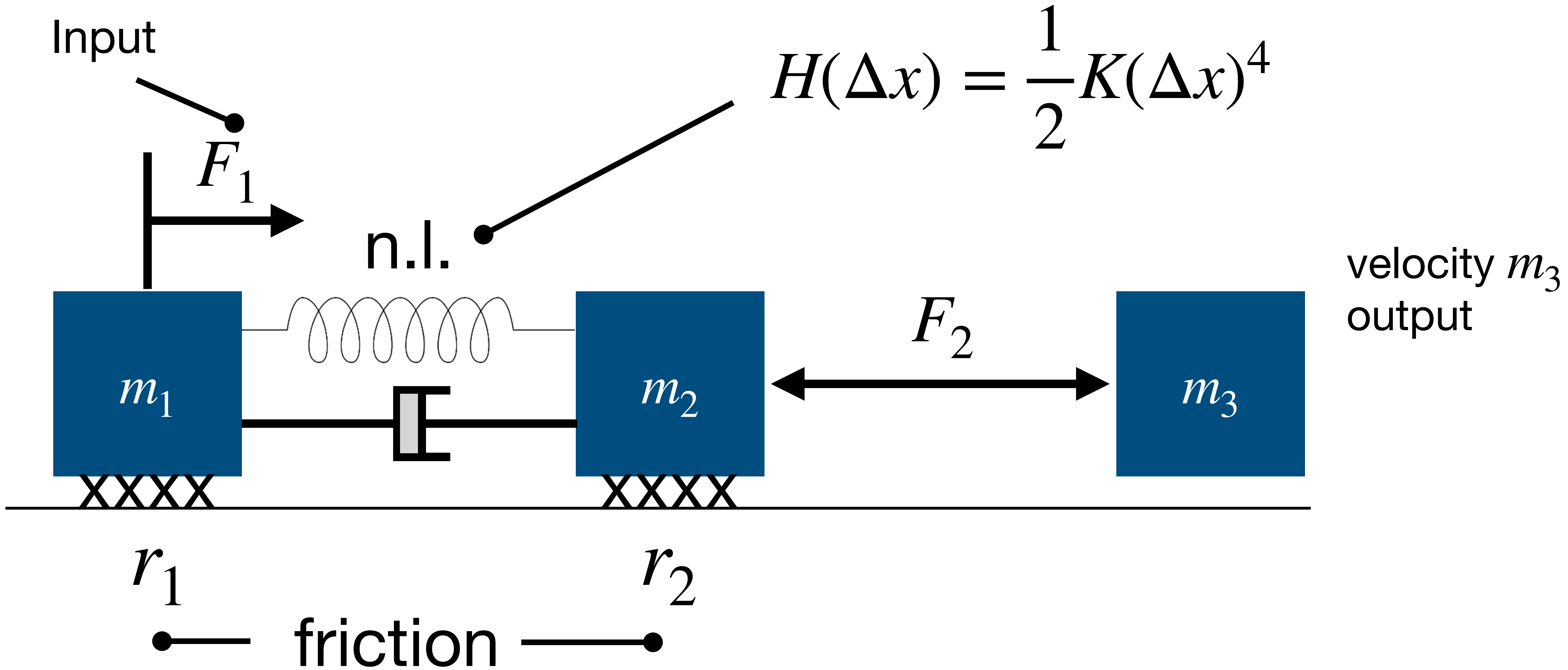
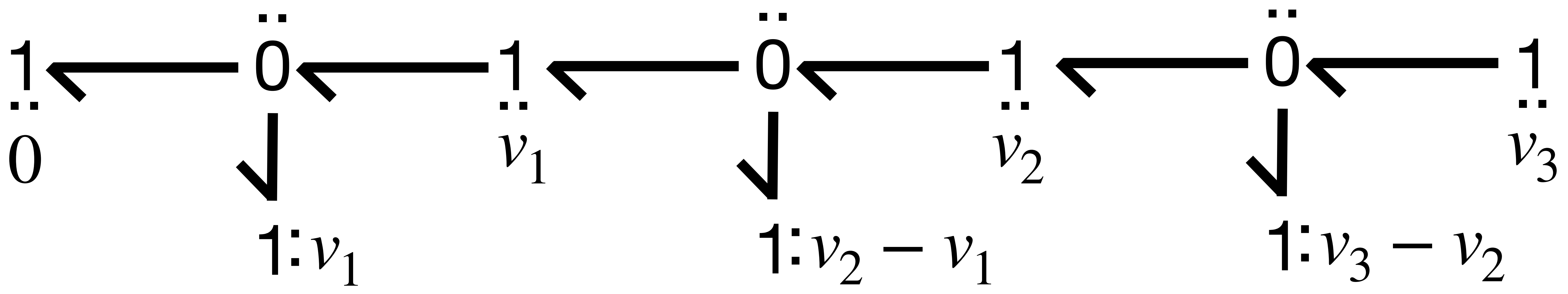
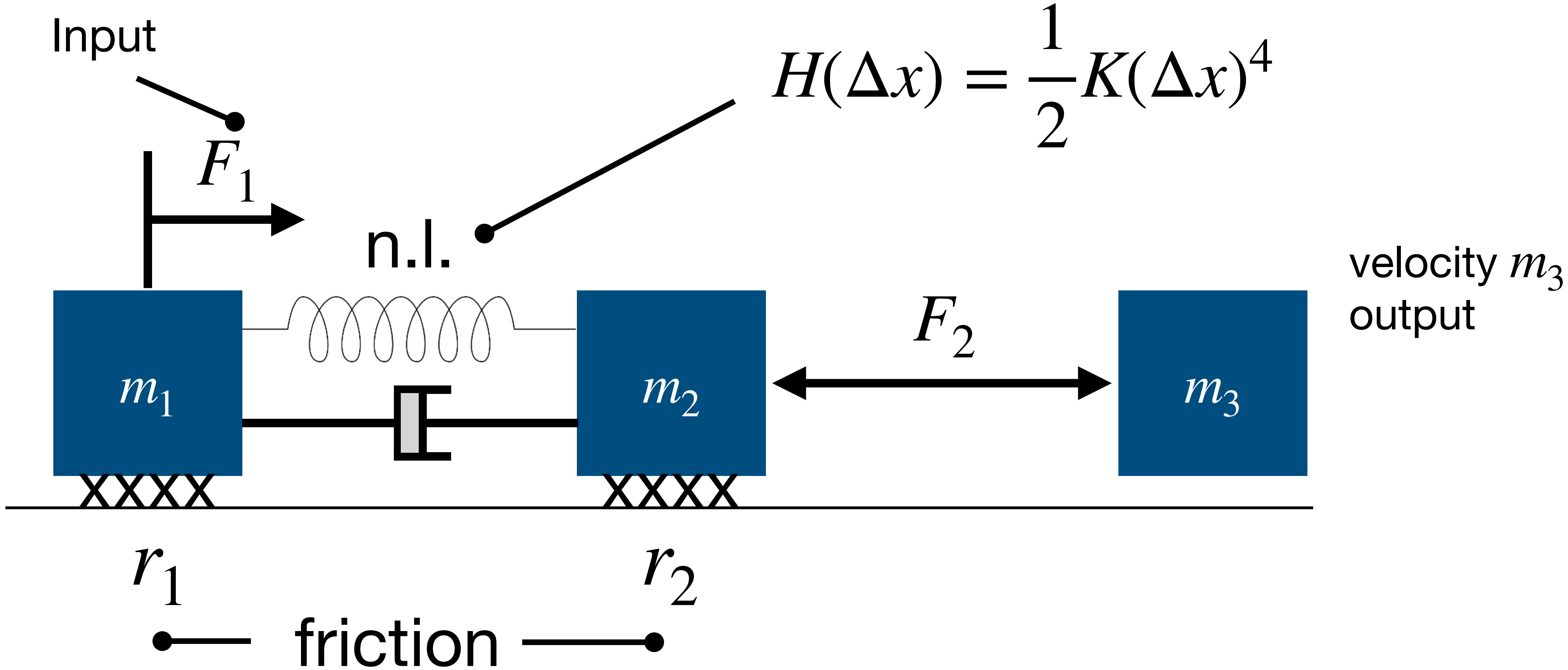
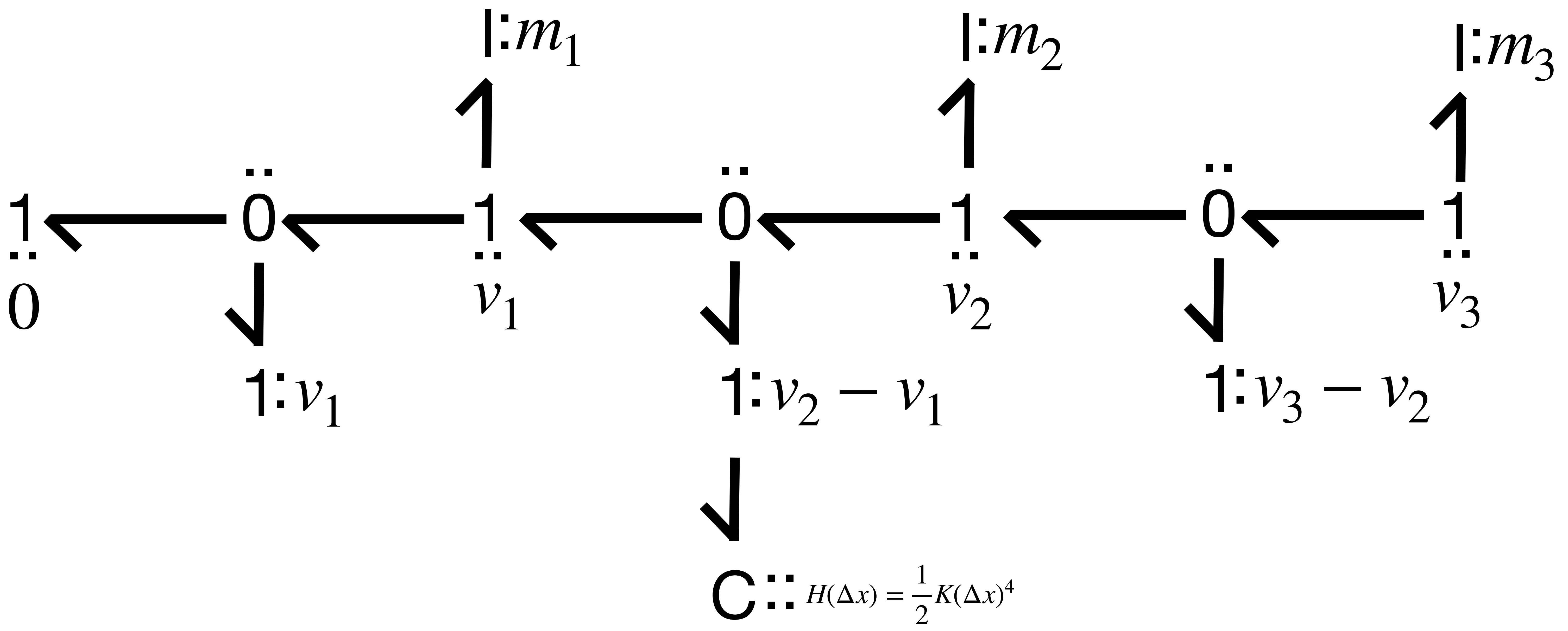
2. Add relative velocities needed



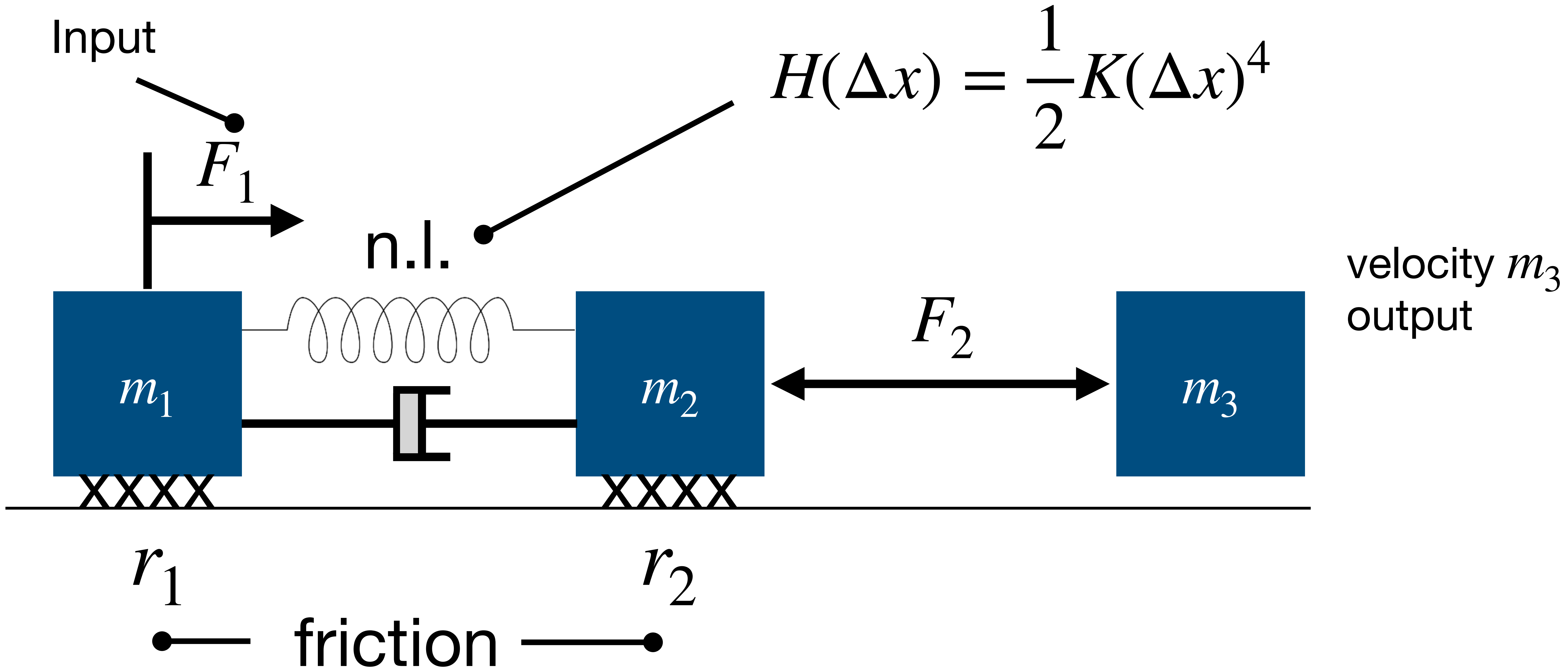


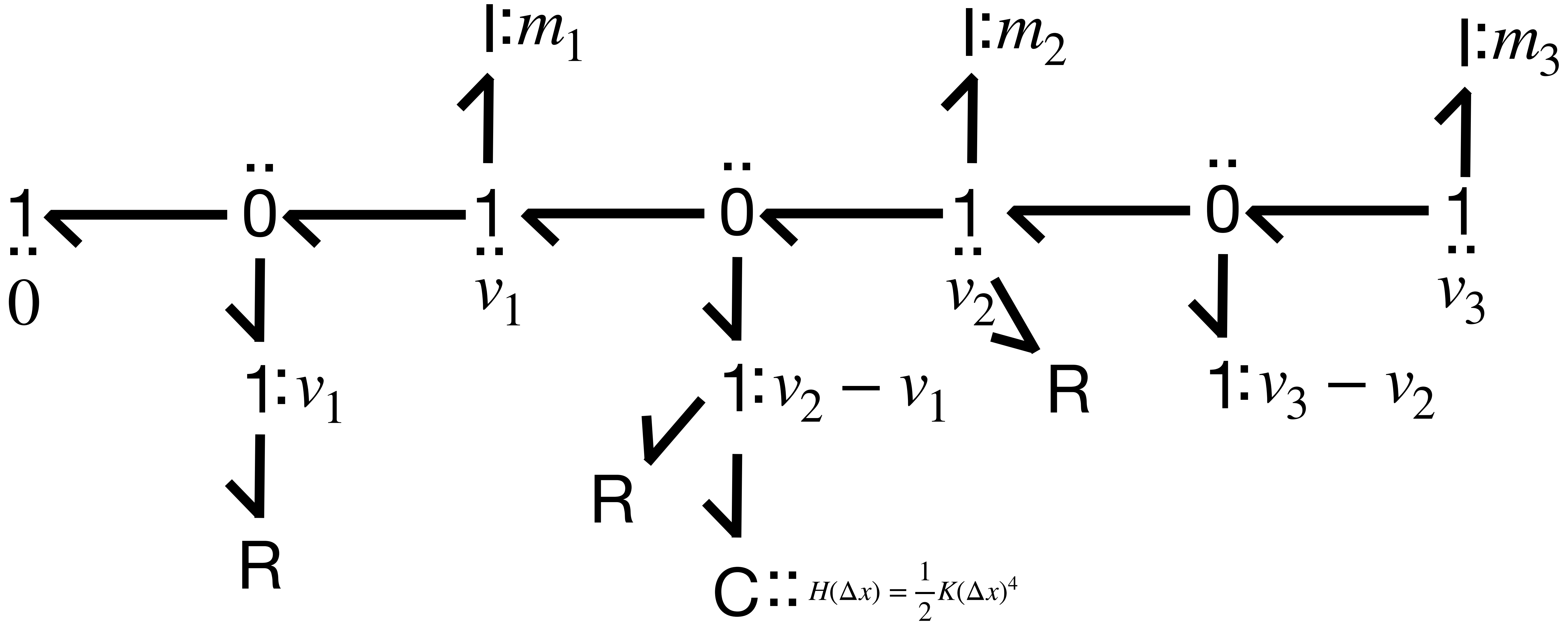
3. Add Storages





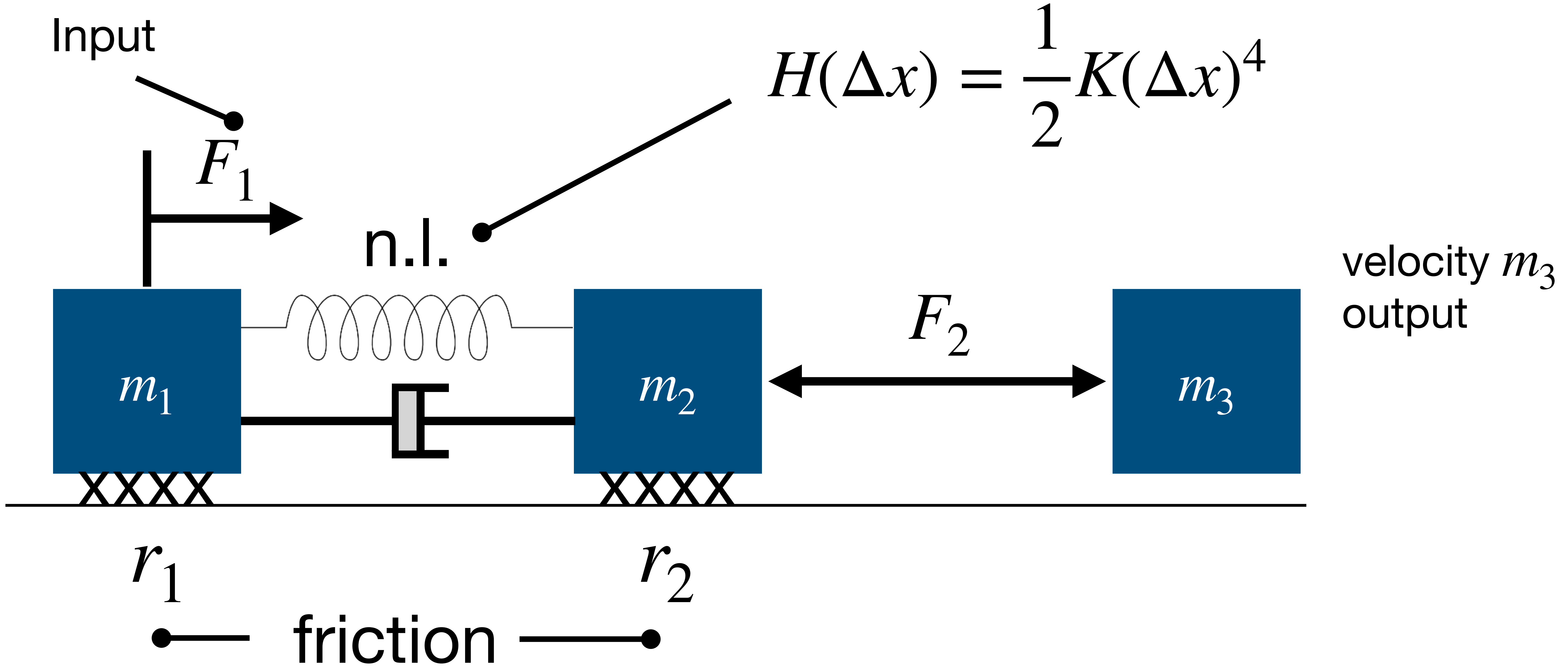
4. Add “Dissipations”

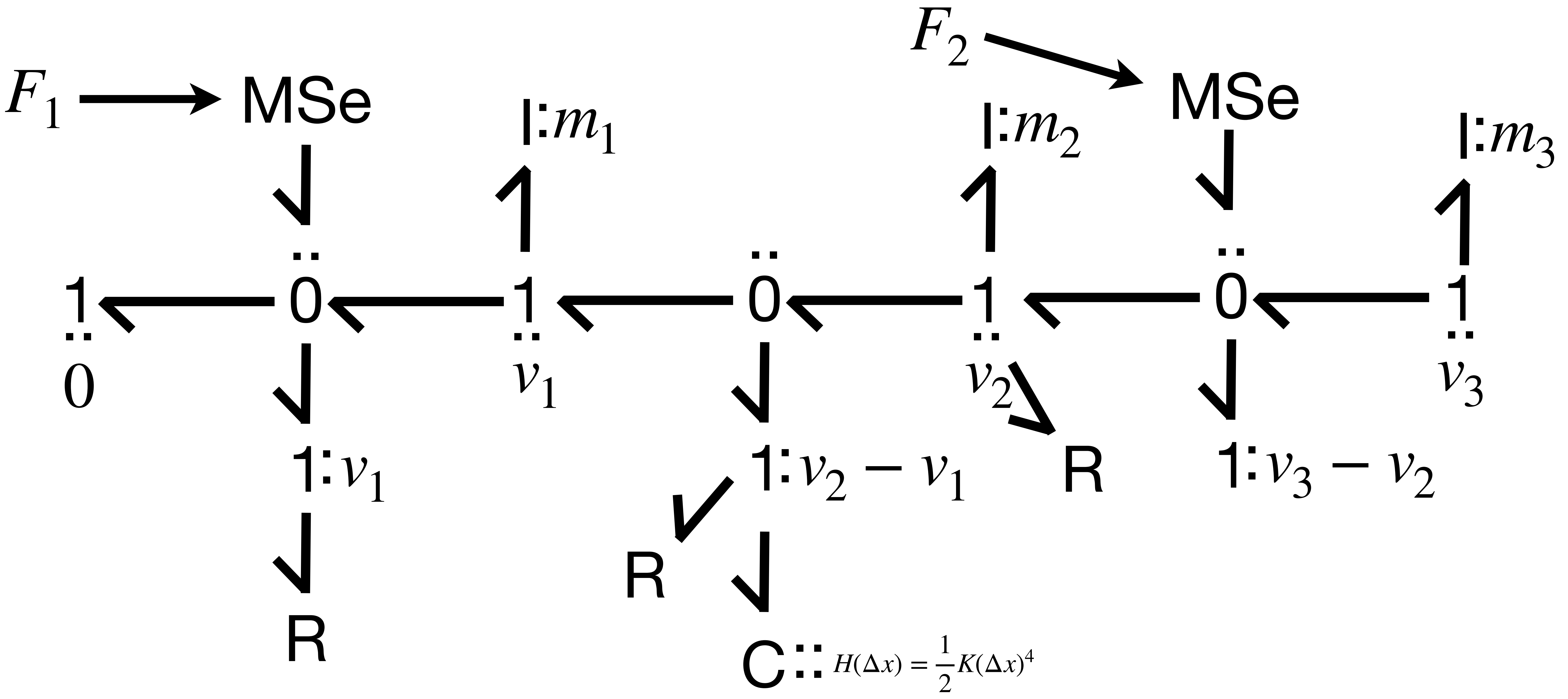




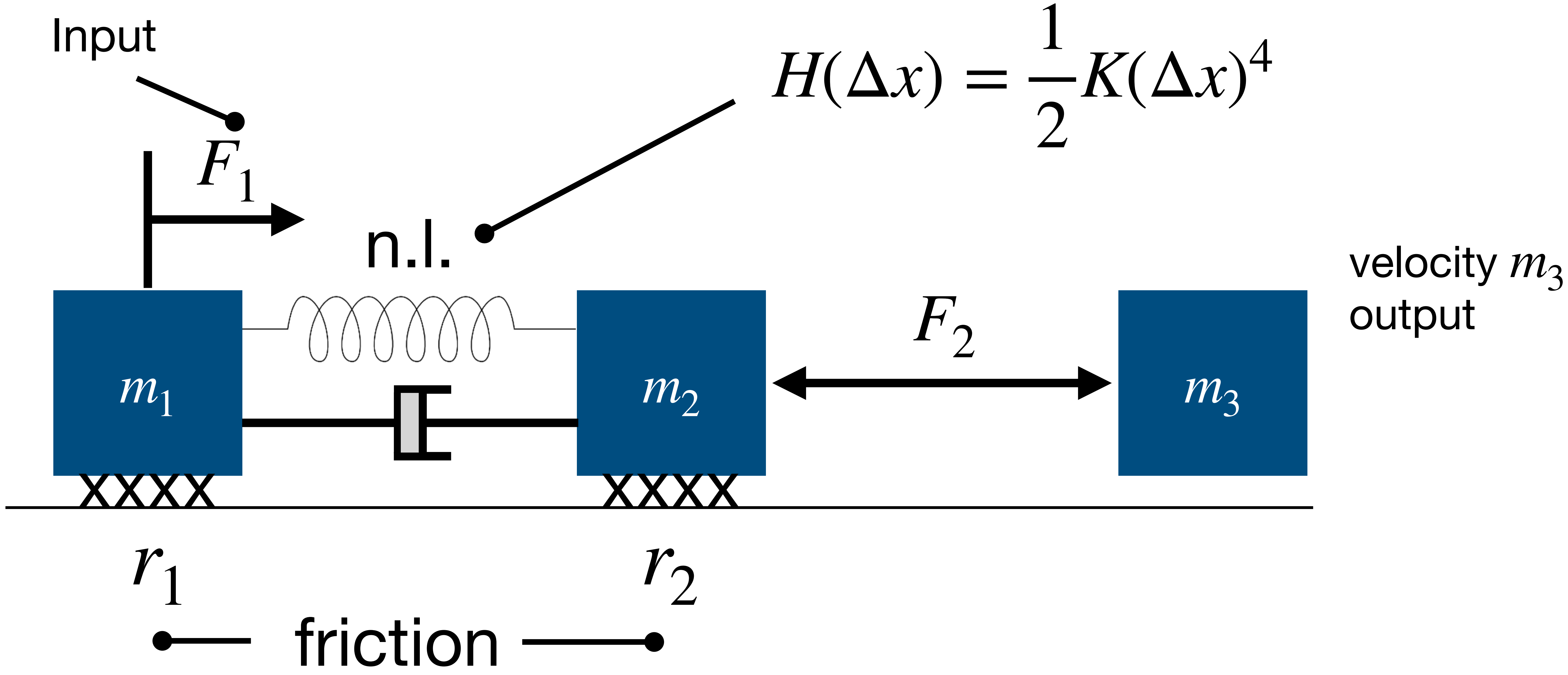
5. Add Sources

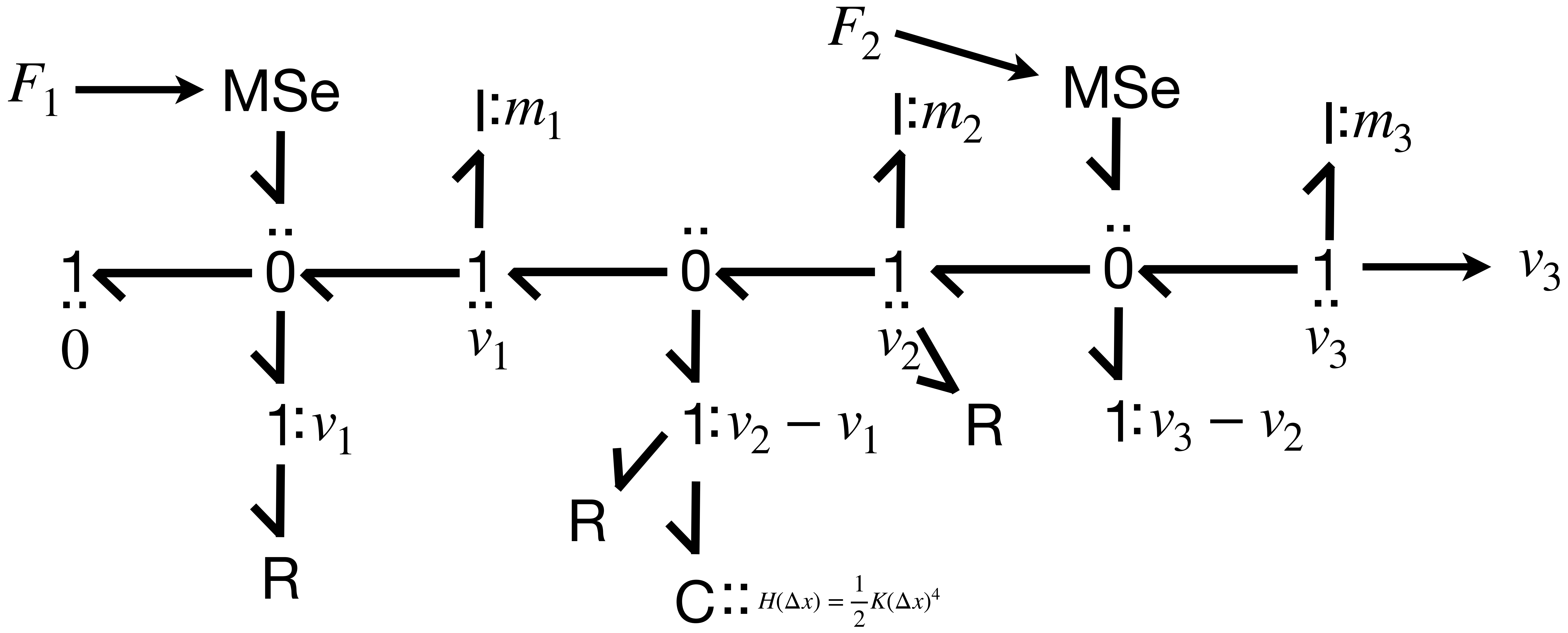
(connection to signal world!)



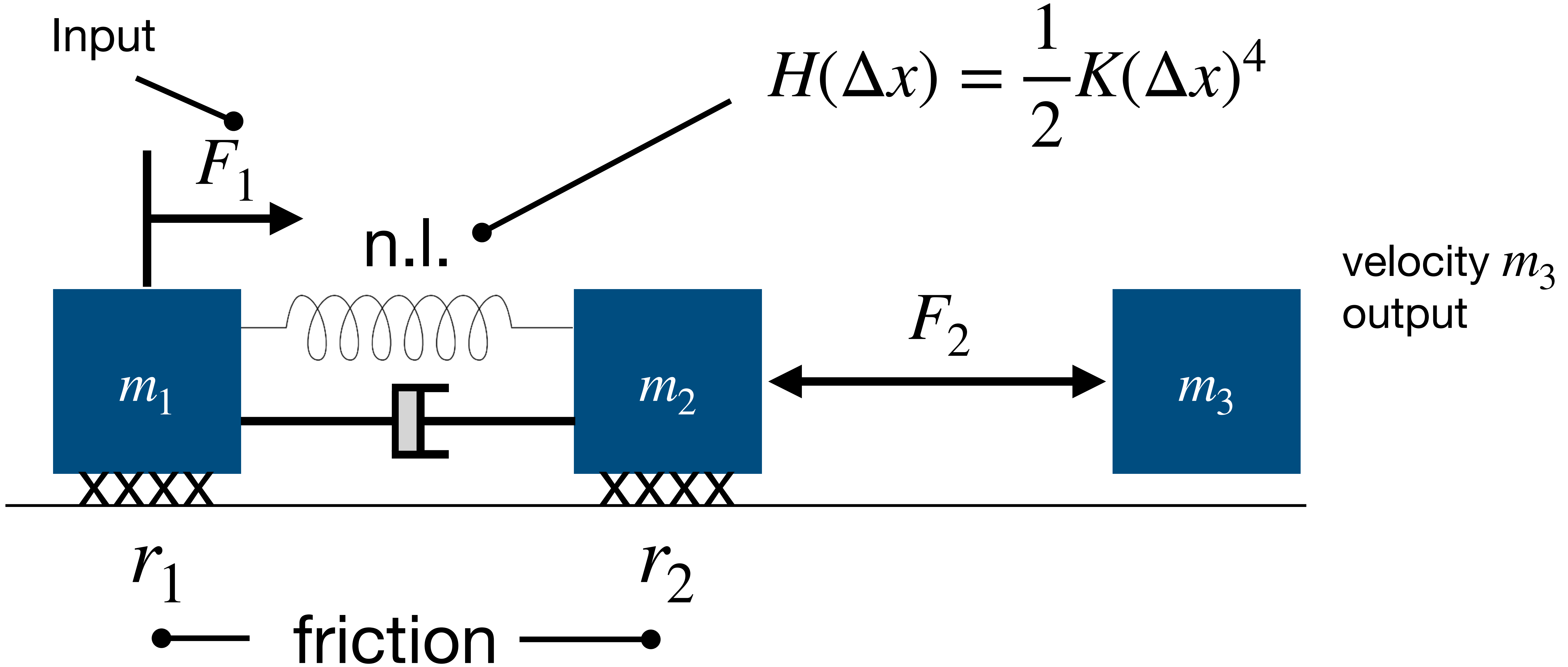


6. Add Outputs
(connection to signal world!)

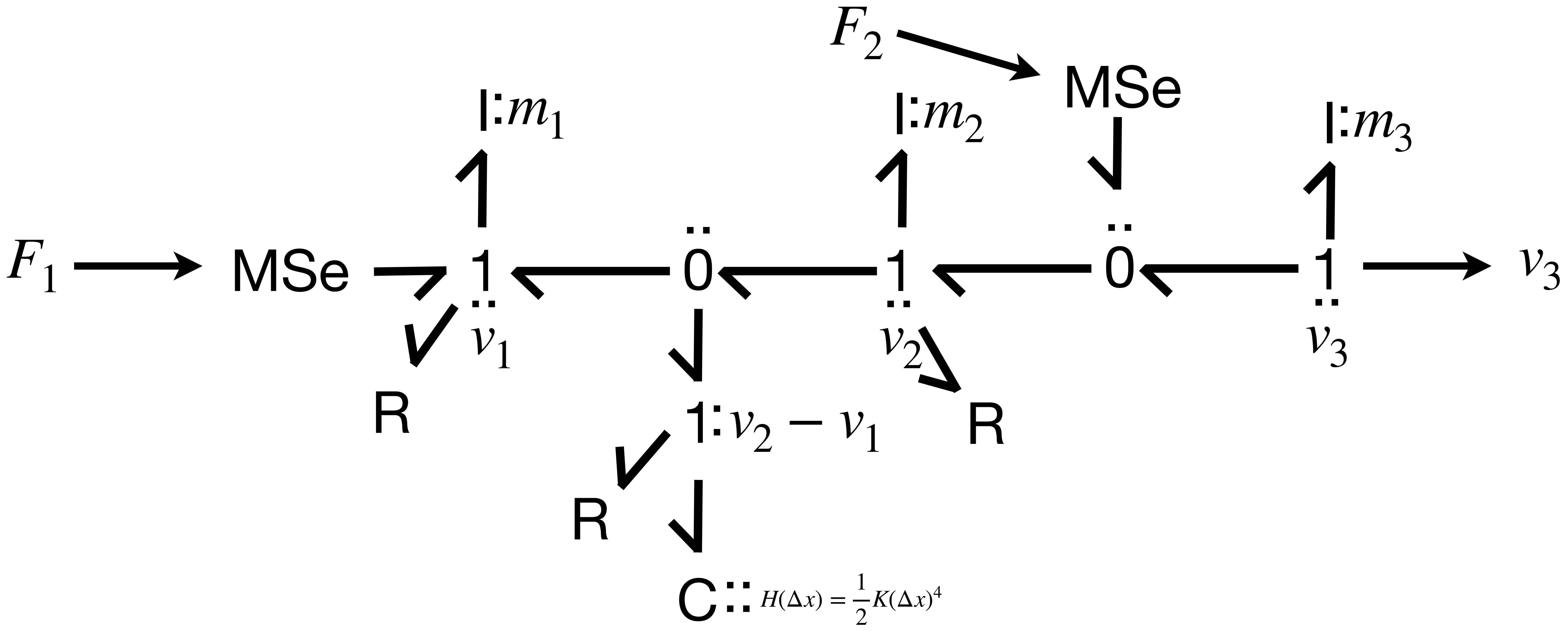




7. Simplify

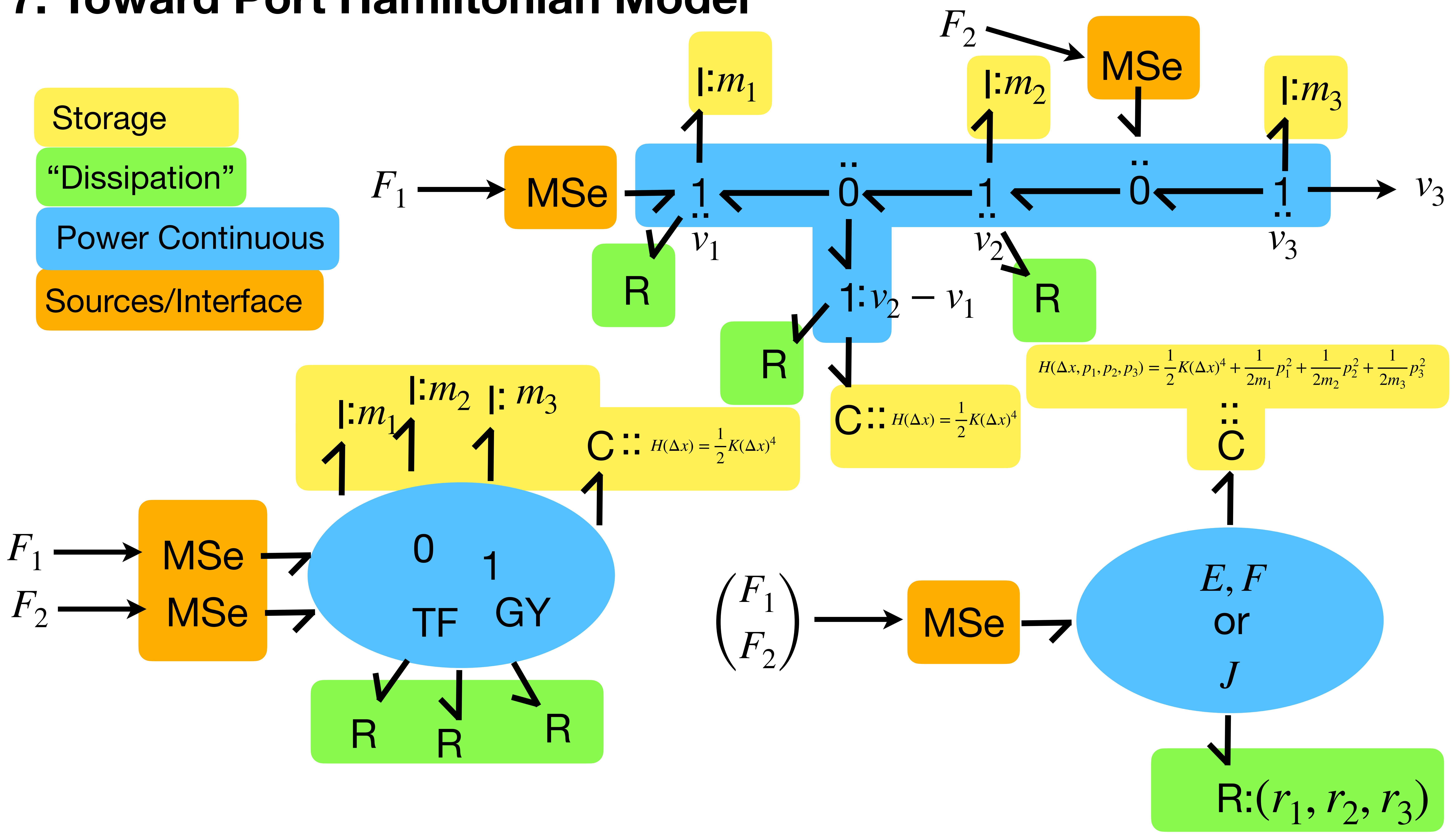


Input
F_1
n.l.
$H(\Delta x) = \frac{1}{2}K(\Delta x)^4$
m_1
m_2
F_2
m_3
velocity m_3
output
r_1
r_2
friction



F_2
MSe
I:m_1
I:m_2
I:m_3
F_1
MSe
1
0
1
0
1
v_3
v_1
v_2
v_3
R
R
1:$v_2 - v_1$
R
C:: $H(\Delta x) = \frac{1}{2}K(\Delta x)^4$

7. Toward Port Hamiltonian Model



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Conclusions

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Conclusions

- **IMPORTANT** and **FUNDAMENTAL** concepts of mathematics have been briefly introduced because **ESSENTIAL** for properly understand what is needed for **PHYSICAL MODELLING** and why.
- Concepts of **Port Based** modelling have been introduced together with basics of the **Bond Graphs** language and ideas of **Port Hamiltonian** systems
- You will see during the course that such concepts will be used again and again and generalise to work with **Serious Robotics**
- Thanks to such tools **Multibody 6D will almost be a trivial extension of 1D** because the structures will be leading!